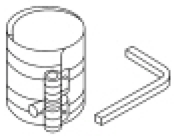
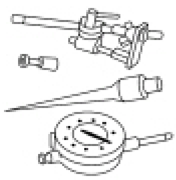
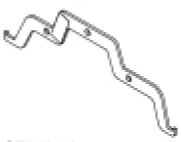
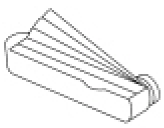
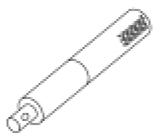
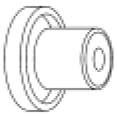
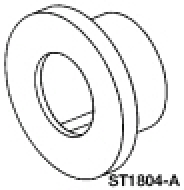
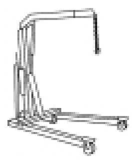
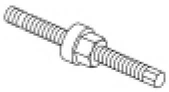
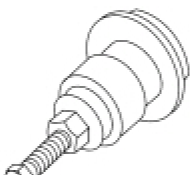
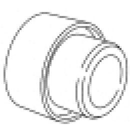
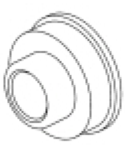
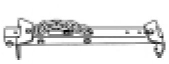
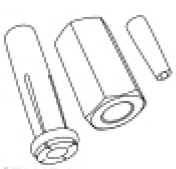
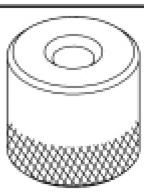
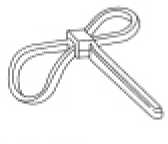
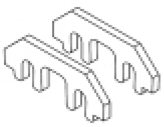
Engine

Ford Ecat Electronic Parts Catalog

Special Tool(s)

 ST1376-A	Compressor, Piston Ring 303-D032 (D81L-6002-C)
 ST1214-A	Dial Indicator Gauge with Holding Fixture 100-002 (TOOL-4201-C)
 ST3101-A	Bar, Engine Spreader 303-1454
 ST1271-A	Feeler Gauge Set 303-D027 (D81L-4201-A) or equivalent
 ST1326-A	Handle 205-153 (T80T-4000-W)
 ST3234-A	Hold Down, Secondary Chain 303-1530
 ST2212-A	Installer, Differential Bearing Cone 205-142 (T80T-4000-J)

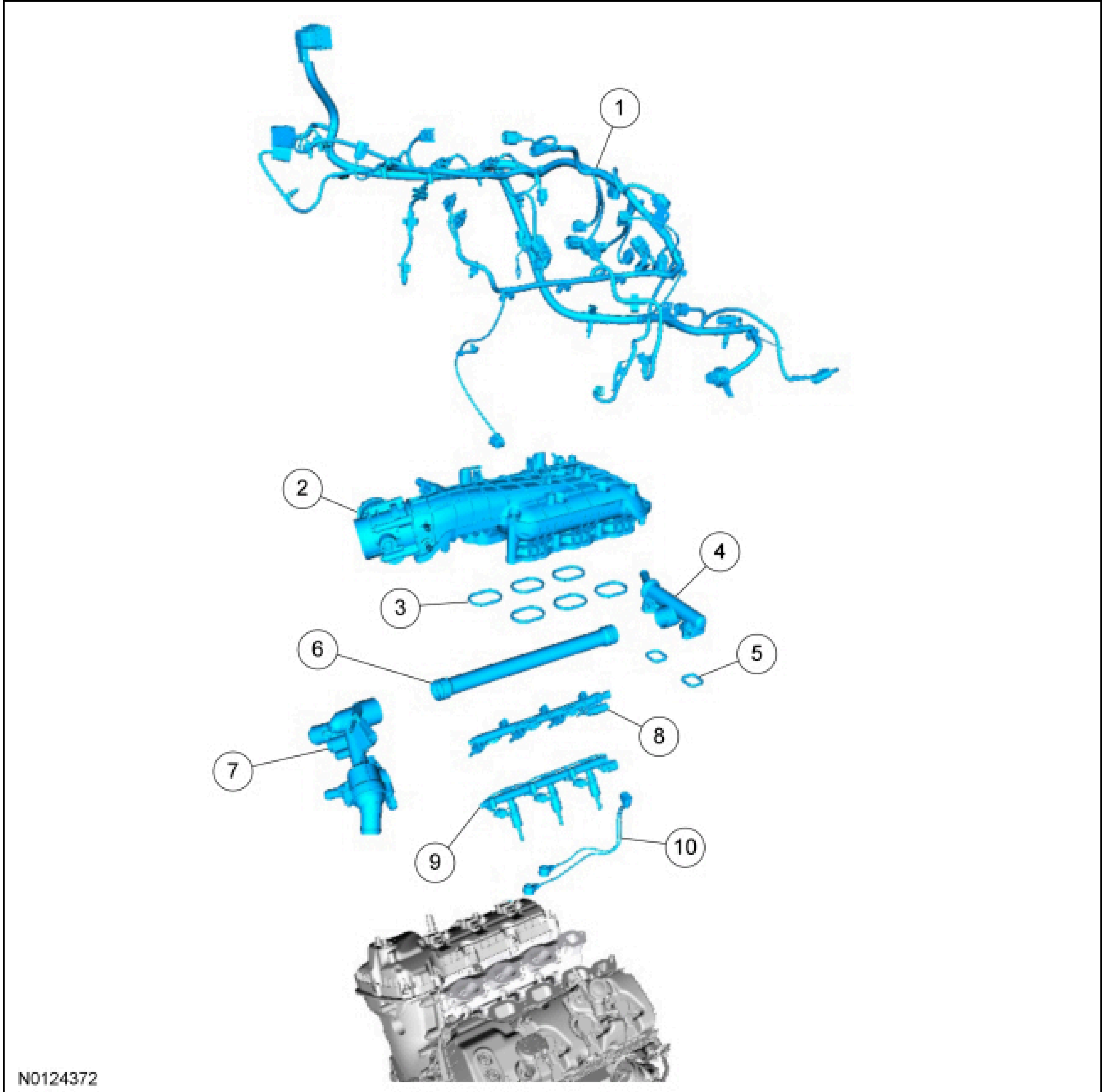
 ST1804-A	Installer, Spindle Bearing 205-149 (T80T-4000-R)
 ST1341-A	2,200# Floor Crane, Fold Away 300-OTC1819E or equivalent
 ST1287-A	Installer, Crankshaft Vibration Damper 303-102 (T74P-6316-B)
 ST2296-A	Installer, Front Cover Oil Seal 303-335
 ST2981-A	Installer, Front Crankshaft Seal 303-1251
 ST2983-A	Installer, VCT Spark Plug Tube Seal 303-1247/2
 ST1377-A	Lifting Bracket, Engine 303-F047 (014-00073) or equivalent
 ST3097-A	Seal Installer, Fuel Injector 310-207
 ST3266-A	Sizer, Teflon® Seal 303-1567

 ST1438-A	Strap Wrench 303-D055 (D85L-6000-A)
 ST2979-A	Tool, Camshaft Holding 303-1248

Material

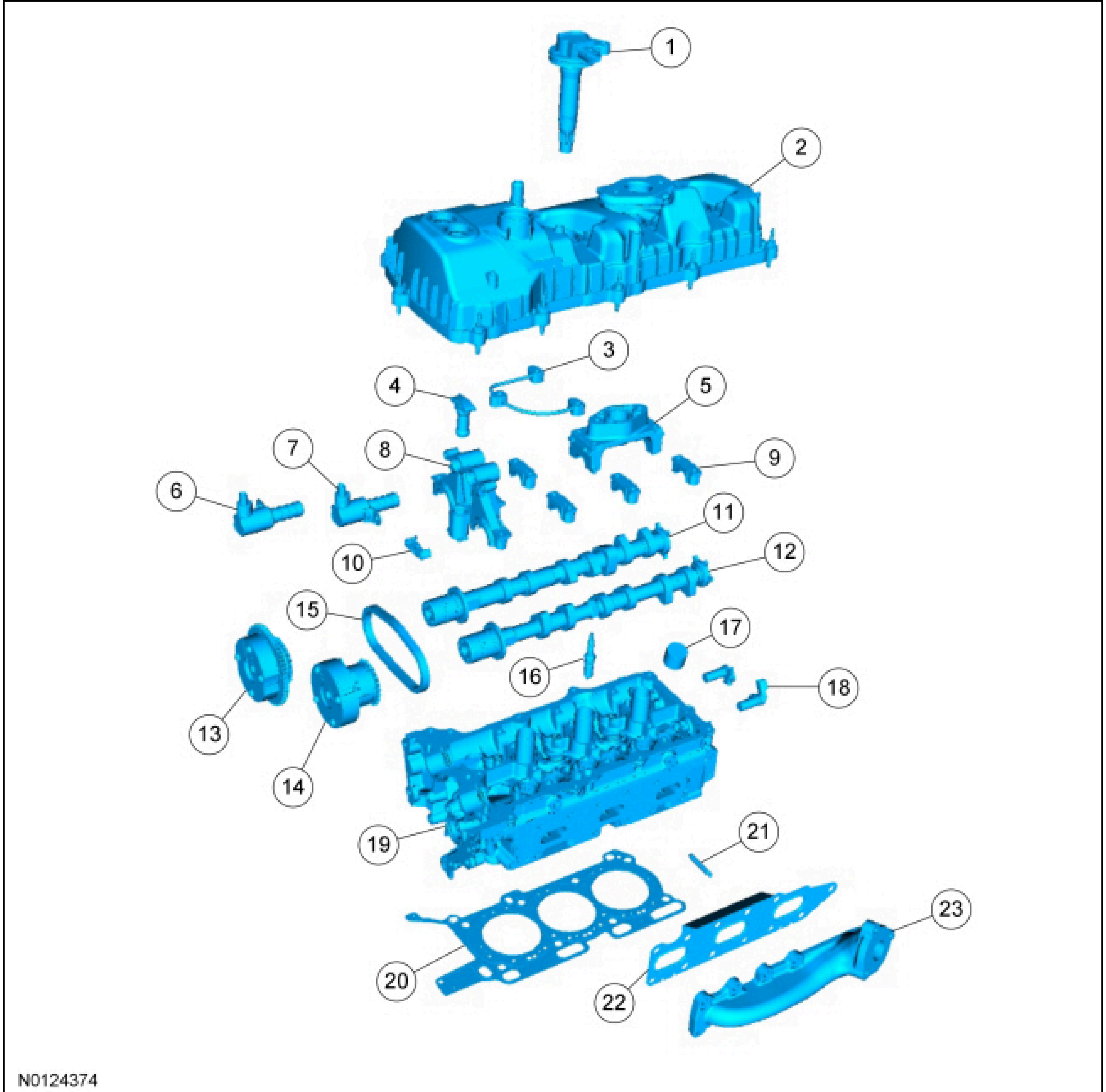
Item	Specification
Motorcraft® Gasket Maker TA-16	WSK-M2G348-A5
Motorcraft® High Performance Engine RTV Silicone TA-357	WSE-M4G323-A6
Motorcraft® SAE 5W-30 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada)	WSS-M2C946-A
Motorcraft® Orange Concentrated Antifreeze/Coolant VC-3-B (US); CVC-3-B2 (Canada)	WSS-M97B44-D
Motorcraft® Thread Sealant with PTFE TA-24-B	WSK-M2G350-A2

Engine Upper



Item	Part Number	Description
1	12A58112A581	Engine control harness
2	94249424	Intake manifold
3	94399439	Intake manifold gasket
4	8C3688C368	Coolant crossover
5	—	Coolant crossover gasket (2 required)
6	8A5058A505	Thermostat housing inlet tube
7	85928592	Thermostat housing
8	9D2809D280	RH Fuel rail
9	9D2809D280	LH Fuel rail
10	12A69912A699	Knock Sensor (KS)

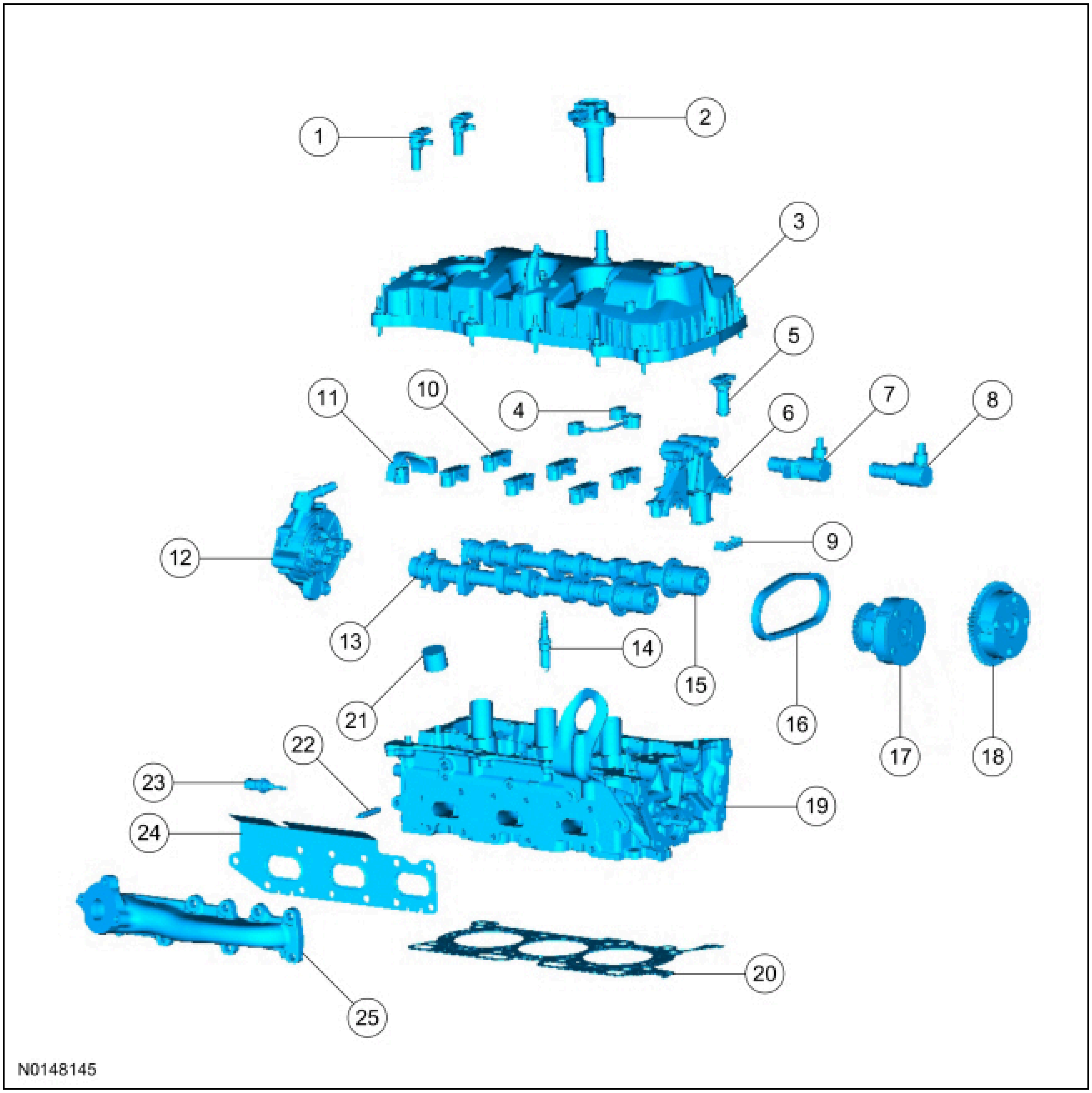
Engine Upper — LH Cylinder Head



Item	Part Number	Description
1	1202912029	Ignition coil-on-plug (3 required)
2	65826582	LH valve cover
3	6K6026K602	Valve train oil feed tube
4	6K2546K254	LH secondary timing chain tensioner
5	6A2586A258	LH camshaft cap
6	6M2806M280	Intake camshaft Variable Camshaft Timing (VCT) oil control solenoid
7	6M2806M280	Exhaust camshaft VCT oil control solenoid
8	6B2806B280	Camshaft mega cap
9	6A2586A258	LH camshaft cap (4 required)
10	6K2976K297	LH secondary tensioner shoe
11	62506250	LH intake camshaft
12	62506250	LH exhaust camshaft
13	62566256	Intake camshaft VCT assembly

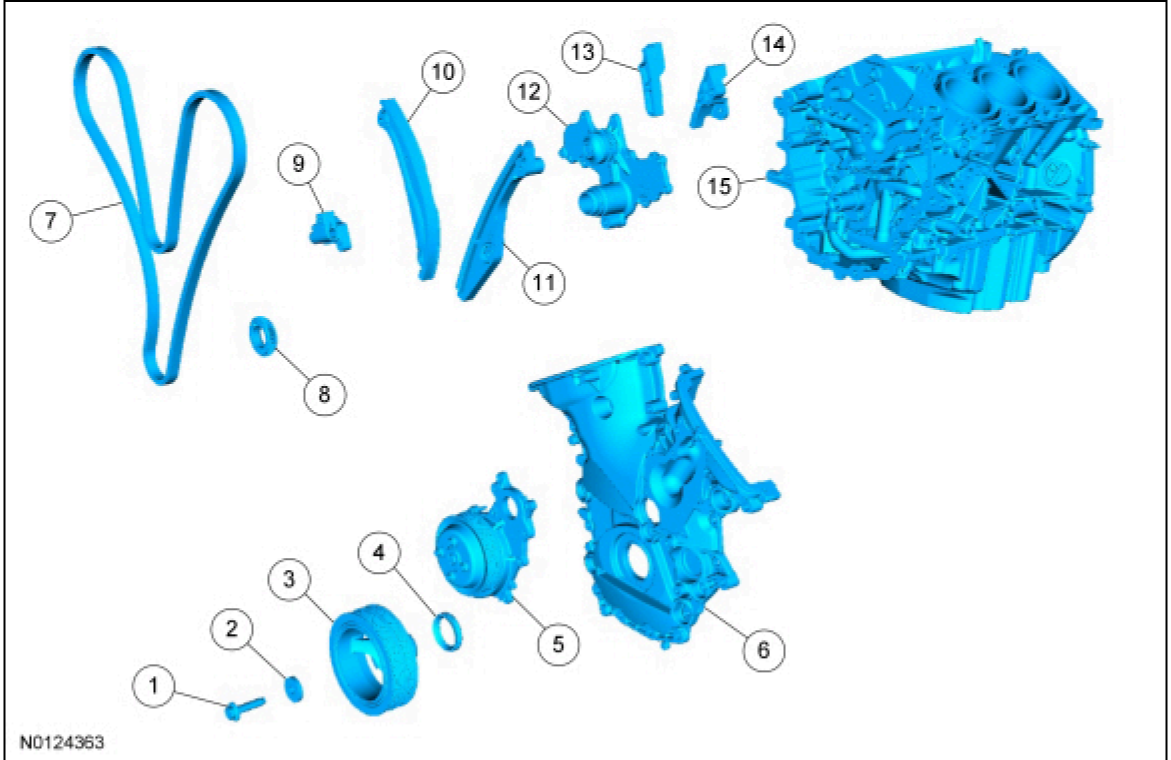
14	6C5256C525	Exhaust camshaft VCT assembly
15	62686268	LH secondary timing chain
16	1240512405	Spark plug (3 required)
17	65006500	Valve tappet (16 required)
18	6B2886B288	LH cylinder head Camshaft Position (CMP) sensor (2 required)
19	60496049	LH cylinder head
20	60516051	LH cylinder head gasket
21	W712244W712244	LH exhaust manifold stud (8 required)
22	94489448	LH exhaust manifold gasket
23	94319431	LH exhaust manifold

Engine Upper — RH Cylinder Head



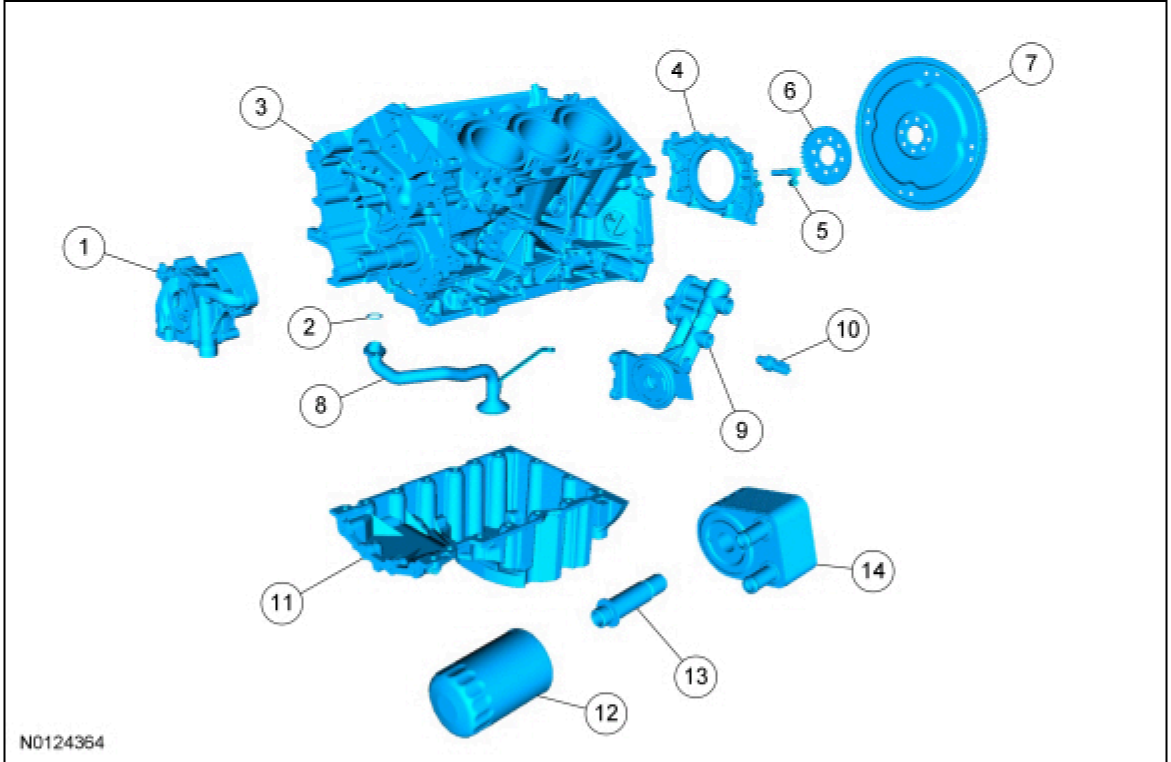
Item	Part Number	Description
1	6B2886B288	RH cylinder head Camshaft Position (CMP) sensor (2 required)
2	1202912029	Ignition coil-on-plug (3 required)
3	65826582	RH valve cover
4	6K6026K602	Valve train oil feed tube
5	6K2546K254	RH secondary timing chain tensioner
6	6B2806B280	Camshaft mega cap
7	6M2806M280	Exhaust camshaft Variable Camshaft Timing (VCT) oil control solenoid
8	6M2806M280	Intake camshaft VCT oil control solenoid
9	6K2976K297	RH secondary tensioner shoe
10	6A2586A258	RH camshaft cap (6 required)
11	6B2846B284	RH rear camshaft bearing cap assembly
12	2A2512A251	Vacuum pump
13	62506250	RH exhaust camshaft
14	1240512405	Spark plug (3 required)
15	62506250	RH intake camshaft
16	62686268	RH secondary timing chain
17	6C5256C525	Exhaust camshaft VCT assembly
18	62566256	Intake camshaft VCT assembly
19	60496049	RH cylinder head
20	60516051	RH cylinder head gasket
21	65006500	Valve tappet (16 required)
22	W712244W712244	RH exhaust manifold stud (8 required)
23	6G0046G004	Cylinder Head Temperature (CHT) sensor
24	94489448	RH exhaust manifold gasket
25	94309430	RH exhaust manifold

Engine Front



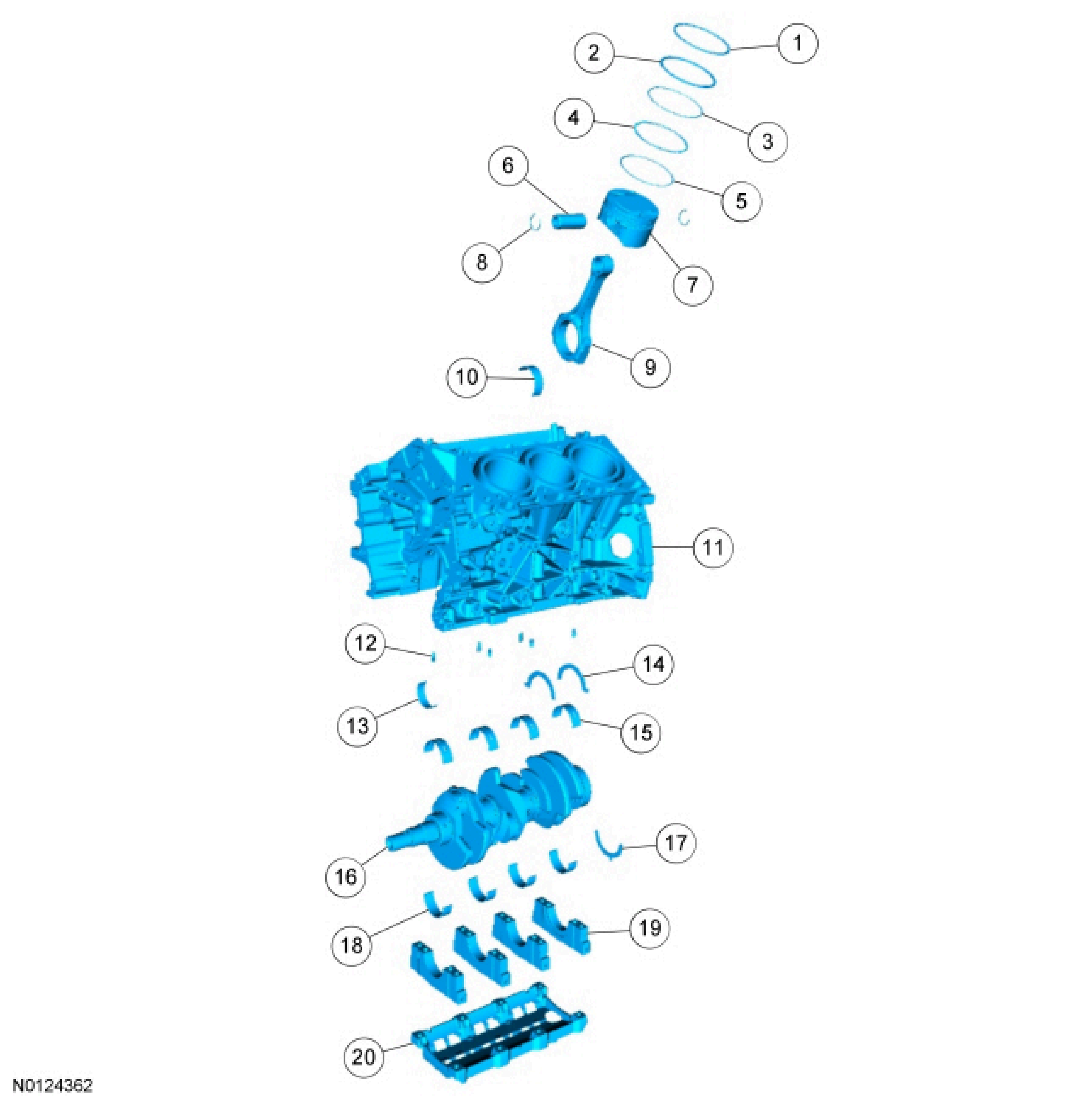
Item	Part Number	Description
1	6A3406A340	Crankshaft pulley bolt
2	63786378	Crankshaft pulley washer
3	63126312	Crankshaft pulley
4	67006700	Crankshaft front seal
5	85018501	Coolant pump
6	60196019	Engine front cover
7	62686268	Timing chain
8	63066306	Crankshaft timing sprocket
9	6K2546K254	Primary timing chain tensioner
10	6K2556K255	Primary timing chain tensioner arm
11	6B2746B274	LH lower primary timing chain guide
12	85018501	Timing chain gear
13	6M2566M256	RH primary timing chain guide
14	6K2976K297	LH upper primary timing chain guide
15	60106010	Cylinder block

Lower Engine Block (View 1)



Item	Part Number	Description
1	66006600	Oil pump
2	66256625	Oil pump screen and pickup tube O-ring seal
3	60106010	Cylinder block
4	6K3016K301	Crankshaft rear seal retainer
5	6C3156C315	Crankshaft Position (CKP) sensor
6	12A22712A227	Ignition pulse ring
7	63756375	Flexplate
8	66226622	Oil pump screen and pickup tube
9	68816881	Oil filter adapter
10	92789278	Engine Oil Pressure (EOP) switch
11	66756675	Oil pan
12	67316731	Engine oil filter
13	6L6266L626	Oil cooler insert bolt
14	6A6426A642	Oil cooler

Lower Engine Block (View 2)



Item	Part Number	Description
1	—	Piston compression upper ring (part of 6148) (6 required)
2	—	Piston compression lower ring (part of 6148) (6 required)
3	—	Piston oil control upper segment ring (part of 6148) (6 required)
4	—	Piston oil control spacer (part of 6148) (6 required)
5	—	Piston oil control lower segment ring (part of 6148) (6 required)
6	61406140	Piston pin (6 required)
7	61106110	Piston (6 required)
8	61356135	Piston pin retainer (12 required)
9	62006200	Connecting rod (6 required)
10	62116211	Connecting rod upper bearing (6 required)
11	60106010	Cylinder block
12	6K8686K868	Piston oil cool valve (6 required)
13	62116211	Connecting rod lower bearing (6 required)
14	6A3416A341	Crankshaft upper thrust washer (2 required)

15	6D3096D309	Cylinder block crankshaft main bearing (4 required)
16	63036303	Crankshaft
17	6K3026K302	Crankshaft lower thrust washer
18	6D3096D309	Lower crankshaft main bearing (4 required)
19	63256325	Lower crankshaft main bearing cap (4 required)
20	6C3646C364	Main bearing cap support brace

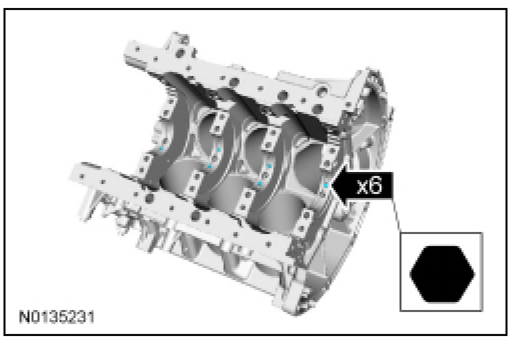
NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

NOTICE: The engine is equipped with a cast aluminum (early build) or stamped steel (late build) crankshaft rear seal retainer plate. All crankshaft rear seal retainer plates must be discarded and a new stamped steel rear seal retainer plate installed.

NOTE: *Assembly of the engine requires various inspections/measurements of the engine components (engine block, crankshaft, connecting rods, pistons and piston rings). These inspections/measurements will aid in determining if the engine components will require replacement. For additional information, refer to Section 303-00.*

All vehicles

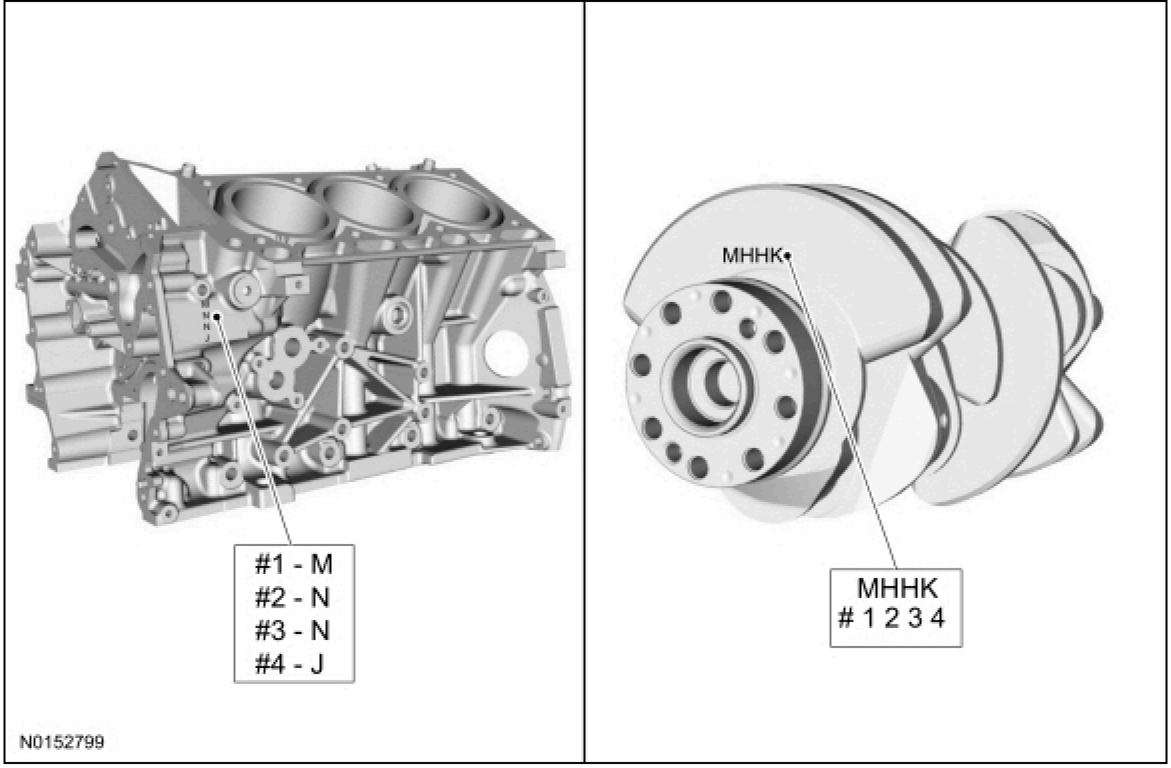
1. Using a hexagonal screwdriver, install the 6 piston oil cool valves.
    - Tighten to 4 Nm (35 lb-in).



2. **NOTE:** *This step is for selecting the correct main bearings.*

    NOTE: *The #1 main bearing is at the front of the engine block.*

    Select the crankshaft main bearings for each crankshaft journal.
    - Record the code that is on the cylinder block face.
    - Record the code that is on the crankshaft flange.
    - The first letter is for main No. 1 and the next letters are for mains 2, 3 and 4.



3. **NOTE:** *This chart is continued in the next step.*

   NOTE: *This chart is for selecting main bearings 1 and 4 only.*

   Using the data recorded earlier and the Bearing Select Fit Chart, Standard Bearings, determine the required bearing grade for main bearings 1 and 4.

   - Read the first letter of the engine block main bearing code and the first letter of the crankshaft main bearing code.
   - Read down the column below the engine block main bearing code letter and across the row next to the crankshaft main bearing code letter, until the 2 intersect. This is the required bearing grade(s) for the No. 1 crankshaft main bearing.
   - As an example, if the engine block code letter is "M" and the crankshaft code letter is "M", the correct bearing grade for this main bearing is a "2" for the upper bearing and a "2" for the lower bearing.
   - Repeat the above steps using the fourth letter of the block and crankshaft codes to select the No. 4 bearing.

MAIN HOUSING BORE DIAMETER

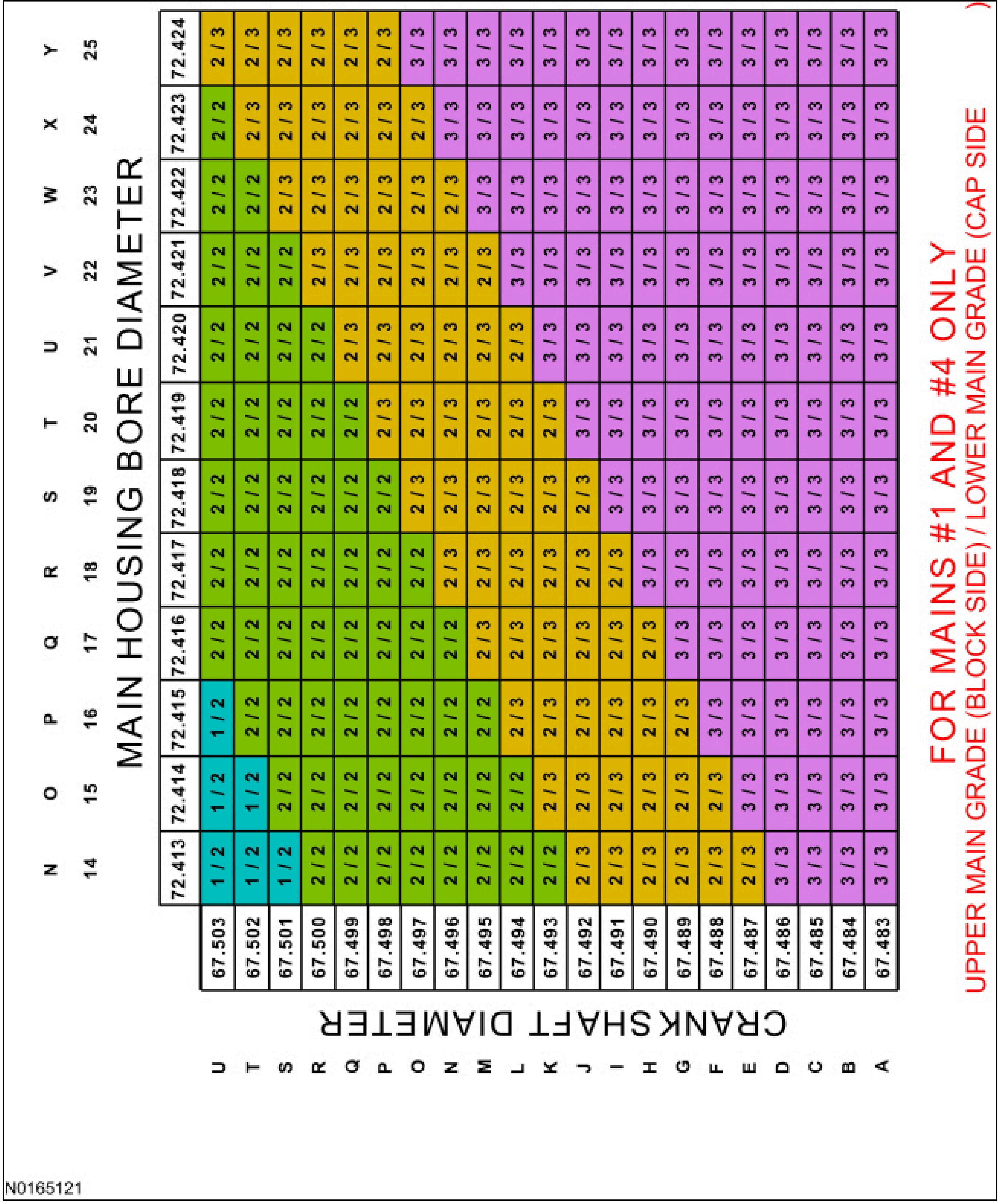
FOR MAINS #1 AND #4 ONLY

UPPER MAIN GRADE (BLOCK SIDE) / LOWER MAIN GRADE (CAP SIDE)

			A	B	C	D	E	F	G	H	I	J	K	L	M
			1	2	3	4	5	6	7	8	9	10	11	12	13
CRANKSHAFT DIAMETER			72.400	72.401	72.402	72.403	72.404	72.405	72.406	72.407	72.408	72.409	72.410	72.411	72.412
	U	67.503	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2
	T	67.502	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2
	S	67.501	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2
	R	67.500	1 / 1	1 / 1	1 / 1	1 / 1	1 / 1	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2
	Q	67.499	1 / 1	1 / 1	1 / 1	1 / 1	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2
	P	67.498	1 / 1	1 / 1	1 / 1	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2
	O	67.497	1 / 1	1 / 1	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2
	N	67.496	1 / 1	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2
	M	67.495	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2
	L	67.494	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2
	K	67.493	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2
	J	67.492	1 / 2	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2
	I	67.491	1 / 2	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3
	H	67.490	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3
	G	67.489	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3
	F	67.488	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3
	E	67.487	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3
	D	67.486	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3
	C	67.485	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3
	B	67.484	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3
	A	67.483	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3

N0165120

4. **NOTE:** *This chart is for selecting main bearings 1 and 4 only.*



FOR MAINS #1 AND #4 ONLY

UPPER MAIN GRADE (BLOCK SIDE) / LOWER MAIN GRADE (CAP SIDE)

MAIN HOUSING BORE DIAMETER

CRANKSHAFT DIAMETER		N 14 72.413	O 15 72.414	P 16 72.415	Q 17 72.416	R 18 72.417	S 19 72.418	T 20 72.419	U 21 72.420	V 22 72.421	W 23 72.422	X 24 72.423	Y 25 72.424
U	67.503	1 / 2	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3
T	67.502	1 / 2	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3
S	67.501	1 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3
R	67.500	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3
Q	67.499	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3
P	67.498	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3
O	67.497	2 / 2	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3
N	67.496	2 / 2	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3
M	67.495	2 / 2	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3
L	67.494	2 / 2	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3	3 / 3
K	67.493	2 / 2	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
J	67.492	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
I	67.491	2 / 3	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
H	67.490	2 / 3	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
G	67.489	2 / 3	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
F	67.488	2 / 3	2 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
E	67.487	2 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
D	67.486	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
C	67.485	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
B	67.484	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3
A	67.483	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3	3 / 3

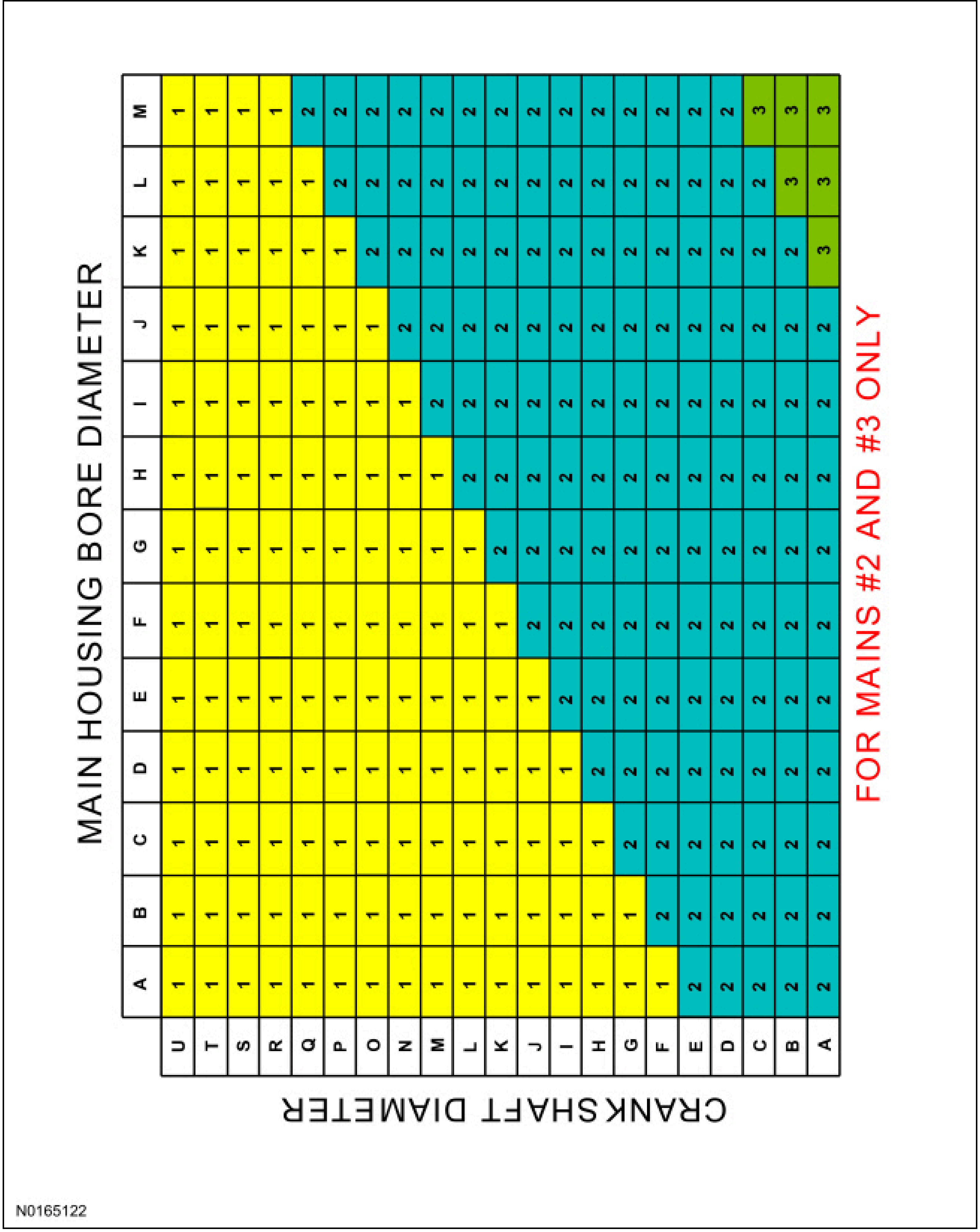
N0165121

5. **NOTE:** *This chart is continued in the next step.*

   NOTE: *This chart is for selecting main bearings 2 and 3 only.*

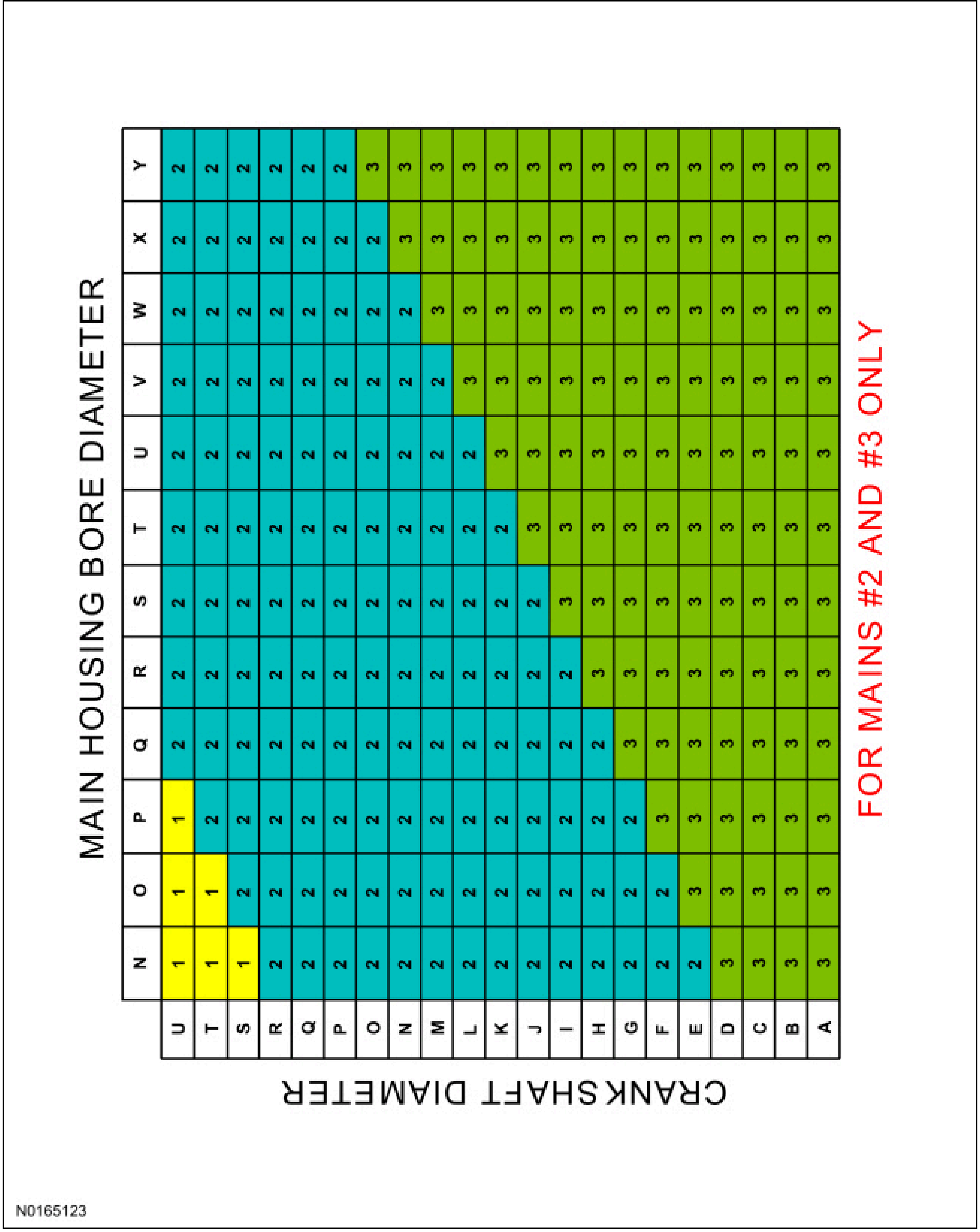
   Using the data recorded earlier and the Bearing Select Fit Chart, Standard Bearings, determine the required bearing grade for main bearings 2 and 3.

   - Read the second letter of the engine block main bearing code and the second letter of the crankshaft main bearing code.
   - Read down the column below the engine block main bearing code letter and across the row next to the crankshaft main bearing code letter, until the 2 intersect. This is the required bearing grade for the No. 2 crankshaft main bearing.
   - As an example, if the engine block code letter is "N" and the crankshaft code letter is "H", the correct bearing grade for this main bearing is "2".
   - Repeat the above steps using the third letter of the block and crankshaft codes to select the No. 3.



Crankshaft Diameter \ Main Housing Bore Diameter	A	B	C	D	E	F	G	H	I	J	K	L	M
U	1	1	1	1	1	1	1	1	1	1	1	1	1
T	1	1	1	1	1	1	1	1	1	1	1	1	1
S	1	1	1	1	1	1	1	1	1	1	1	1	1
R	1	1	1	1	1	1	1	1	1	1	1	1	1
Q	1	1	1	1	1	1	1	1	1	1	1	1	2
P	1	1	1	1	1	1	1	1	1	1	1	2	2
O	1	1	1	1	1	1	1	1	1	1	2	2	2
N	1	1	1	1	1	1	1	1	1	2	2	2	2
M	1	1	1	1	1	1	1	1	2	2	2	2	2
L	1	1	1	1	1	1	1	2	2	2	2	2	2
K	1	1	1	1	1	1	2	2	2	2	2	2	2
J	1	1	1	1	1	2	2	2	2	2	2	2	2
I	1	1	1	1	2	2	2	2	2	2	2	2	2
H	1	1	1	2	2	2	2	2	2	2	2	2	2
G	1	1	2	2	2	2	2	2	2	2	2	2	2
F	1	2	2	2	2	2	2	2	2	2	2	2	2
E	2	2	2	2	2	2	2	2	2	2	2	2	2
D	2	2	2	2	2	2	2	2	2	2	2	2	2
C	2	2	2	2	2	2	2	2	2	2	2	2	3
B	2	2	2	2	2	2	2	2	2	2	2	3	3
A	2	2	2	2	2	2	2	2	2	2	3	3	3

6. **NOTE:** *This chart is for selecting main bearings 2 and 3 only.*



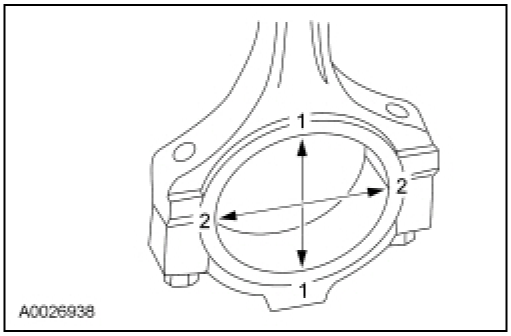
Crankshaft diameter	N	O	P	Q	R	S	T	U	V	W	X	Y
U	1	1	1	2	2	2	2	2	2	2	2	2
T	1	1	2	2	2	2	2	2	2	2	2	2
S	1	2	2	2	2	2	2	2	2	2	2	2
R	2	2	2	2	2	2	2	2	2	2	2	2
Q	2	2	2	2	2	2	2	2	2	2	2	2
P	2	2	2	2	2	2	2	2	2	2	2	2
O	2	2	2	2	2	2	2	2	2	2	2	3
N	2	2	2	2	2	2	2	2	2	2	3	3
M	2	2	2	2	2	2	2	2	2	3	3	3
L	2	2	2	2	2	2	2	2	3	3	3	3
K	2	2	2	2	2	2	2	3	3	3	3	3
J	2	2	2	2	2	2	3	3	3	3	3	3
I	2	2	2	2	2	3	3	3	3	3	3	3
H	2	2	2	2	3	3	3	3	3	3	3	3
G	2	2	2	3	3	3	3	3	3	3	3	3
F	2	2	3	3	3	3	3	3	3	3	3	3
E	2	3	3	3	3	3	3	3	3	3	3	3
D	3	3	3	3	3	3	3	3	3	3	3	3
C	3	3	3	3	3	3	3	3	3	3	3	3
B	3	3	3	3	3	3	3	3	3	3	3	3
A	3	3	3	3	3	3	3	3	3	3	3	3

7. **NOTICE: The rod cap installation must keep the same orientation as marked during disassembly or engine damage may occur.**

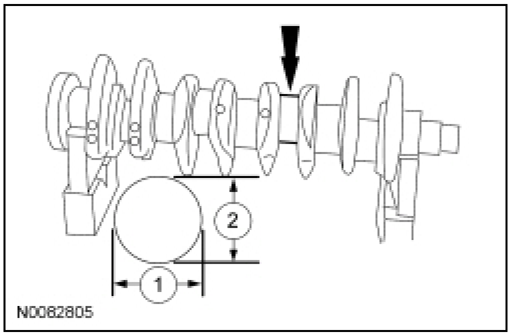
   Using the original connecting rod cap bolts, install the connecting rod caps and bolts.
   - Tighten the bolts in 3 stages.
   - Stage 1: Tighten to 23 Nm (17 lb-ft).
   - Stage 2: Tighten to 43 Nm (32 lb-ft).
   - Stage 3: Tighten an additional 90 degrees.

8. Measure the connecting rod large end bore in 2 directions. For additional information, refer to Specifications in this section.
   - Remove the bolts and the rod cap.

- Discard the connecting rod cap bolts.



9. Measure each of the crankshaft connecting rod bearing journal diameters in at least 2 directions. For additional information, refer to Specifications in this section.



10. Using the chart, select the correct connecting rod bearings for each crankshaft connecting rod journal.

CONNECTING ROD LARGE END BORE (59.XXX)

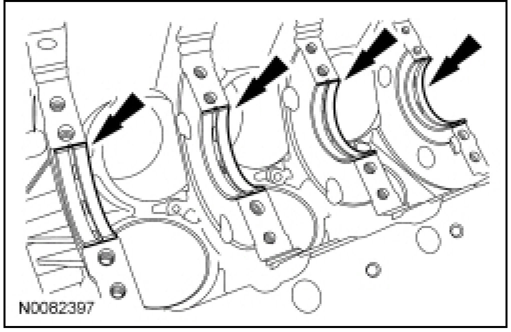
CRANK JOURNAL	.866	.867	.868	.869	.870	.871	.872	.873	.874	.875	.876	.877	.878	.879	.880	.881	.882	.883	.884	.885	.886
	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
56.003	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
56.002	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
56.001	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
56.000	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.999	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.998	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.998	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.998	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.995	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.994	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.993	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.992	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.991	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.990	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.989	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.988	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.987	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.986	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.985	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.984	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3
55.983	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3

WALL GRADES	MINIMUM	MAXIMUM	NOMINAL
1	1.918 mm	1.924 mm	1.921 mm
2	1.921 mm	1.927 mm	1.924 mm
3	1.925 mm	1.931 mm	1.928 mm

E228835

11. **NOTE:** *Before assembling the cylinder block, all sealing surfaces must be free of chips, dirt, paint and foreign material. Also, make sure the coolant and oil passages are clear.*

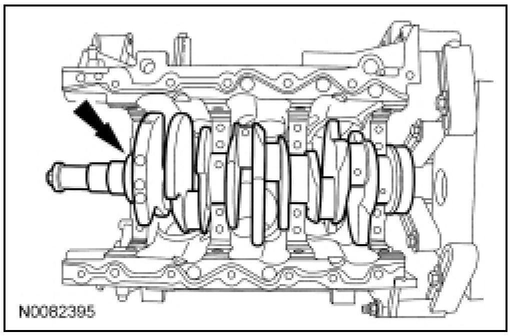
    Lubricate the upper crankshaft main bearings with clean engine oil and install the 4 crankshaft main bearings in the cylinder block.



12. **NOTE:** *Do not install the upper thrust bearings until the crankshaft is installed.*

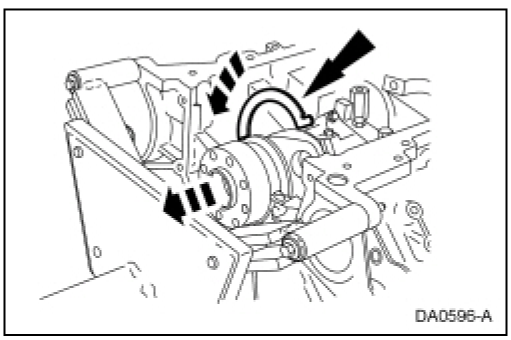
    NOTE: *Lubricate the thrust surfaces of the crankshaft with clean engine oil.*

    Install the crankshaft onto the upper main bearings.



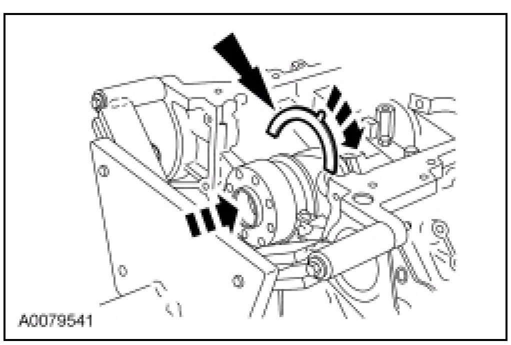
13. **NOTE:** *Make sure the side of the thrust washer, with the wide oil grooves, faces the crankshaft thrust surface.*

    Push the crankshaft rearward and install the rear crankshaft upper thrust washer at the back of the No. 4 rear bulkhead.

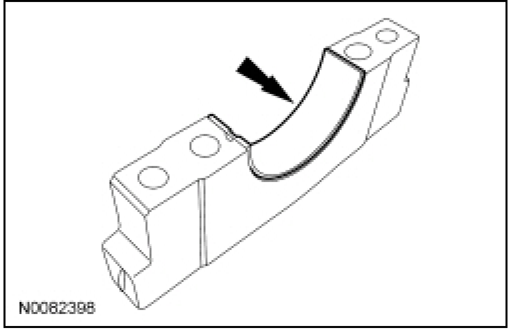


14. **NOTE:** *Make sure the side of the thrust washer, with the wide oil grooves, faces the crankshaft thrust surface.*

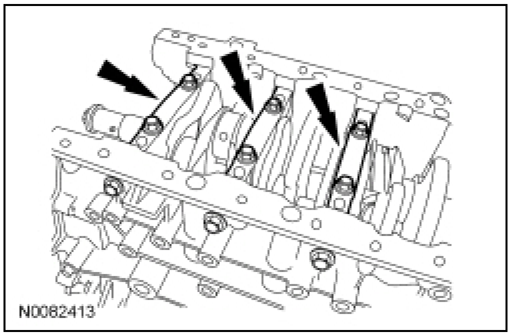
    Push the crankshaft forward and install the front crankshaft upper thrust washer at the front of the No. 4 rear bulkhead.



15. Lubricate the crankshaft lower main bearings with clean engine oil and install them into the main bearing caps. Visually check seating and squareness of the bearings to make sure of proper seating in caps.



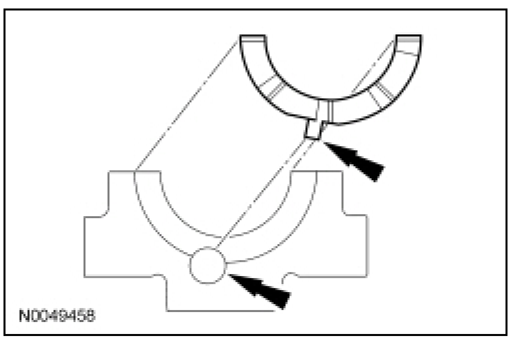
16. Position the No. 1, No. 2 and No. 3 main bearing caps on the cylinder block and, keeping the caps as square as possible, alternately draw the caps down evenly using the new bolts until the main bearing caps are seated.



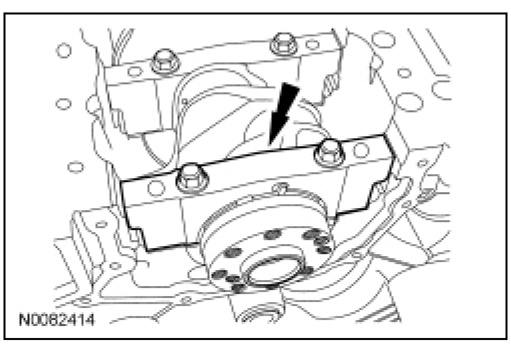
17. **NOTE:** *Make sure the side of the thrust washer, with the wide oil grooves, faces the crankshaft thrust surface.*

    NOTE: *To aid in assembly, apply petroleum jelly to the back of the crankshaft thrust washer.*

    Install the lower crankshaft thrust washer to the back side of the No. 4 rear main bearing cap, with the tab aligned with the cutout in the main bearing cap.



18. Position main bearing cap No. 4 on the cylinder block and keeping the cap as square as possible, alternately draw the cap down evenly using the new bolts until the main bearing cap is seated.

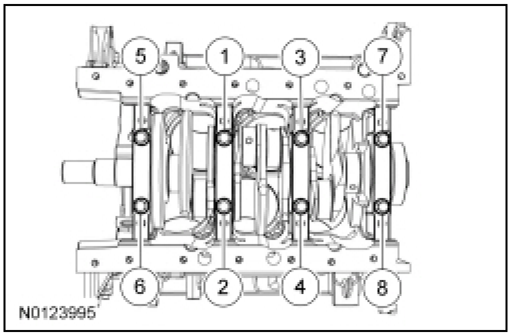


19. Loosen the No. 4 main bearing cap bolts.

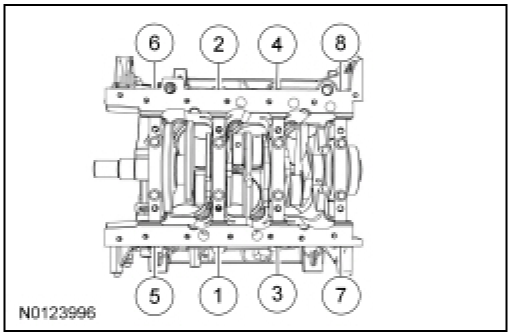
20. **NOTE:** *While tightening the main bearing vertical bolts, push the crankshaft forward and the No. 4 main bearing cap rearward to seat the crankshaft thrust washers.*

    Tighten the main bearing bolts in 2 stages in the sequence shown.

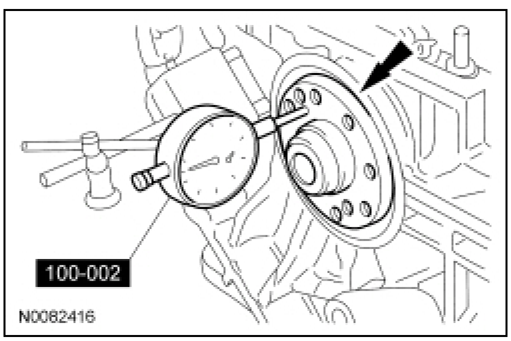
- Stage 1: Tighten fasteners 1 through 8 to 33 Nm (24 lb-ft).
- Stage 2: Tighten fasteners 1 through 8 an additional 90 degrees.



21. Install the new 8 main bearing cap side bolts. Tighten in 2 stages in the sequence shown.
    - Stage 1: Tighten fasteners 1 through 8 to 45 Nm (33 lb-ft).
    - Stage 2: Tighten fasteners 1 through 8 an additional 90 degrees.

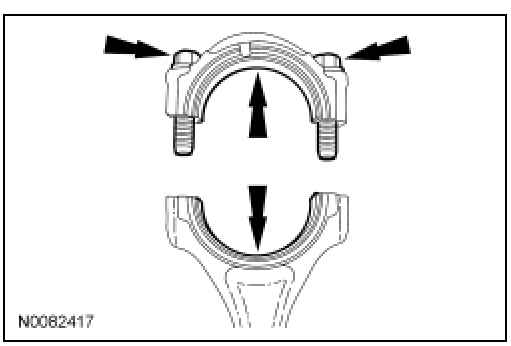


22. Using the Dial Indicator Gauge with Holding Fixture, measure crankshaft end play.
    - Position the crankshaft to the rear of the cylinder block.
    - Zero the Dial Indicator Gauge.
    - Move the crankshaft to the front of the cylinder block. Note and record the crankshaft end play.

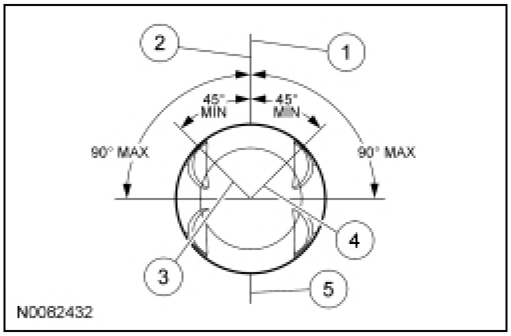


23. **NOTICE: The rod cap installation must keep the same orientation as marked during disassembly or engine damage may occur.**

    Prepare the connecting rod and cap.
    - Insert the new bolts in the rod cap.
    - Insert the upper and lower rod bearings into the rod and cap.



24. Before installing the pistons into the cylinder block, verify proper ring gap location.
    1. Center line of the piston parallel to the wrist pin bore
    2. Upper compression ring gap location
    3. Upper oil control segment ring gap location
    4. Lower oil control segment ring gap location
    5. Expander ring and lower compression ring gap location



25. **NOTICE: Be sure not to scratch the cylinder wall or crankshaft journal with the connecting rod. Push the piston down until the connecting rod bearing seats on the crankshaft journal.**

    NOTE: *The next 3 steps are for all 6 connecting rods, rod caps and pistons. Only 1 connecting rod, rod cap and piston is shown.*

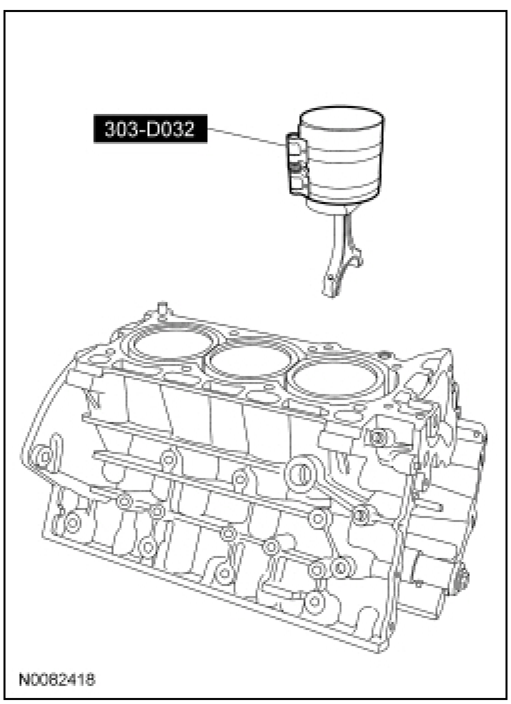
    NOTE: *Lubricate the pistons, piston rings, connecting rod bearings and the entire cylinder bores with clean engine oil.*

    NOTE: *Make sure the piston rings are positioned to specifications for installation. For additional information, refer to Disassembly and Assembly of Subassemblies - Piston in this section.*

    NOTE: *If the piston and/or connecting rod are being installed new, the piston rod orientation marks and the arrow on the top of the dome of the piston should be facing toward the front of the engine block.*

    NOTE: *If the piston and connecting rod are to be reinstalled, they must be installed in the same orientation as disassembled.*

    Using the Piston Ring Compressor, install the piston and connecting rod assemblies.

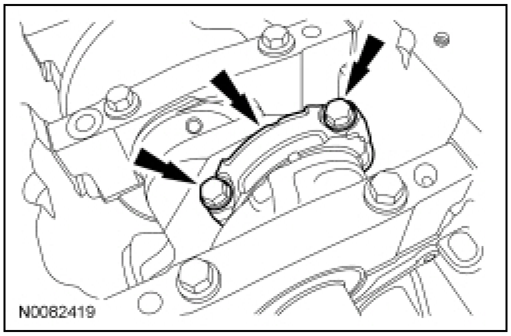


26. Seat the connecting rod on the crankshaft journal.

27. **NOTICE: The rod cap installation must keep the same orientation as marked during disassembly or engine damage may occur.**

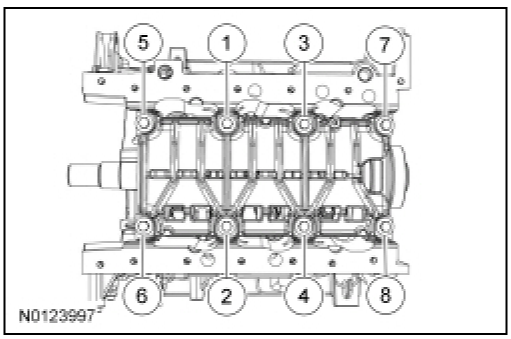
    NOTE: *After installation of each piston, connecting rod, rod cap and bolts, rotate the crankshaft to verify smooth operation.*

    Install the connecting rod cap and bolts. Tighten the bolts in 3 stages.
    - Stage 1: Tighten to 23 Nm (17 lb-ft).
    - Stage 2: Tighten to 43 Nm (32 lb-ft).
    - Stage 3: Tighten an additional 90 degrees.



28. Repeat the previous 3 steps until all 6 piston, connecting rod and connecting rod cap assemblies are installed.

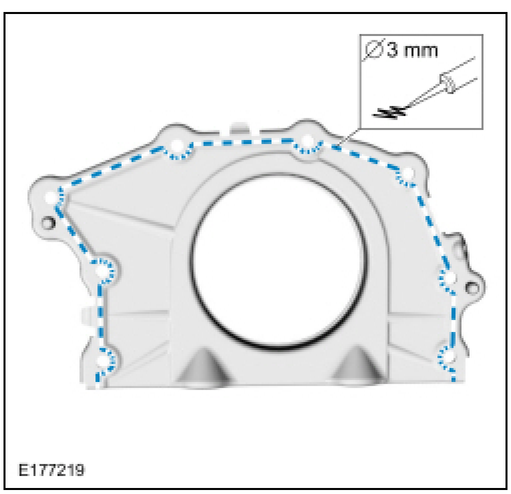
29. Install the main bearing cap support brace and the new bolts. Tighten in 2 stages in the sequence shown.
    - Stage 1: Tighten fasteners to 24 Nm (18 lb-ft).
    - Stage 2: Tighten fasteners an additional 180 degrees.



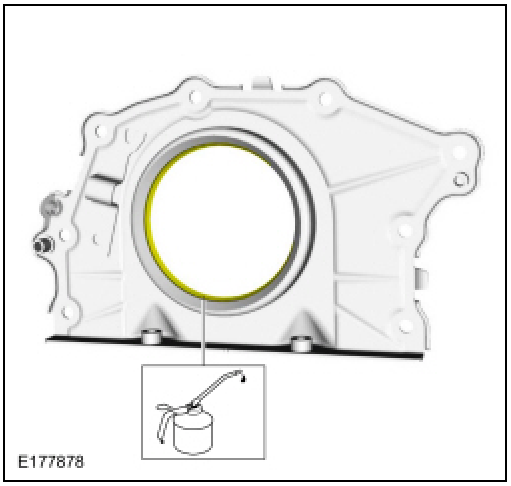
30. **NOTICE: Failure to use Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.**

    NOTE: *The crankshaft rear seal retainer plate must be installed and the bolts tightened within 10 minutes of sealant application.*

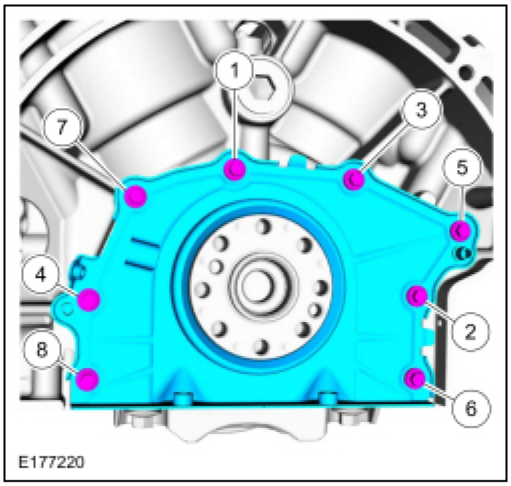
    Apply a 3 mm (0.1181 in) bead of Motorcraft® High Performance Engine RTV Silicone.



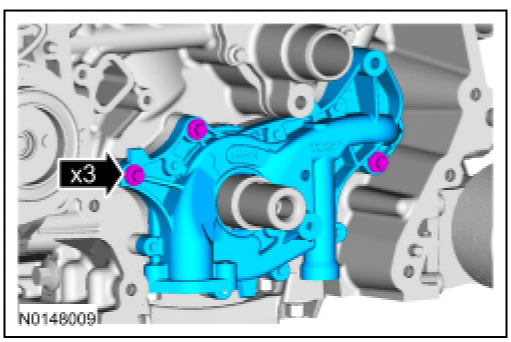
31. Lubricate the crankshaft rear seal with clean engine oil.



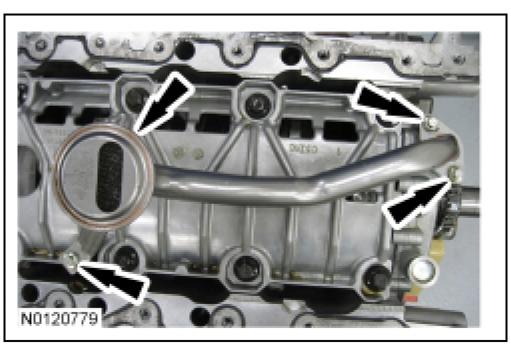
32. Install the crankshaft rear seal retainer plate and the bolts.
   - Tighten to 10 Nm (89 lb-in).



33. Install the oil pump and the 3 bolts.
   - Tighten to 10 Nm (89 lb-in).



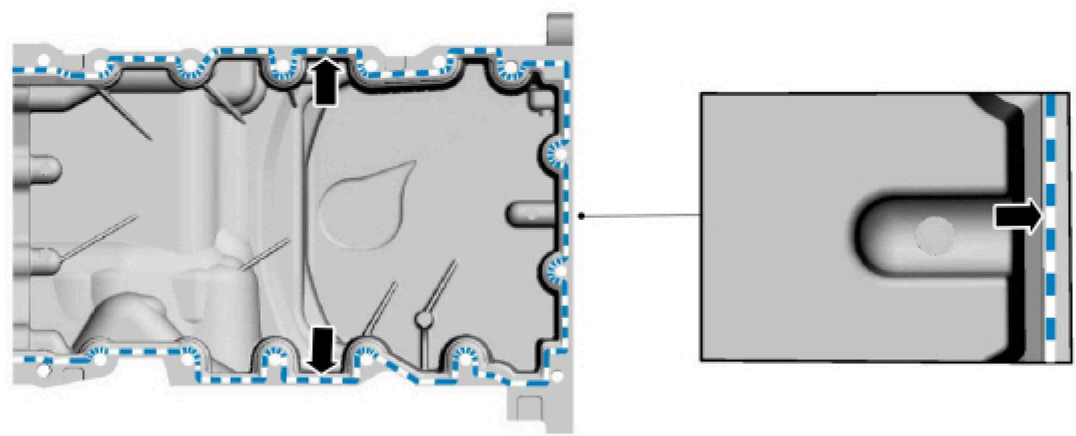
34. Using a new O-ring seal, install the oil pump screen and pickup tube and the 3 bolts.
   - Tighten to 10 Nm (89 lb-in).



35. **NOTICE: Failure to use Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.**

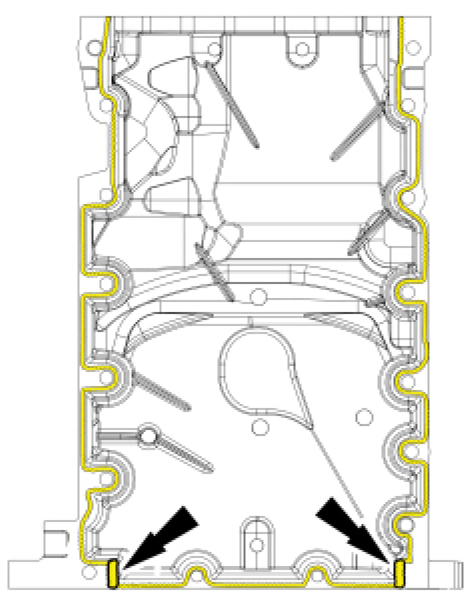
    NOTE: *The oil pan and the 4 specified bolts must be installed and the oil pan aligned to the cylinder block within 4 minutes of sealant application. Final tightening of the oil pan bolts must be carried out within 60 minutes of sealant application.*

    Apply a 3 mm (0.11 in) bead of Motorcraft® High Performance Engine RTV Silicone on the inside edge of the oil pan to the sealing surface of the oil pan-to-engine block mating surface.



N0130514

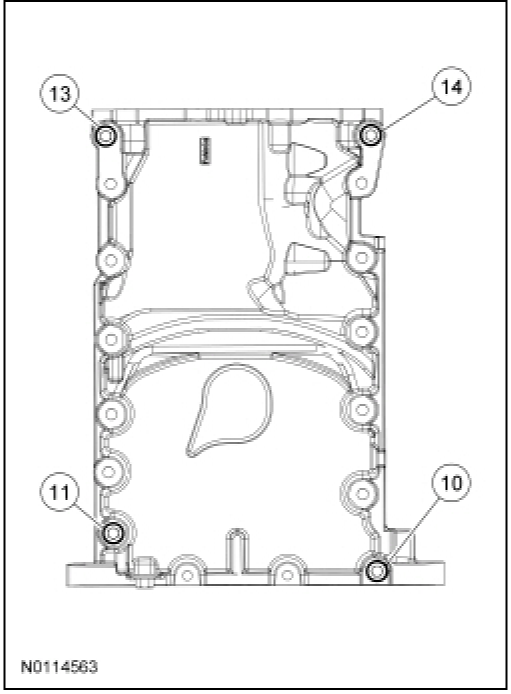
36. Apply a 5.5 mm (0.21 in) bead of Motorcraft® High Performance Engine RTV Silicone to the 2 crankshaft seal retainer plate-to-cylinder block joint areas on the sealing surface of the oil pan.



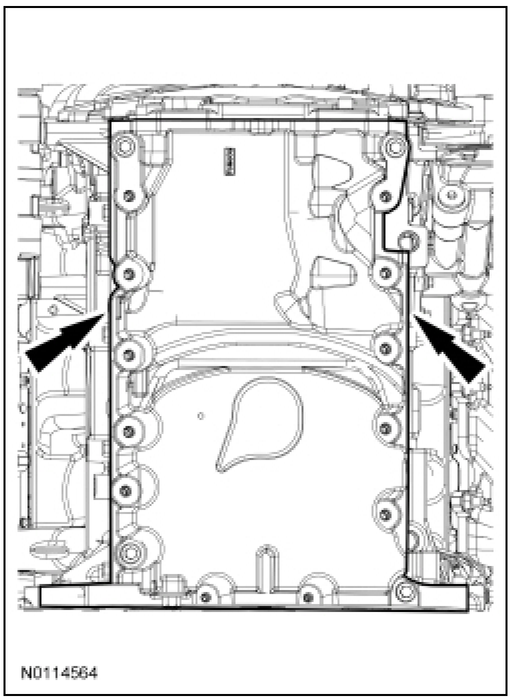
N0117186

37. **NOTE:** *The oil pan and the 4 specified bolts must be installed within 4 minutes of the start of sealant application.*

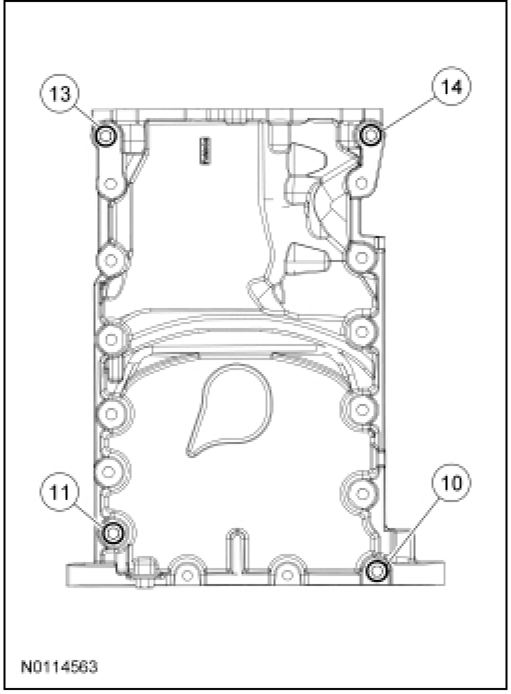
    Install the oil pan and bolts 10, 11, 13 and 14 finger tight.



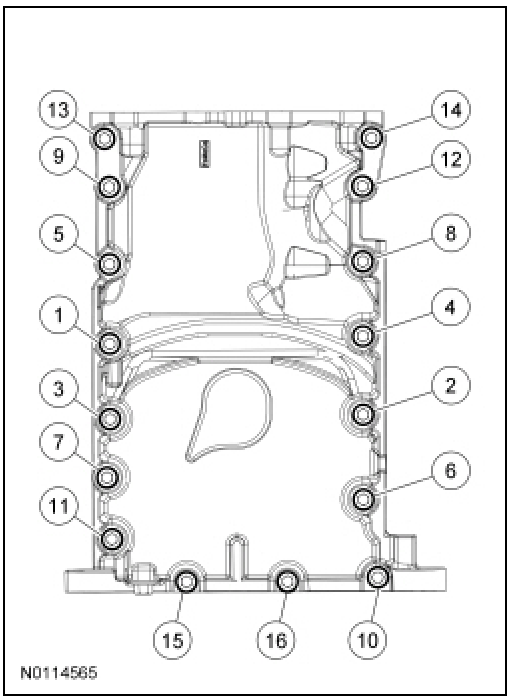
38. Laterally align the oil pan to the block. Make sure the oil pan is flush with the block at these 2 points.



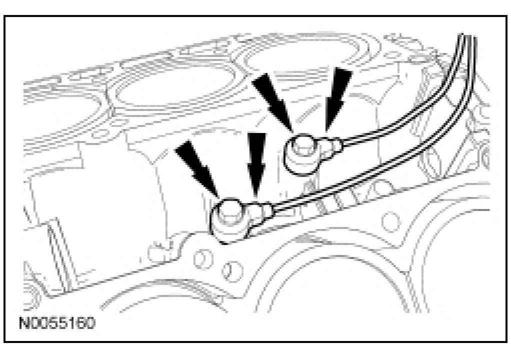
39. Tighten bolts 10, 11, 13 and 14 in the sequence shown.
    - Tighten to 3 Nm (27 lb-in).



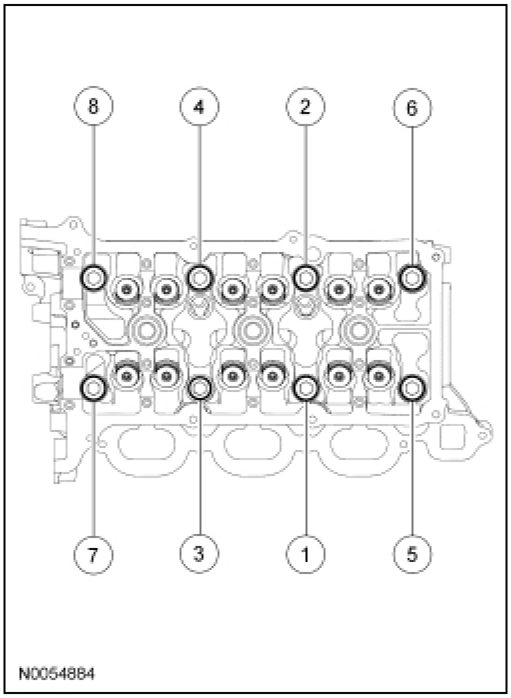
40. Install the remaining oil pan bolts and tighten in the sequence shown.
    - Tighten bolts 1-9 and 11-14 to 20 Nm (177 lb-in), then rotate an additional 45 degrees.
    - Tighten bolts 15 and 16 to 10 Nm (89 lb-in), then rotate an additional 45 degrees.
    - Tighten bolt 10 to 20 Nm (177 lb-in), then rotate an additional 90 degrees.



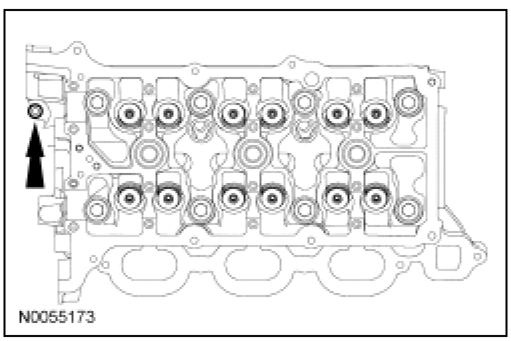
41. Install the Knock Sensor (KS) and the 2 bolts.
    - Tighten to 20 Nm (177 lb-in).



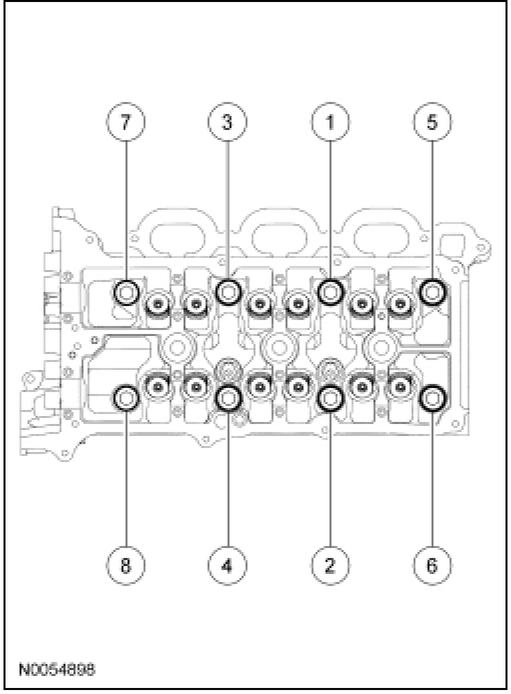
42. Install a new gasket, the RH cylinder head and 8 new bolts. Tighten in 5 stages in the sequence shown.
    - Stage 1: Tighten to 20 Nm (177 lb-in).
    - Stage 2: Tighten to 35 Nm (26 lb-ft).
    - Stage 3: Tighten 90 degrees.
    - Stage 4: Tighten 90 degrees.
    - Stage 5: Tighten 45 degrees.



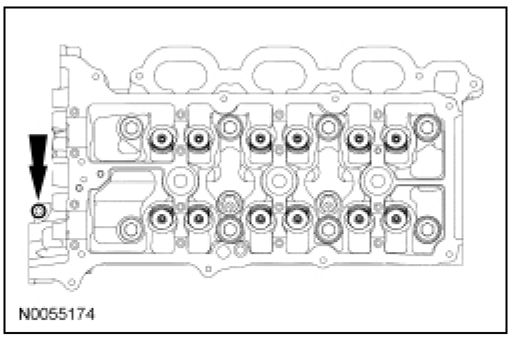
43. Install the new M6 bolt.
    - Tighten to 10 Nm (89 lb-in).



44. Install a new gasket, the LH cylinder head and 8 new bolts. Tighten in 5 stages in the sequence shown.
    - Stage 1: Tighten to 20 Nm (177 lb-in).
    - Stage 2: Tighten to 35 Nm (26 lb-ft).
    - Stage 3: Tighten 90 degrees.
    - Stage 4: Tighten 90 degrees.
    - Stage 5: Tighten 45 degrees.



45. Install the new M6 bolt.
    - Tighten to 10 Nm (89 lb-in).

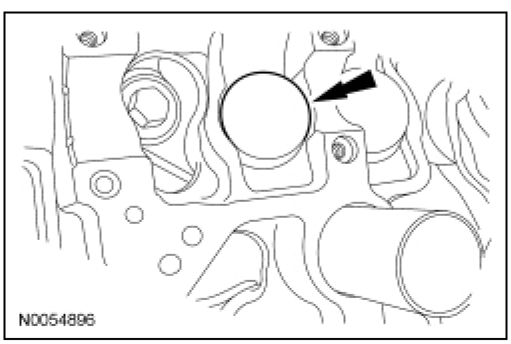


46. **NOTE:** *The valve tappets must be installed in their original positions.*

    NOTE: *Coat the valve tappets with clean engine oil prior to installation.*

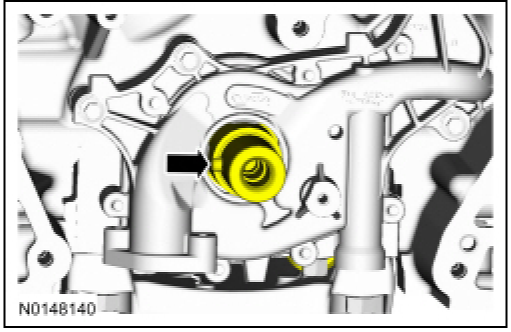
    NOTE: *LH shown, RH similar.*

    Install the valve tappets.



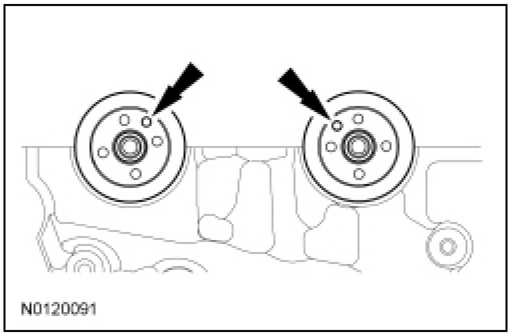
47. **NOTICE: The crankshaft must remain in the freewheeling position (crankshaft dowel pin at 9 o'clock) until after the camshafts are installed and the valve clearance is checked/adjusted. Do not turn the crankshaft until instructed to do so. Failure to follow this process will result in severe engine damage.**

    Position the crankshaft dowel pin in the 9 o'clock position.



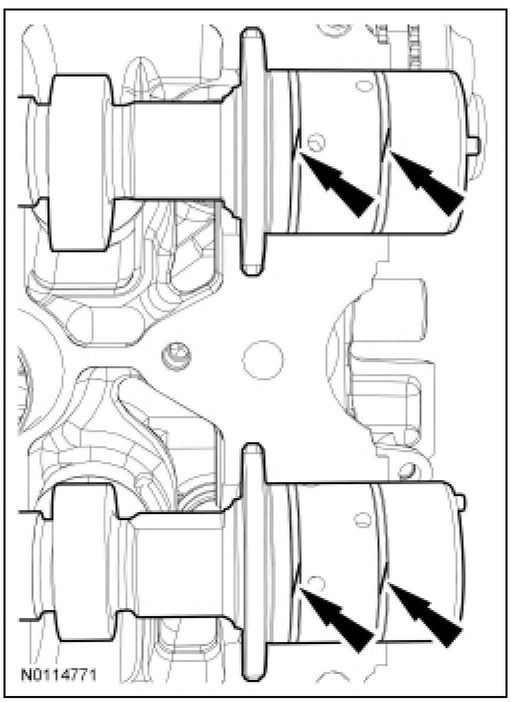
48. **NOTE:** *Coat the camshafts with clean engine oil prior to installation.*

    Position the camshafts onto the RH cylinder head in the neutral position as shown.



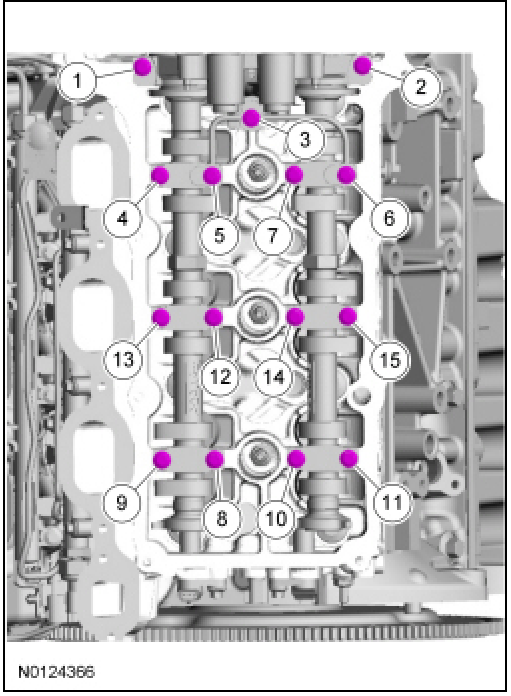
49. **NOTICE: The camshaft seal gaps must be at the 12 o'clock position or damage to the engine may occur.**

    Position the 4 camshaft seals gaps as shown.

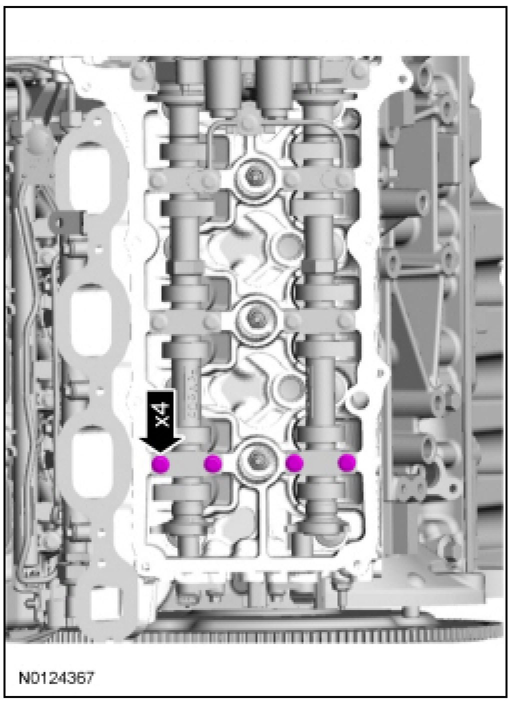


50. **NOTE:** *Cylinder head camshaft bearing caps are numbered to verify that they are assembled in their original positions.*

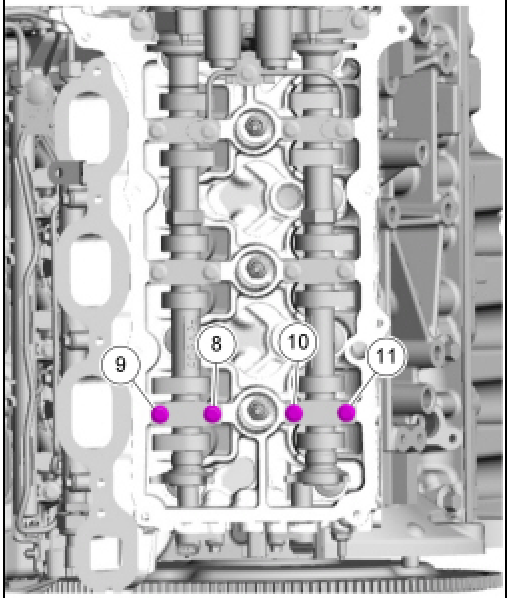
    Install the 6 camshaft caps, mega cap, valve train oil tube and the 15 bolts in the sequence shown.
    - Tighten to 8 Nm (71 lb-in) then an additional 45 degrees.



51. Loosen the 4 camshaft caps bolts.



52. Tighten the 4 camshaft caps bolts in the sequence shown.
    - Tighten bolts 8, 9, 10 and 11 to 8 Nm (71 lb-in) then an additional 45 degrees.

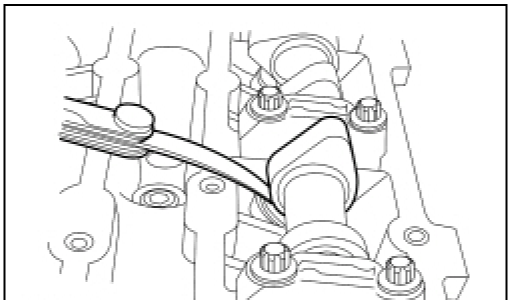


N0124368

53. **NOTICE: If any components are installed new, the engine valve clearance must be checked/adjusted or engine damage may occur.**

    NOTE: *Use a camshaft sprocket bolt to turn the camshafts.*

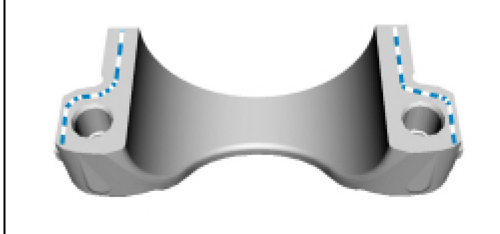
    Using a feeler gauge, confirm that the valve tappet clearances are within specification. If valve tappet clearances are not within specification, the clearance must be adjusted by installing new valve tappet(s) of the correct size. For additional information, refer to Valve Clearance Check in this section.



A0004277

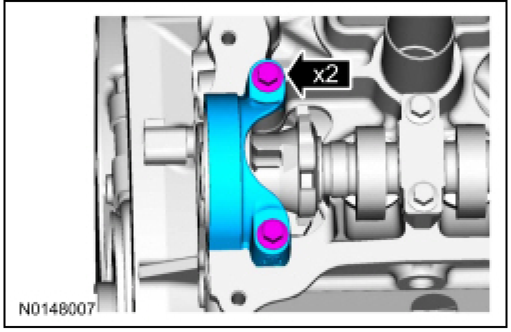
54. **NOTE:** *The camshaft bearing cap assembly must be secured within 10 minutes of the Gasket Maker application. If the camshaft bearing cap assembly is not secured within 10 minutes, the sealant must be removed and the sealing area cleaned with Motorcraft® Metal Surface Prep.*

    Apply a 3 mm (0.118 in) bead of sealant to the camshaft bearing cap assembly.

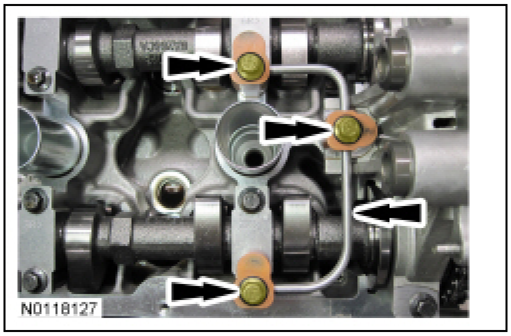


N0147496

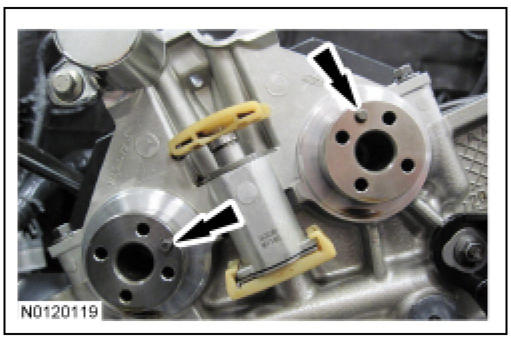
55. Install the camshaft bearing cap assembly and the 2 bolts.
    - Tighten to 8 Nm (71 lb-in) then an additional 45 degrees.
    - Remove the excess gasket maker.



56. Remove the 3 bolts and the RH valve train oil tube.

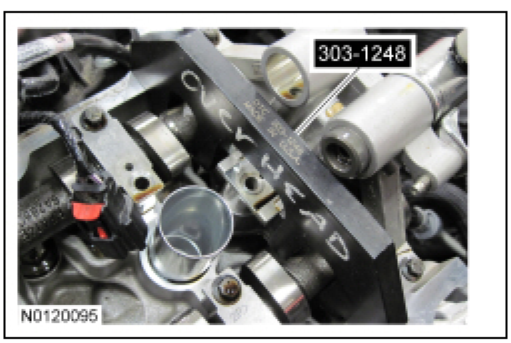


57. Rotate the RH camshafts to the Top Dead Center (TDC) position as shown.



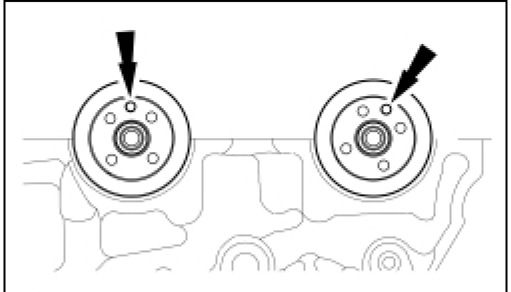
58. **NOTE:** *The Camshaft Holding Tool will hold the camshafts in the TDC position.*

    Install the Camshaft Holding Tool onto the flats of the RH camshafts.



59. **NOTE:** *Coat the camshafts with clean engine oil prior to installation.*

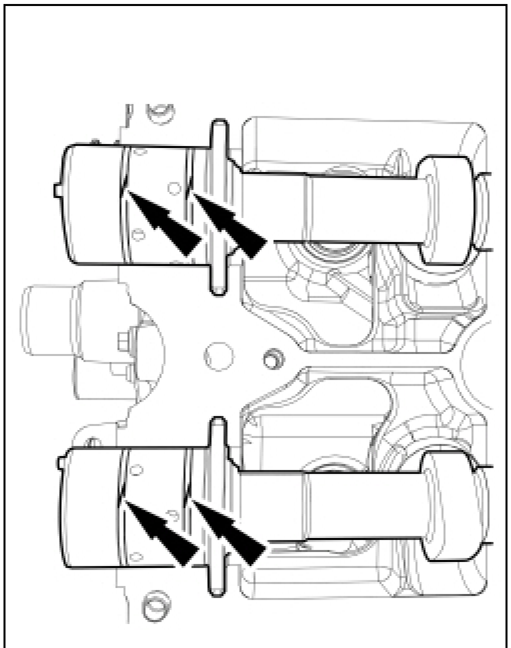
    Position the camshafts onto the LH cylinder head in the neutral position as shown.



N0120090

60. **NOTICE: The camshaft seal gaps must be at the 12 o'clock position or damage to the engine may occur.**

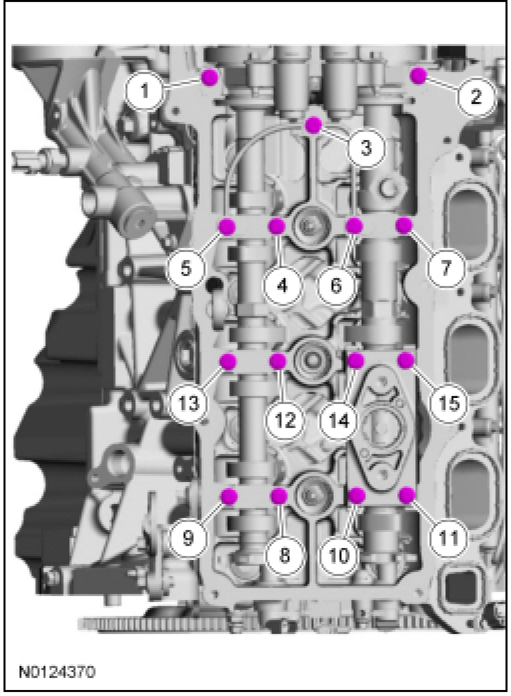
    Position the 4 camshaft seals gaps as shown.



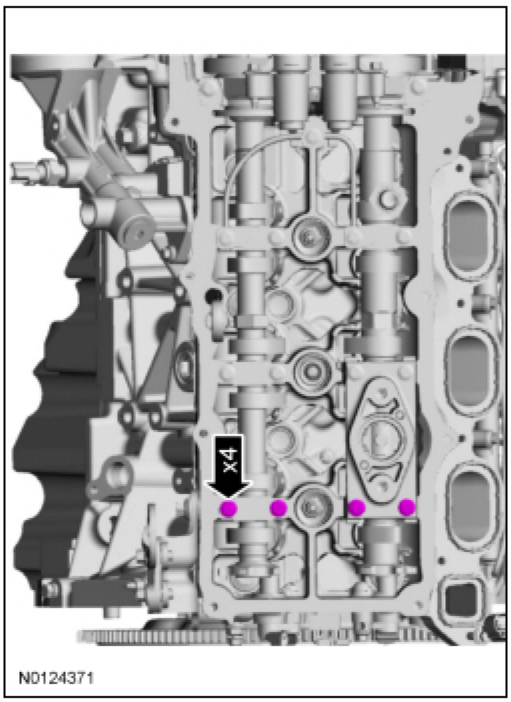
N0114770

61. **NOTE:** *Cylinder head camshaft bearing caps are numbered to verify that they are assembled in their original positions.*

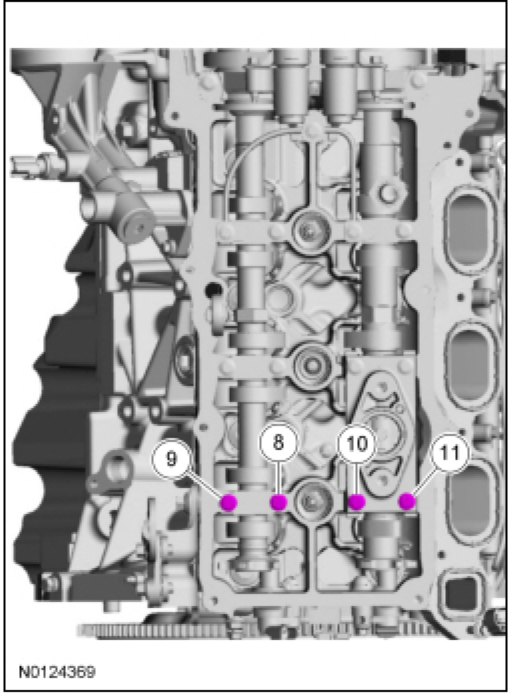
    Install the 6 camshaft caps, mega cap, valve train oil tube and the 15 bolts in the sequence shown.
    - Tighten to 8 Nm (71 lb-in) then an additional 45 degrees.



62. Loosen the 4 camshaft caps bolts.



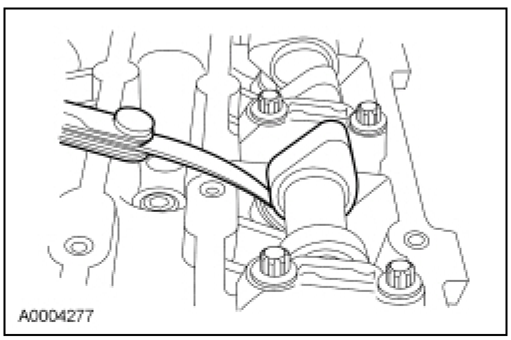
63. Tighten the 4 camshaft caps bolts in the sequence shown.
    - Tighten bolts 8, 9, 10 and 11 to 8 Nm (71 lb-in) then an additional 45 degrees.



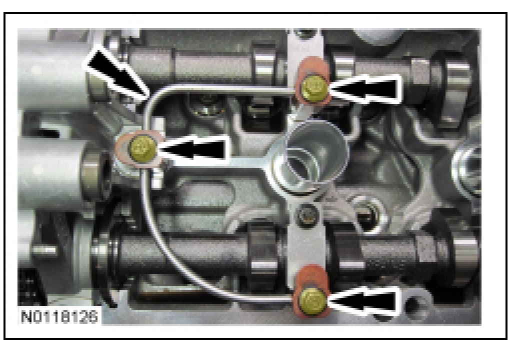
64. **NOTICE: If any components are installed new, the engine valve clearance must be checked/adjusted or engine damage may occur.**

    NOTE: *Use a camshaft sprocket bolt to turn the camshafts.*

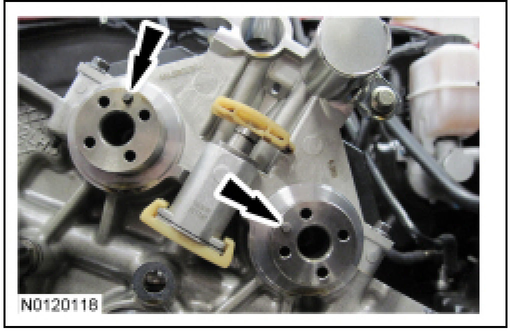
    Using a feeler gauge, confirm that the valve tappet clearances are within specification. If valve tappet clearances are not within specification, the clearance must be adjusted by installing new valve tappet(s) of the correct size. For additional information, refer to Valve Clearance Check in this section.



65. Remove the 3 bolts and the LH valve train oil tube.

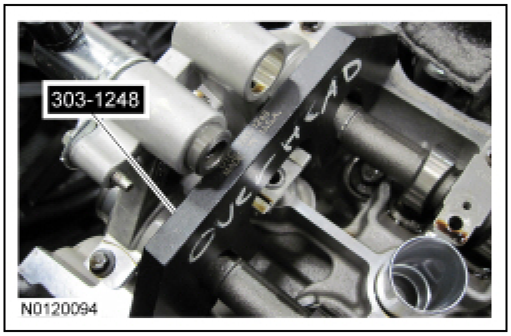


66. Rotate the LH camshafts to the TDC position as shown.


N0120118

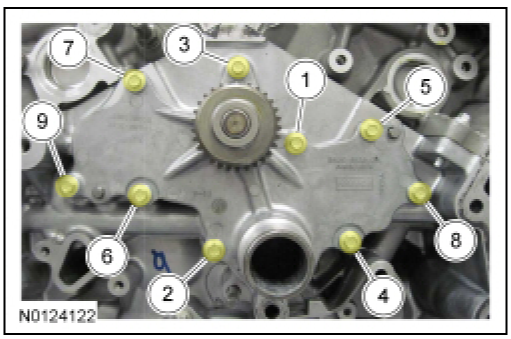
67. **NOTE:** *The Camshaft Holding Tool will hold the camshafts in the TDC position.*

    Install the Camshaft Holding Tool onto the flats of the LH camshafts.



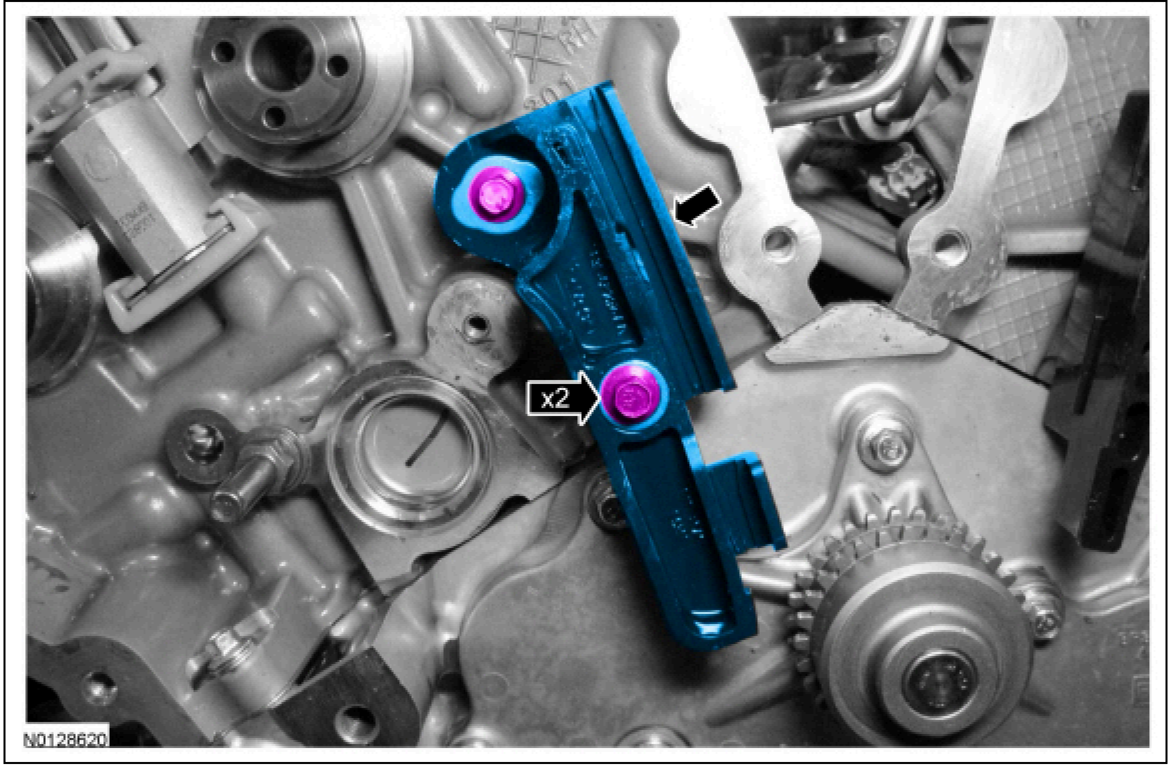
N0120094

68. Using a new gasket, install the timing chain gear and the 9 bolts. Tighten in 2 stages in the sequence shown.
    - Stage 1: Tighten to 10 Nm (89 lb-in).
    - Stage 2: Tighten an additional 45 degrees.



N0124122

69. Install the RH primary timing chain guide and the 2 bolts.
    - Tighten to 10 Nm (89 lb-in).



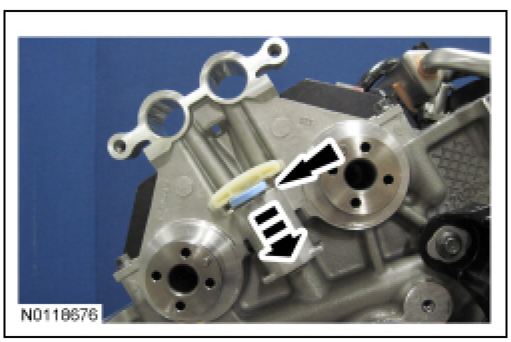
Engines with secondary timing chain tensioner removed

NOTICE: The following steps are only for the replacement of the secondary timing chain tensioners. Do not reuse the secondary timing chain tensioners if removed or damage to the engine may occur.

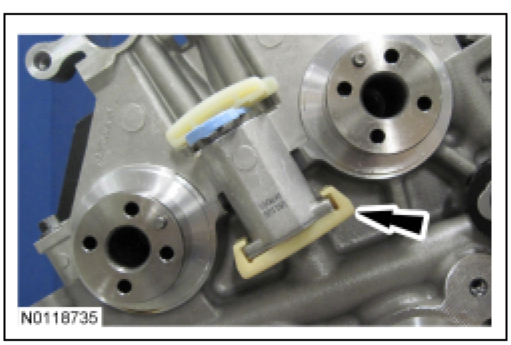
70. **NOTE:** *Apply clean engine oil to the secondary timing chain tensioner O-ring seals and mega cap bore.*

    NOTE: *Do not remove the secondary timing chain tensioner shipping clip, until instructed to do so.*

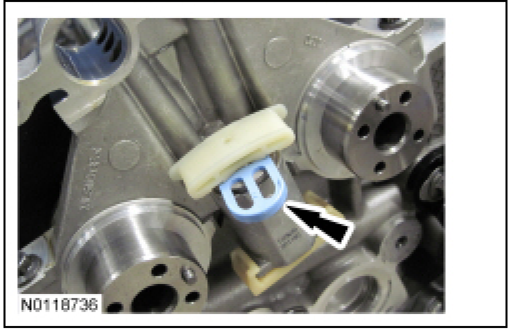
    Install the RH secondary timing chain tensioner by pushing it down all the way until a snap is heard and the tensioner is seated all the way down the mega cap bore.



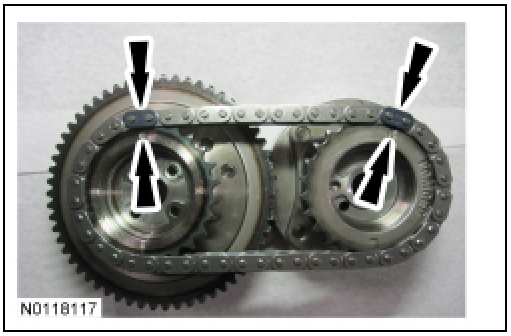
71. Install the RH secondary timing chain tensioner shoe.



72. Remove and discard the RH secondary timing chain tensioner shipping clip.

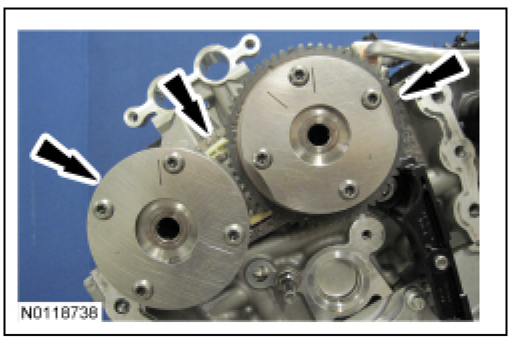


73. Assemble the RH Variable Camshaft Timing (VCT) assembly, the RH exhaust camshaft sprocket and the RH secondary timing chain.
    - Align the colored links with the timing marks.

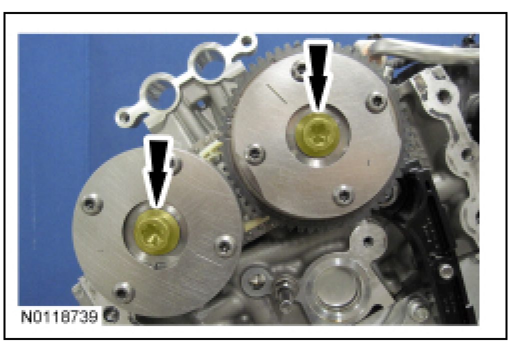


74. **NOTE:** *It may be necessary to rotate the camshafts slightly, to install the RH secondary timing assembly.*

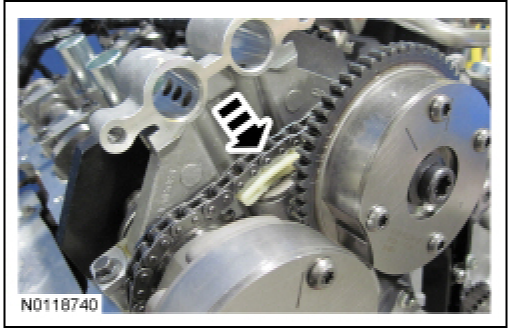
    Position the 2 RH VCT assemblies and secondary timing chain onto the camshafts by aligning the holes in the VCT assemblies with the dowel pins in the camshafts.



75. Install the 2 new RH VCT bolts and tighten in 4 stages.
    - Stage 1: Tighten to 40 Nm (30 lb-ft).
    - Stage 2: Loosen one full turn.
    - Stage 3: Tighten to 25 Nm (18 lb-ft).
    - Stage 4: Tighten an additional 180 degrees.



76. Activate the RH secondary timing chain tensioner by pressing down on the secondary tensioner shoe until it bottoms out, let go of the tensioner and it will spring up putting tension on the chain.

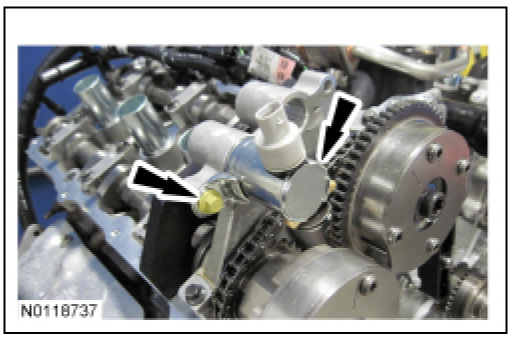


77. **NOTICE: Do not use excessive force when installing the Variable Camshaft Timing (VCT) oil control solenoid. Damage to the mega cap could cause the cylinder head to be inoperable. If difficult to install the VCT oil control solenoid, inspect the bore and VCT oil control solenoid to ensure there are no burrs, sharp edges or contaminants present on the mating surface. Only clean the external surfaces as necessary.**

    NOTE: *A slight twisting motion will aid in the installation of the VCT oil control solenoid.*

    NOTE: *Keep the VCT oil control solenoid clean of dirt and debris.*

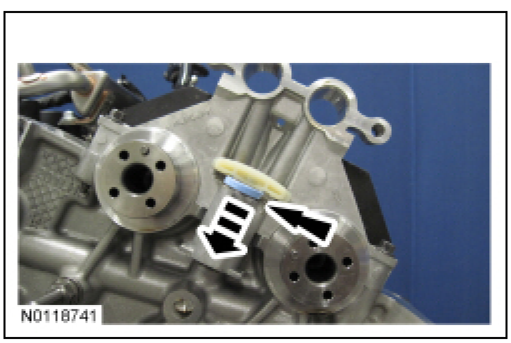
    Install the RH exhaust VCT oil control solenoid.
    - Tighten to 8 Nm (71 lb-in) then an additional 20 degrees.



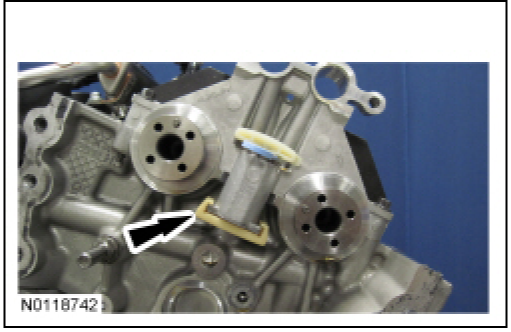
78. **NOTE:** *Apply clean engine oil to the secondary timing chain tensioner O-ring seals and mega cap bore.*

    NOTE: *Do not remove the secondary timing chain tensioner shipping clip, until instructed to do so.*

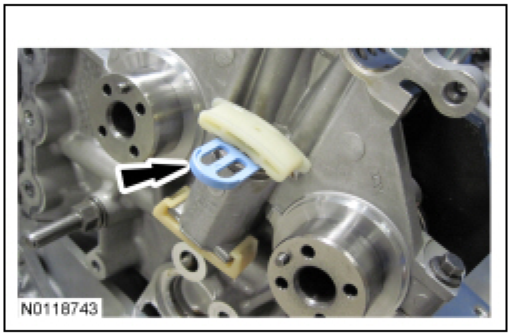
    Install the LH secondary timing chain tensioner by pushing it down all the way until a snap is heard and the tensioner is seated all the way down the mega cap bore.



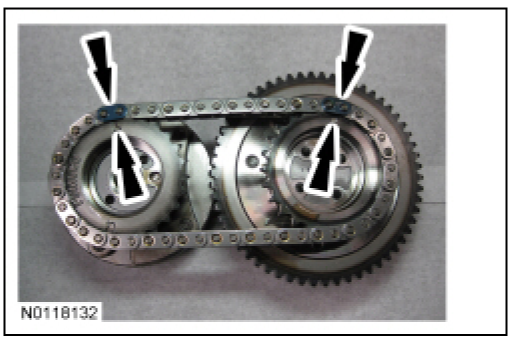
79. Install the LH secondary timing chain tensioner shoe.



80. Remove and discard the LH secondary timing chain tensioner shipping clip.

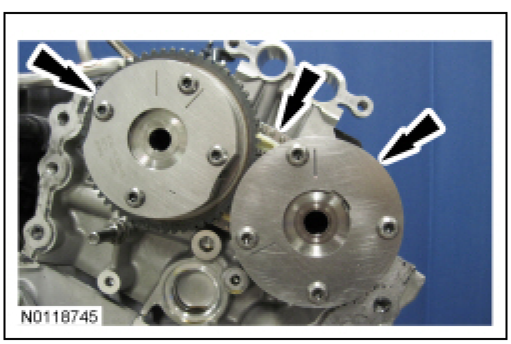


81. Assemble the 2 LH VCT assemblies and the LH secondary timing chain.
    - Align the colored links with the timing marks.

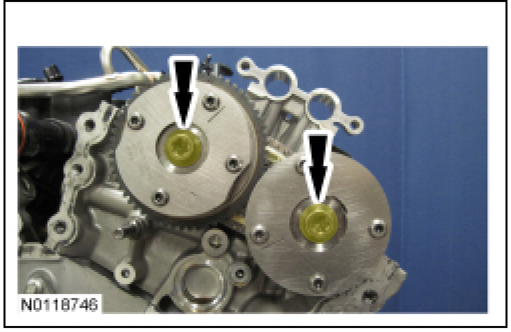


82. **NOTE:** *It may be necessary to rotate the camshafts slightly, to install the LH secondary timing assembly.*

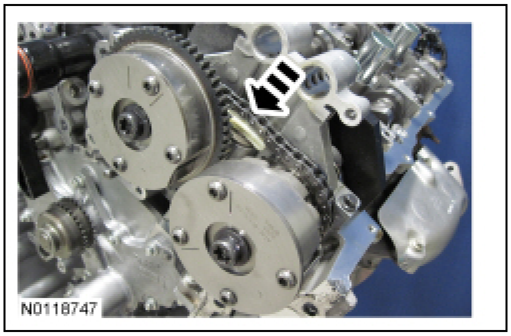
    Position the 2 LH VCT assemblies and secondary timing chain onto the camshafts by aligning the holes in the VCT assemblies with the dowel pins in the camshafts.



83. Install the 2 new LH VCT bolts and tighten in 4 stages.
    - Stage 1: Tighten to 40 Nm (30 lb-ft).
    - Stage 2: Loosen one full turn.
    - Stage 3: Tighten to 25 Nm (18 lb-ft).
    - Stage 4: Tighten an additional 180 degrees.



84. Activate the LH secondary timing chain tensioner by pressing down on the secondary tensioner shoe until it bottoms out, let go of the tensioner and it will spring up putting tension on the chain.

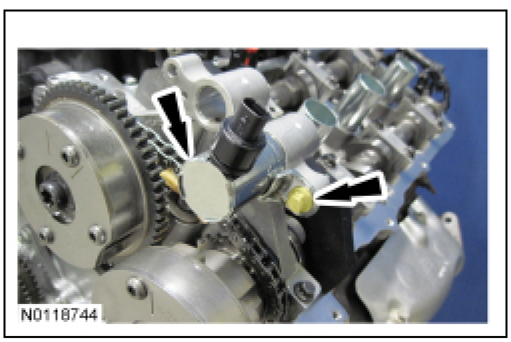


85. **NOTICE: Do not use excessive force when installing the Variable Camshaft Timing (VCT) oil control solenoid. Damage to the mega cap could cause the cylinder head to be inoperable. If difficult to install the VCT oil control solenoid, inspect the bore and VCT oil control solenoid to ensure there are no burrs, sharp edges or contaminants present on the mating surface. Only clean the external surfaces as necessary.**

    NOTE: *A slight twisting motion will aid in the installation of the VCT oil control solenoid.*

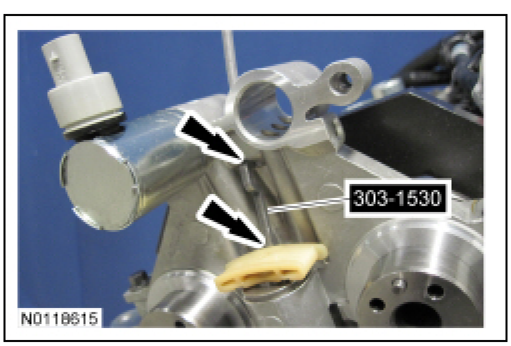
    NOTE: *Keep the VCT oil control solenoid clean of dirt and debris.*

    Install the LH exhaust VCT oil control solenoid.
    - Tighten to 8 Nm (71 lb-in) then an additional 20 degrees.

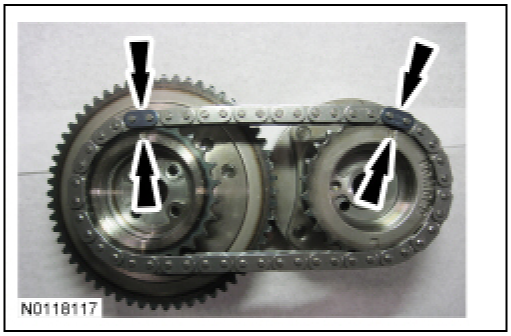


Engines with secondary timing chain tensioner not removed

86. Compress the RH secondary chain tensioner and install the Secondary Chain Hold Down to retain the tensioner in the collapsed position.

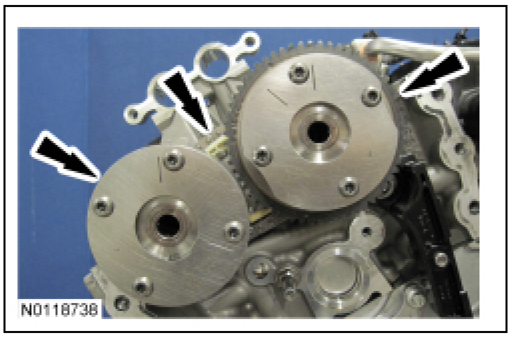


87. Assemble the RH Variable Camshaft Timing (VCT) assembly, the RH exhaust camshaft sprocket and the RH secondary timing chain.
    - Align the colored links with the timing marks.

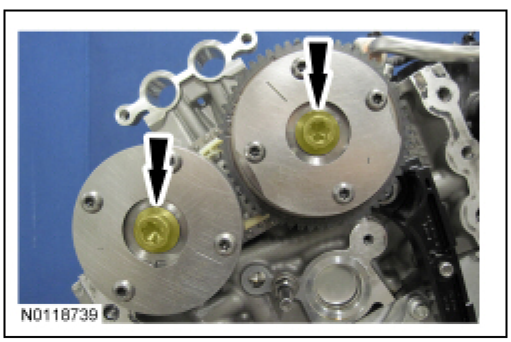


88. **NOTE:** *It may be necessary to rotate the camshafts slightly, to install the RH secondary timing assembly.*

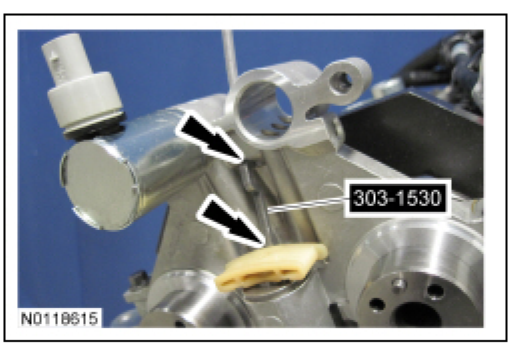
    Position the 2 RH VCT assemblies and secondary timing chain onto the camshafts by aligning the holes in the VCT assemblies with the dowel pins in the camshafts.



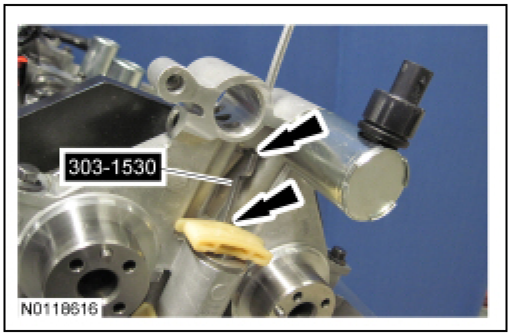
89. Install the 2 new RH VCT bolts and tighten in 4 stages.
    - Stage 1: Tighten to 40 Nm (30 lb-ft).
    - Stage 2: Loosen one full turn.
    - Stage 3: Tighten to 25 Nm (18 lb-ft).
    - Stage 4: Tighten an additional 180 degrees.



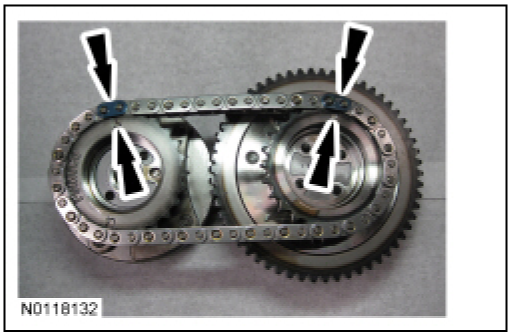
90. Compress the RH secondary timing chain tensioner and remove the Secondary Chain Hold Down.
    - Make sure the secondary timing chain is centered on the timing chain tensioner guides.



91. Compress the LH secondary chain tensioner and install the Secondary Chain Hold Down to retain the tensioner in the collapsed position.

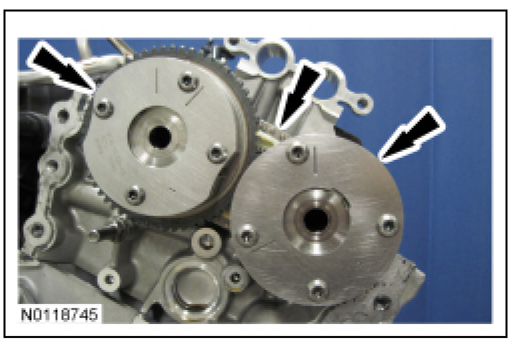


92. Assemble the 2 LH VCT assemblies and the LH secondary timing chain.
   - Align the colored links with the timing marks.

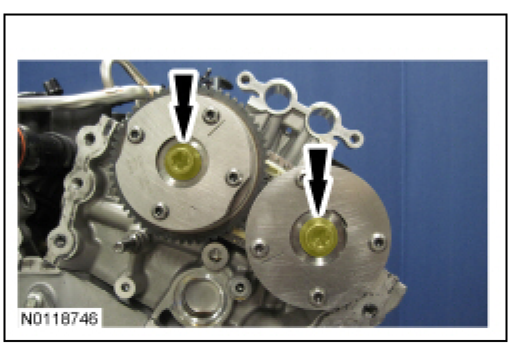


93. **NOTE:** *It may be necessary to rotate the camshafts slightly, to install the LH secondary timing assembly.*

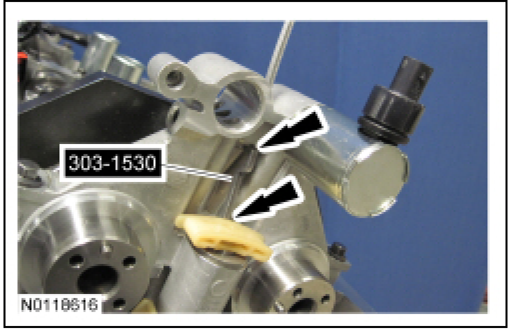
    Position the 2 LH VCT assemblies and secondary timing chain onto the camshafts by aligning the holes in the VCT assemblies with the dowel pins in the camshafts.



94. Install the 2 new LH VCT bolts and tighten in 4 stages.
   - Stage 1: Tighten to 40 Nm (30 lb-ft).
   - Stage 2: Loosen one full turn.
   - Stage 3: Tighten to 25 Nm (18 lb-ft).
   - Stage 4: Tighten an additional 180 degrees.

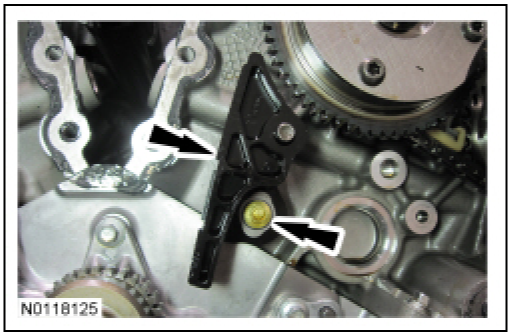


95. Compress the LH secondary timing chain tensioner and remove the Secondary Chain Hold Down.
   - Make sure the secondary timing chain is centered on the timing chain tensioner guides.

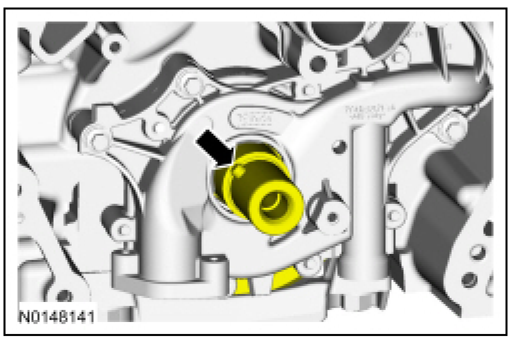


All engines

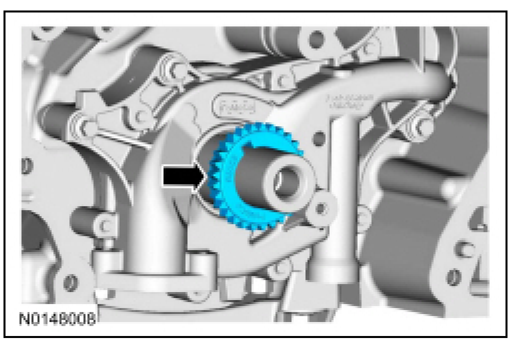
96. Install the upper LH primary timing chain guide and the bolt.
    - Tighten to 10 Nm (89 lb-in).



97. Rotate the crankshaft 60 degrees to the Top Dead Center (TDC) position (crankshaft dowel at 11 o'clock).

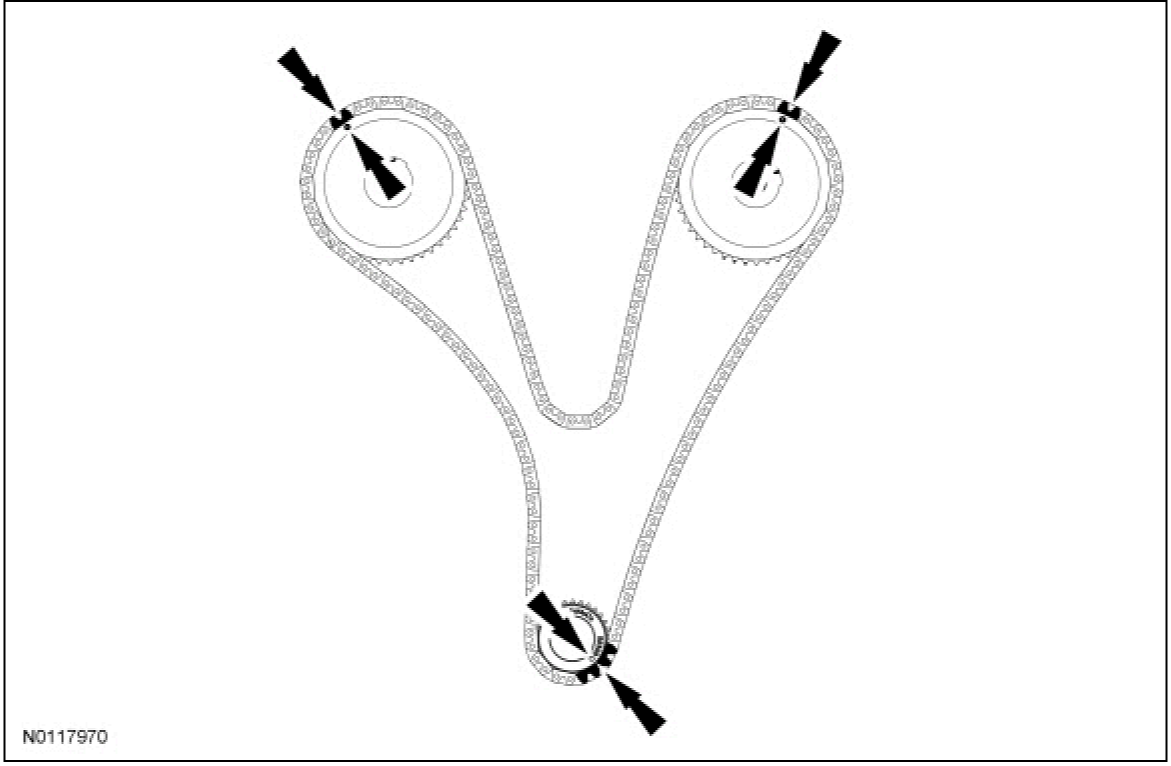


98. Install the crankshaft timing chain sprocket with timing dot mark out.

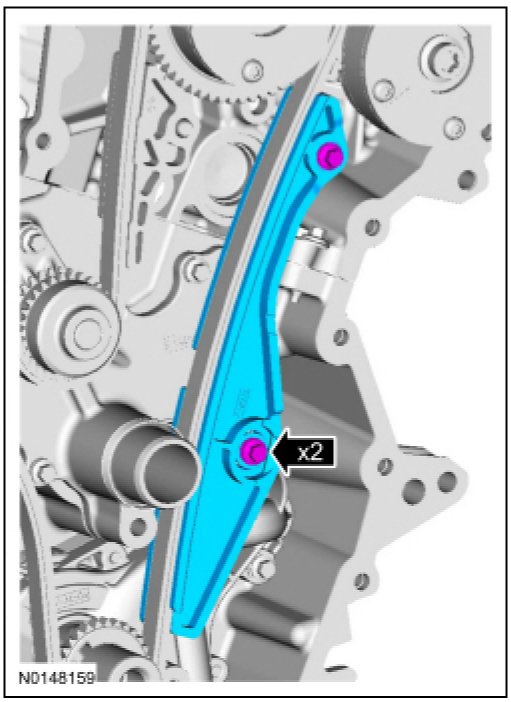


99. **NOTE:** *It may be necessary to rotate the camshafts slightly, to align the timing marks.*

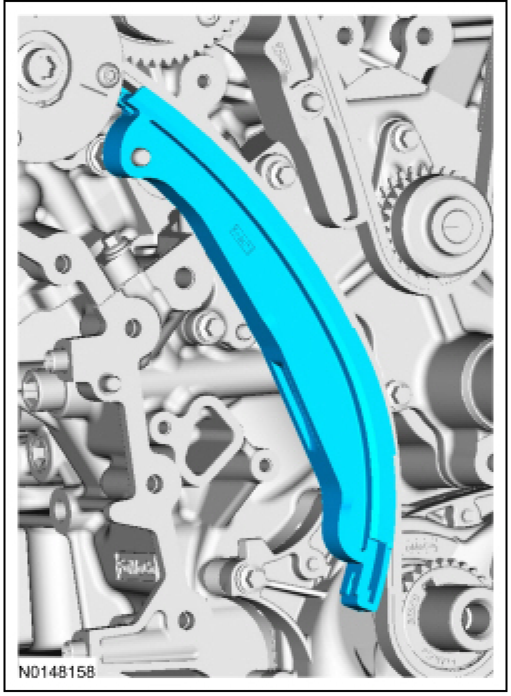
    Install the primary timing chain with the colored links aligned with the timing marks on the VCT assemblies and the crankshaft sprocket.



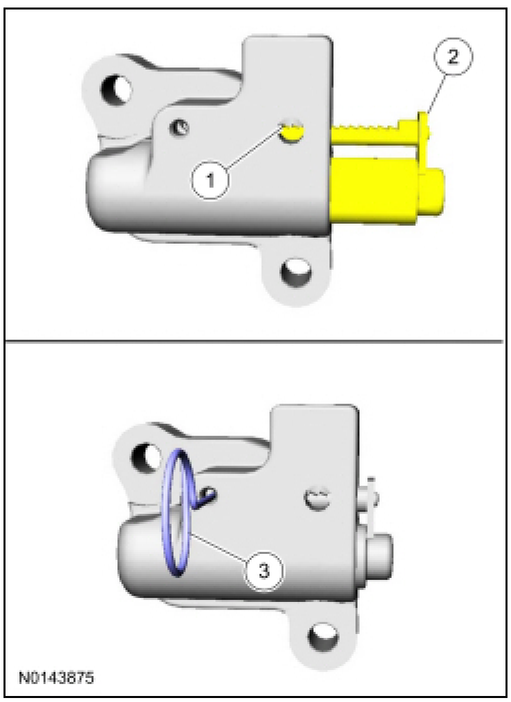
100. Install the lower LH primary timing chain guide and the 2 bolts.
    - Tighten to 10 Nm (89 lb-in).



101. Install the primary timing chain tensioner arm.

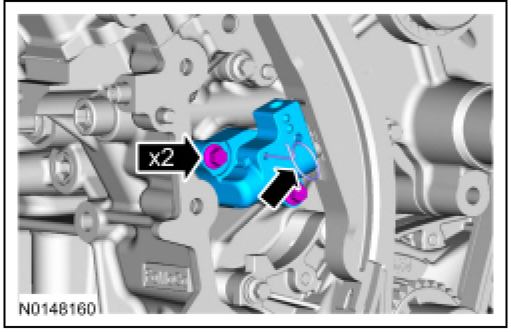


102. Reset the primary timing chain tensioner.
    1. Release the ratchet detent.
    2. Using a soft-jawed vise, compress the ratchet plunger.
    3. Align the hole in the ratchet plunger with the hole in the tensioner housing and install a suitable lockpin.

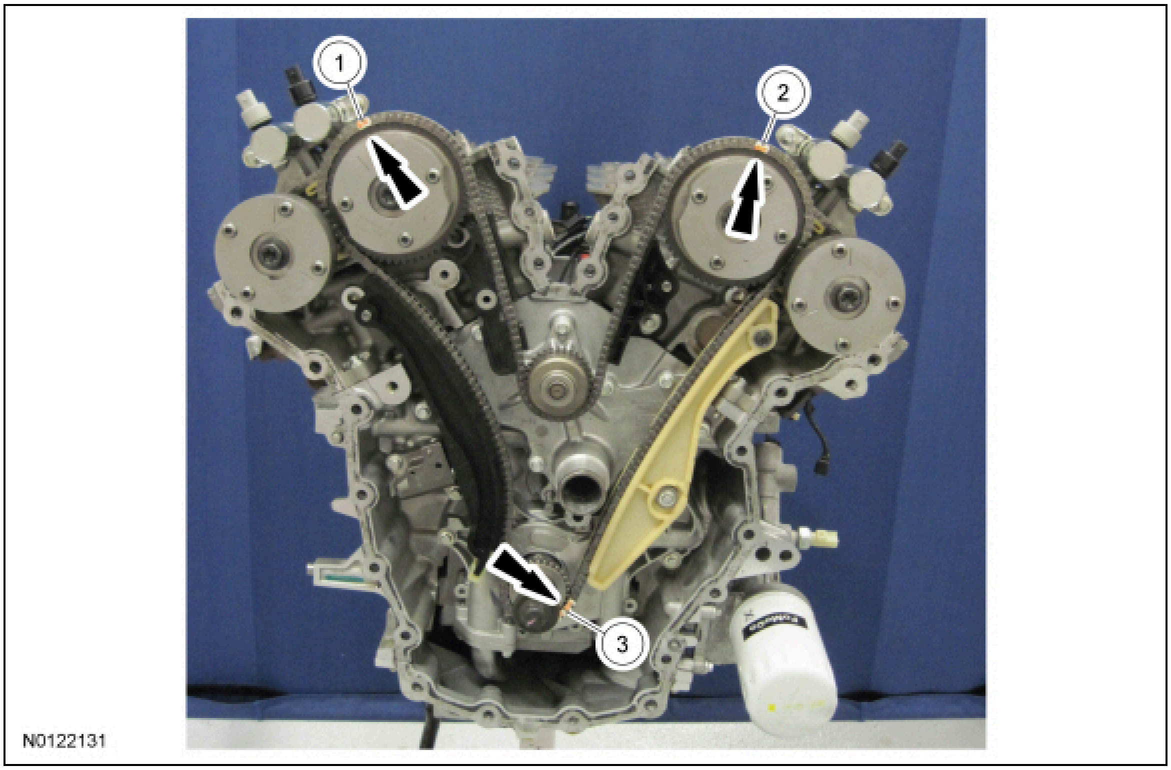


103. **NOTE:** *It may be necessary to rotate the camshafts slightly to remove slack from the timing chain to install the tensioner.*

    Install the primary timing chain tensioner and the 2 bolts.
    - Tighten to 10 Nm (89 lb-in).
    - Remove the lockpin.



104. As a post-check, verify correct alignment of all timing marks.
   - There are 48 links in between the RH intake VCT assembly colored link (1) and the LH intake VCT assembly colored link (2).
   - There are 35 links in between LH intake VCT assembly colored link (2) and the 2 crankshaft sprocket links (3).

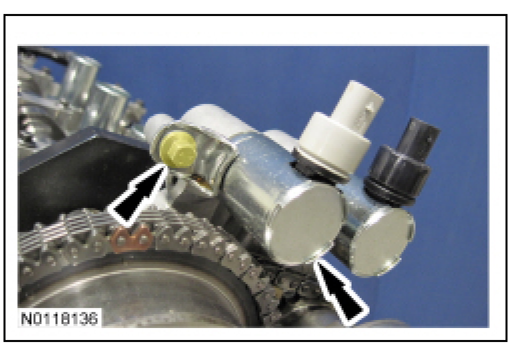


105. **NOTICE: Do not use excessive force when installing the Variable Camshaft Timing (VCT) oil control solenoid. Damage to the mega cap could cause the cylinder head to be inoperable. If difficult to install the VCT oil control solenoid, inspect the bore and VCT oil control solenoid to ensure there are no burrs, sharp edges or contaminants present on the mating surface. Only clean the external surfaces as necessary.**

   NOTE: *A slight twisting motion will aid in the installation of the VCT oil control solenoid.*

   NOTE: *Keep the VCT oil control solenoid clean of dirt and debris.*

   Install the LH intake VCT oil control solenoid and the bolt.
   - Tighten to 8 Nm (71 lb-in) then an additional 20 degrees.

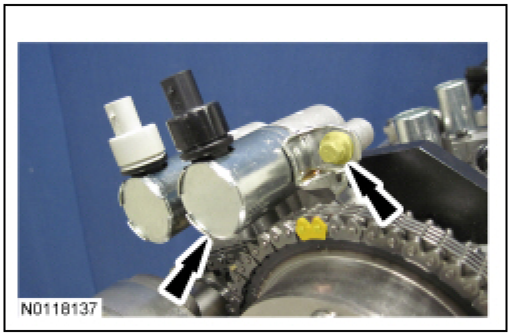


106. **NOTICE: Do not use excessive force when installing the Variable Camshaft Timing (VCT) oil control solenoid. Damage to the mega cap could cause the cylinder head to be inoperable. If difficult to install the VCT oil control solenoid, inspect the bore and VCT oil control solenoid to ensure there are no burrs, sharp edges or contaminants present on the mating surface. Only clean the external surfaces as necessary.**

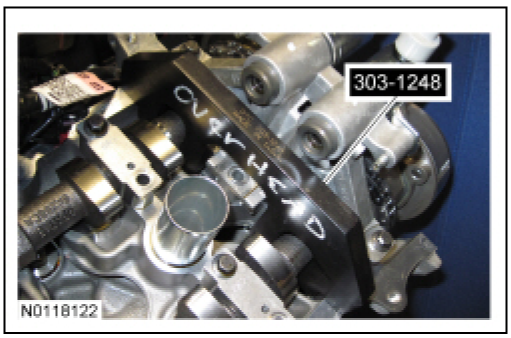
NOTE: *A slight twisting motion will aid in the installation of the VCT oil control solenoid.*

NOTE: *Keep the VCT oil control solenoid clean of dirt and debris.*

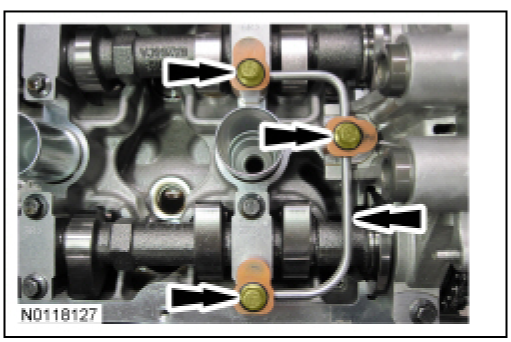
Install the RH intake VCT oil control solenoid and the bolt.
- Tighten to 8 Nm (71 lb-in) then an additional 20 degrees.



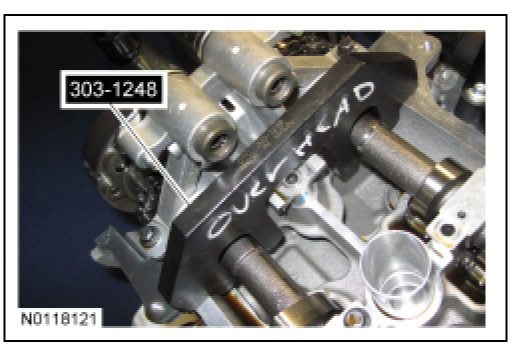
107. Remove the RH Camshaft Holding Tool.



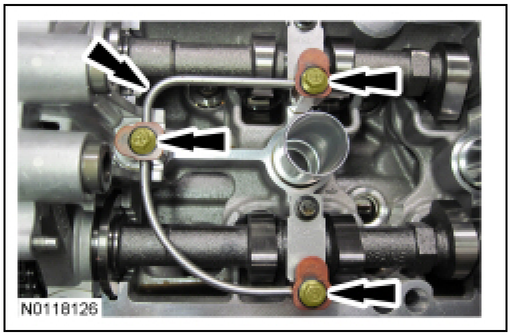
108. Install the RH valve train oil tube and the 3 bolts and tighten in 2 stages.
- Stage 1: Tighten to 8 Nm (71 lb-in).
- Stage 2: Tighten an additional 45 degrees.



109. Remove the LH Camshaft Holding Tool.

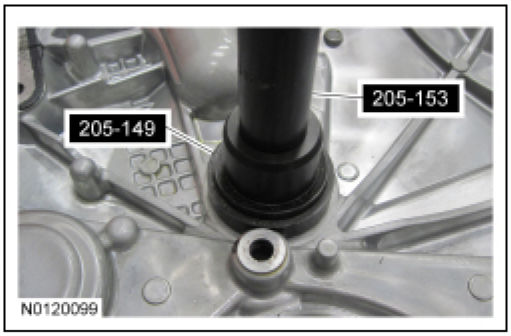


110. Install the LH valve train oil tube and the 3 bolts and tighten in 2 stages.
    - Stage 1: Tighten to 8 Nm (71 lb-in).
    - Stage 2: Tighten an additional 45 degrees.



111. **NOTE:** *Use Spindle Bearing Installer 205-149 to install the seal.*

    Using the Spindle Bearing Installer, install the front cover radial seal from the rear side of the front cover.

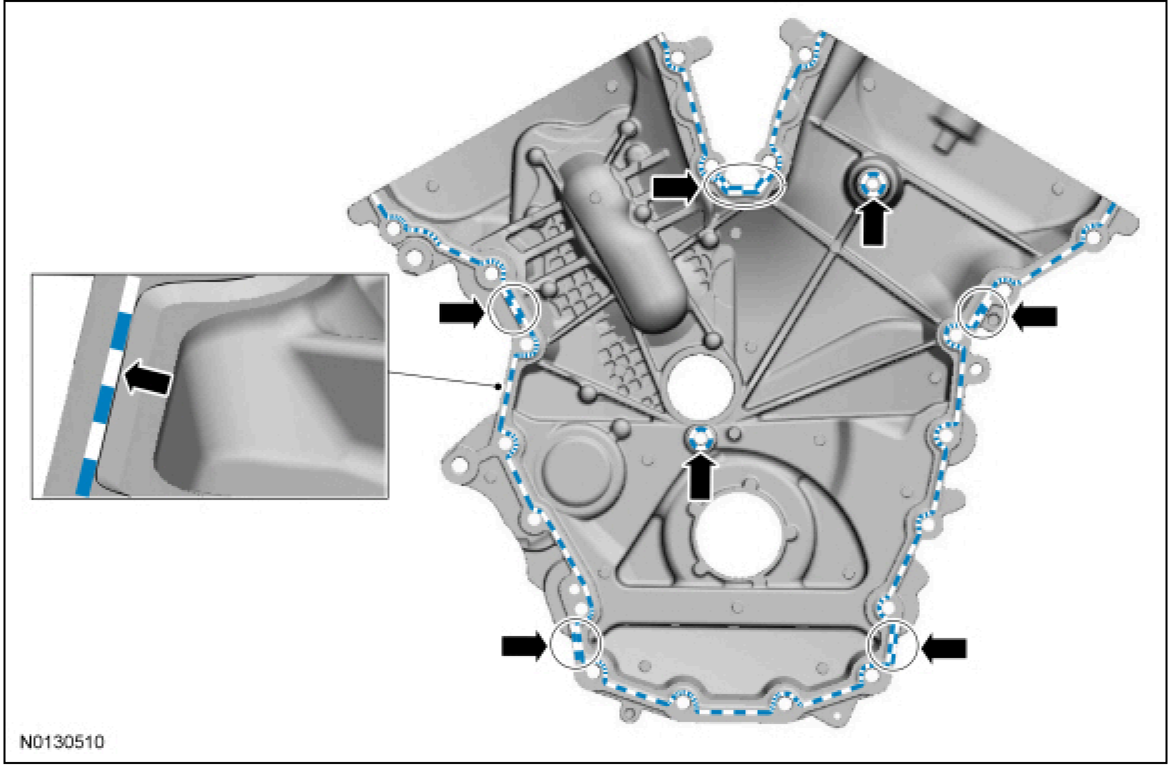


112. **NOTICE: Failure to use Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.**

    NOTICE: Do not expose the Motorcraft® High Performance Engine RTV Silicone to engine oil for at least 90 minutes after installing the engine front cover. Failure to follow this procedure can cause engine oil leakage.

    NOTE: *The engine front cover and bolts 1, 2, 7, 8, 15, 17, 18, 19, 20 and 21 must be installed within 10 minutes of the initial sealant application. The remainder of the engine front cover bolts and the engine mount bracket bolts must be installed and tightened within 35 minutes of the initial sealant application. If the time limits are exceeded, the sealant must be removed, the sealing area cleaned and sealant reapplied. To clean the sealing area, use silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.*

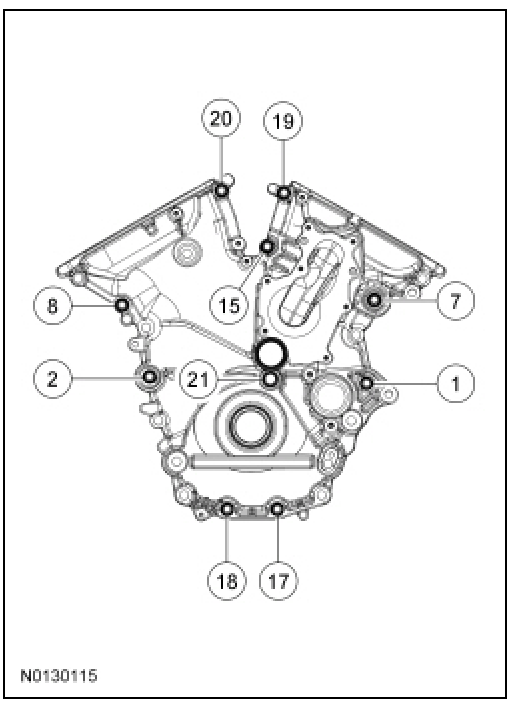
    Apply a 3.0 mm (0.11 in) bead of Motorcraft® High Performance Engine RTV Silicone on the inside edge of the engine front cover to the engine front cover sealing surfaces including the 2 engine front cover bolt bosses.
    - Apply a 5.5 mm (0.21 in) bead of Motorcraft® High Performance Engine RTV Silicone to the oil pan-to-cylinder block joint and the cylinder head-to-cylinder block joint areas of the engine front cover in 5 places as indicated.



113. **NOTE:** *Make sure the 2 locating dowel pins are seated correctly in the cylinder block.*

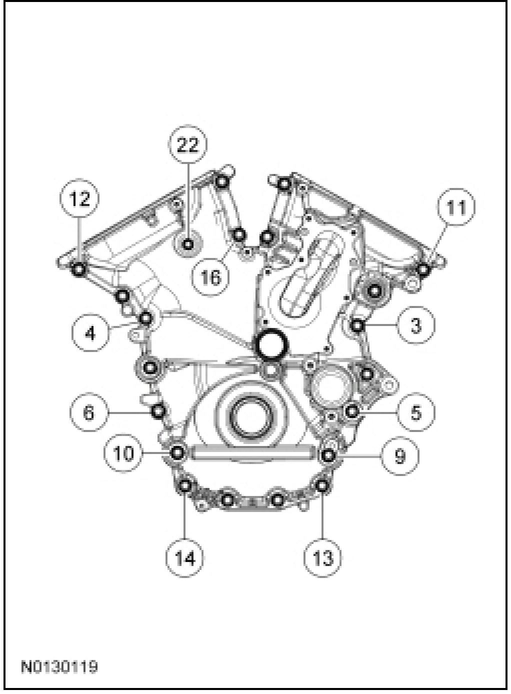
Install the engine front cover and bolts 1, 2, 7, 8, 15, 17, 18, 19, 20 and 21 and tighten in the sequence shown in 3 stages:

- Stage 1: Tighten bolts 1, 2, 7, 8, 15, 17, 18, 19, 20 and 21 in sequence to 10 Nm (89 lb-in).
- Stage 2: Tighten bolts 1, 2, 7, 8, 15, 17, 18, 19 and 20 in sequence to 20 Nm (177 lb-in) then an additional 45 degrees.
- Stage 3: Tighten bolt 21 to 20 Nm (177 lb-in) then an additional 90 degrees.

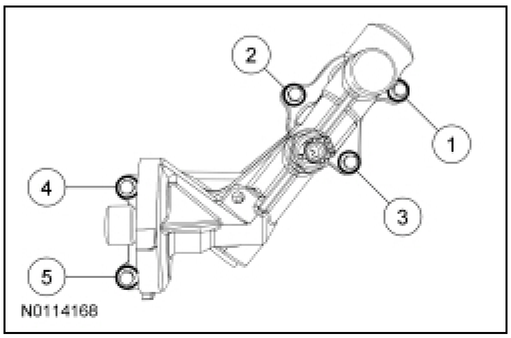


114. Install the remaining 12 engine front cover bolts 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 16, 22 and tighten in the sequence shown in 3 stages:

- Stage 1: Tighten bolts 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, and 16 to 10 Nm (89 lb-in).
- Stage 2: Tighten bolts 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, and 16 to 20 Nm (177 lb-in) then an additional 45 degrees.
- Stage 3: Tighten bolt 22 to 10 Nm (89 lb-in) then an additional 45 degrees.

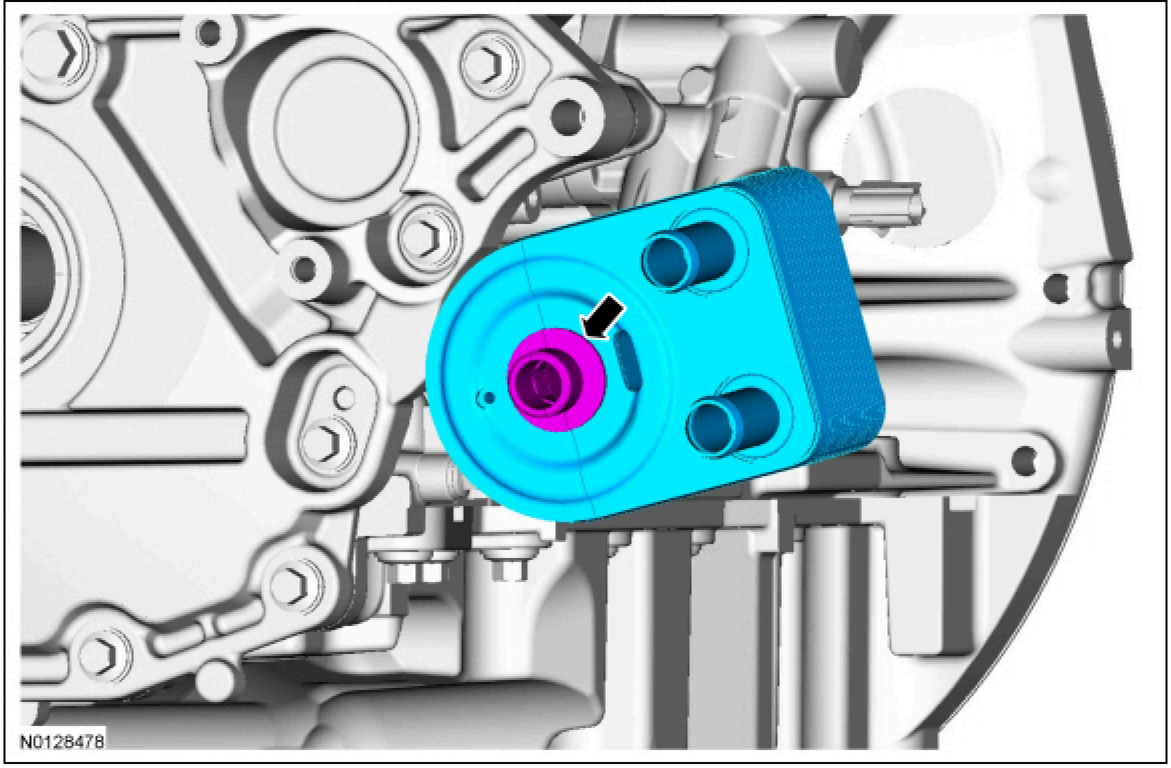


115. Using a new gasket, install the oil filter adapter and the 5 bolts and tighten in sequence shown.
    - Tighten to 10 Nm (89 lb-in) then an additional 45 degrees.

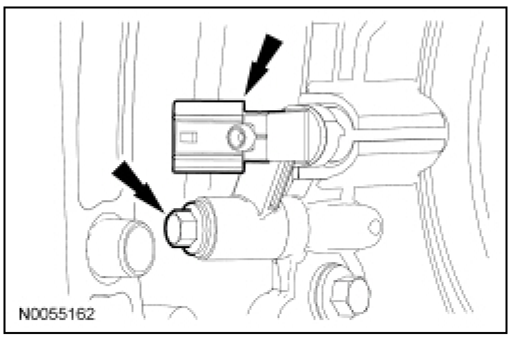


116. **NOTICE: A new oil cooler must be installed or severe damage to the engine can occur.**

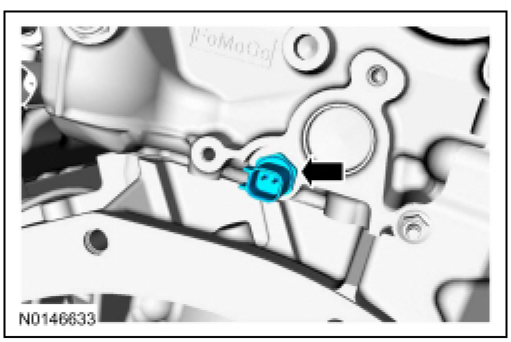
    Install the new oil cooler and the oil cooler insert bolt.
    - Tighten to 58 Nm (43 lb-ft).



117. Install the Crankshaft Position (CKP) sensor and install the bolt.
    - Tighten to 10 Nm (89 lb-in).

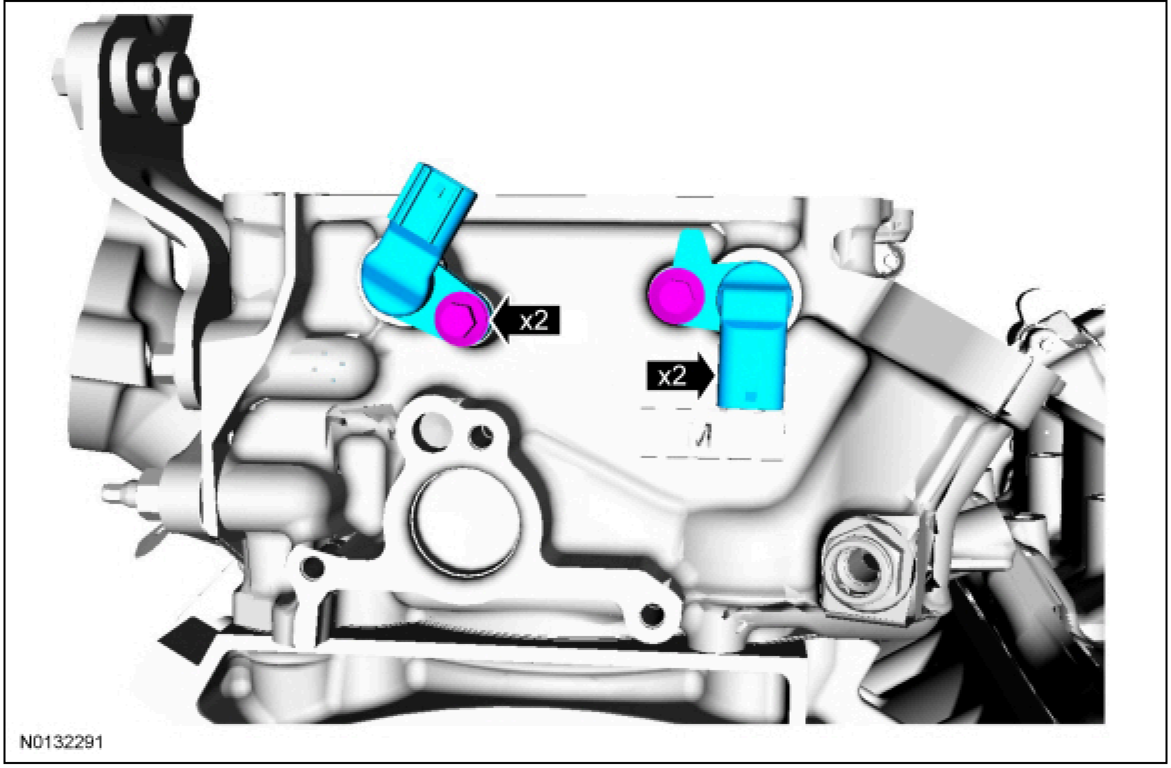


118. Install the new the Cylinder Head Temperature (CHT) sensor.
    - Tighten to 11 Nm (97 lb-in).

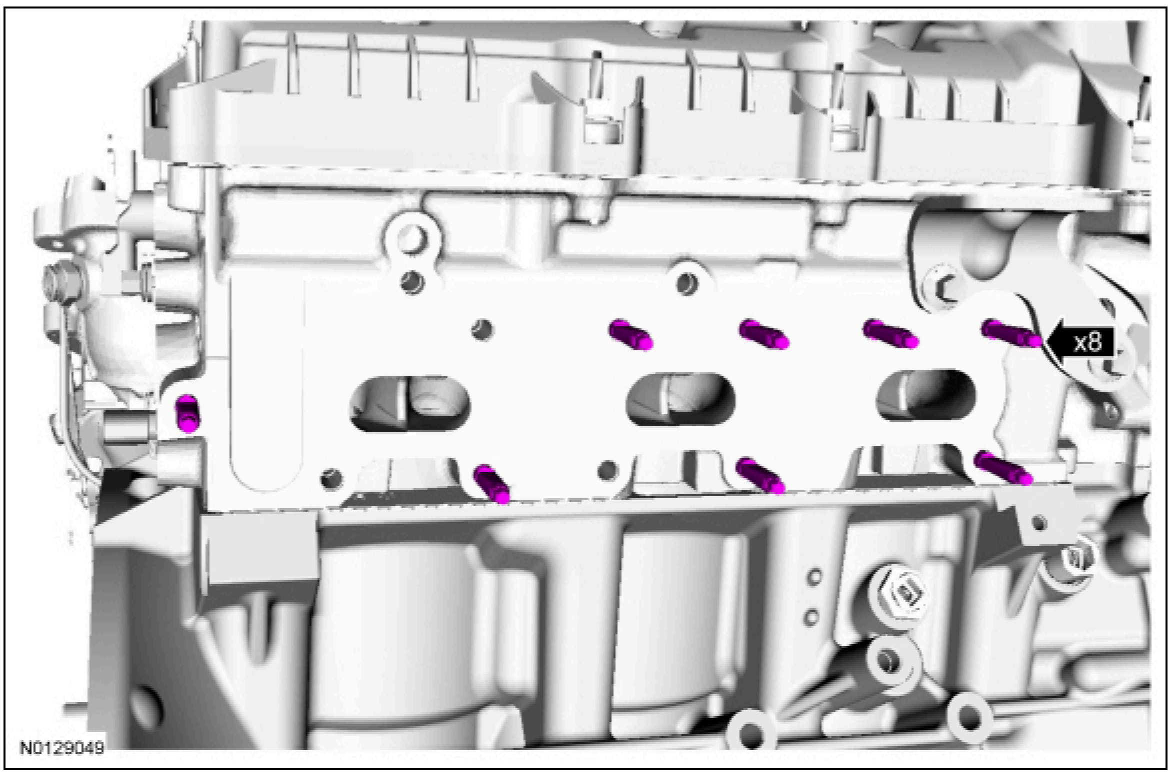


119. **NOTE:** *Lubricate the 2 CMP sensor O-ring seals with clean engine oil.*

    Install the 2 LH CMP sensors and the bolts.
    - Tighten to 10 Nm (89 lb-in).

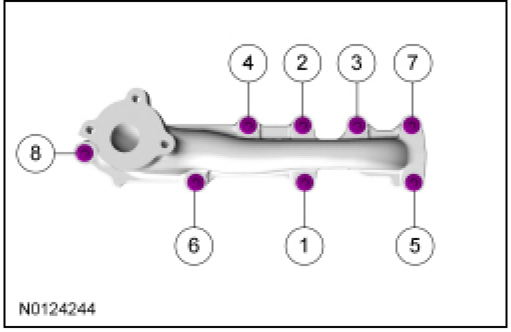


120. Install the 8 new RH exhaust manifold studs.
    - Tighten to 12 Nm (106 lb-in).



121. **NOTICE: Failure to tighten the exhaust manifold nuts to specification a second time will cause the exhaust manifold to develop an exhaust leak.**

    Using a new gasket, install the RH exhaust manifold and the 8 new nuts. Tighten in 2 stages in the sequence shown.
    - Stage 1: Tighten to 19 Nm (168 lb-in).
    - Stage 2: Tighten to 25 Nm (18 lb-ft).

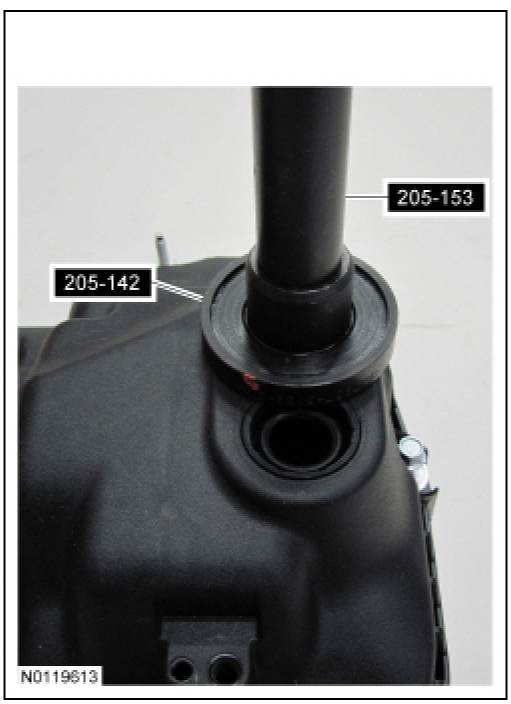


122. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

Using the VCT Spark Plug Tube Seal Installer and Handle, install new spark plug tube seals.

123. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

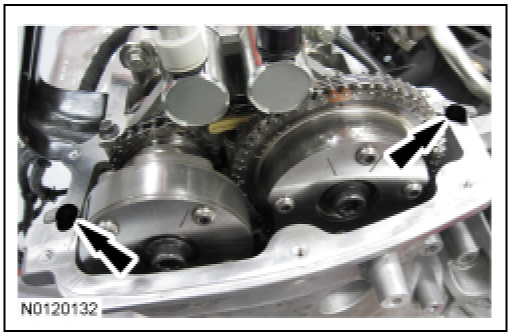
Using the Differential Bearing Cone Installer and Handle, install new VCT solenoid seal(s).



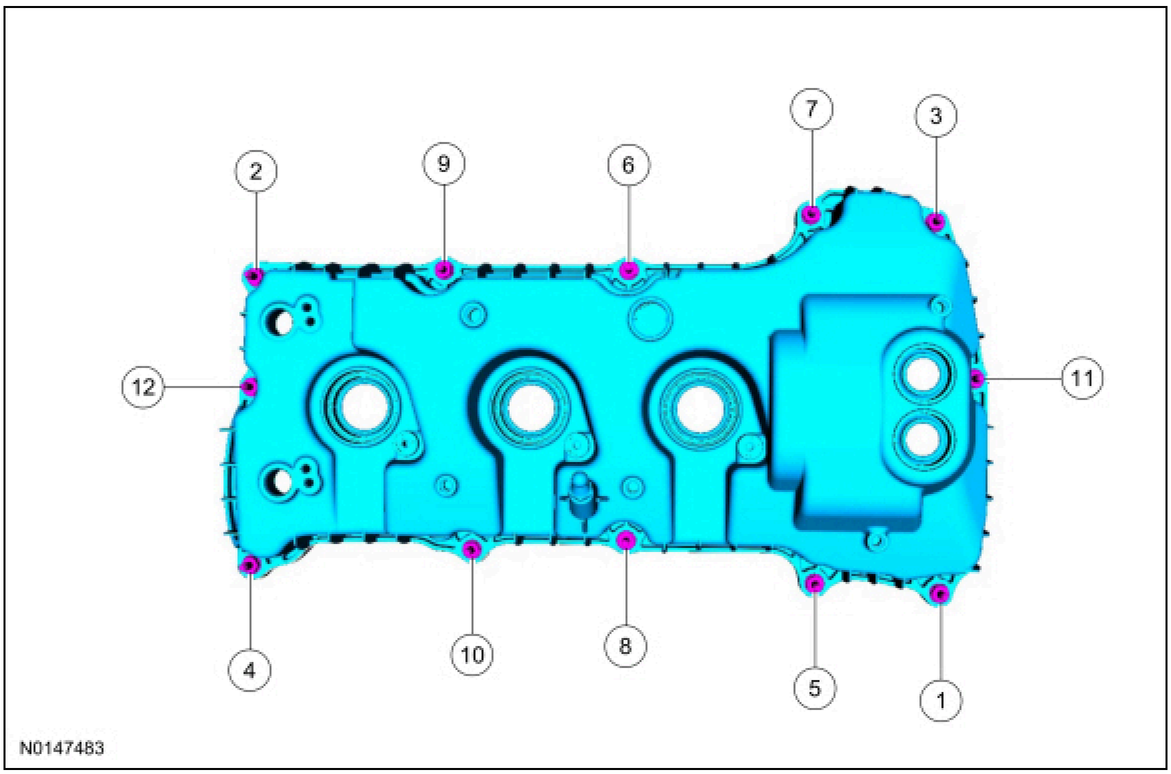
124. **NOTICE: Failure to use the correct Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.**

NOTE: *If the valve cover is not installed and the fasteners tightened within 4 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.*

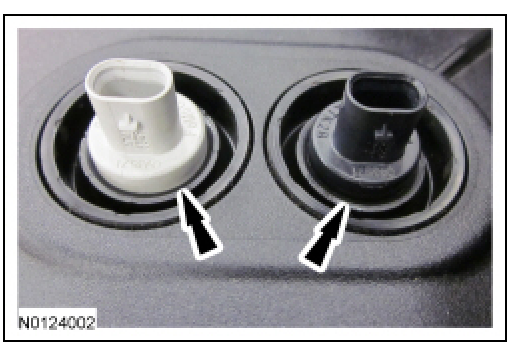
Apply an 8 mm (0.31 in) bead of Motorcraft® High Performance Engine RTV Silicone to the engine front cover-to-RH cylinder head joints.



125. Using a new gasket, install the RH valve cover and tighten the 12 stud bolts in the sequence shown.
   - Tighten to 10 Nm (89 lb-in).



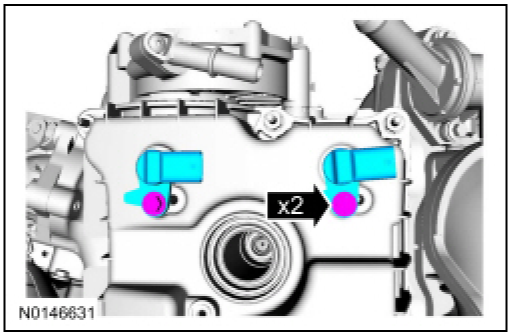
126. Ensure the VCT seals in the valve cover are below the top of the VCT oil control solenoid electrical connector or the VCT seal may leak oil.



127. **NOTE:** *Lubricate the 2 CMP sensor O-ring seals with clean engine oil.*

Install the 2 RH CMP sensors and the bolts.

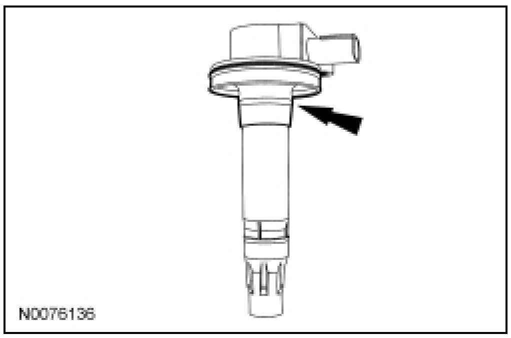
- Tighten to 7 Nm (62 lb-in).



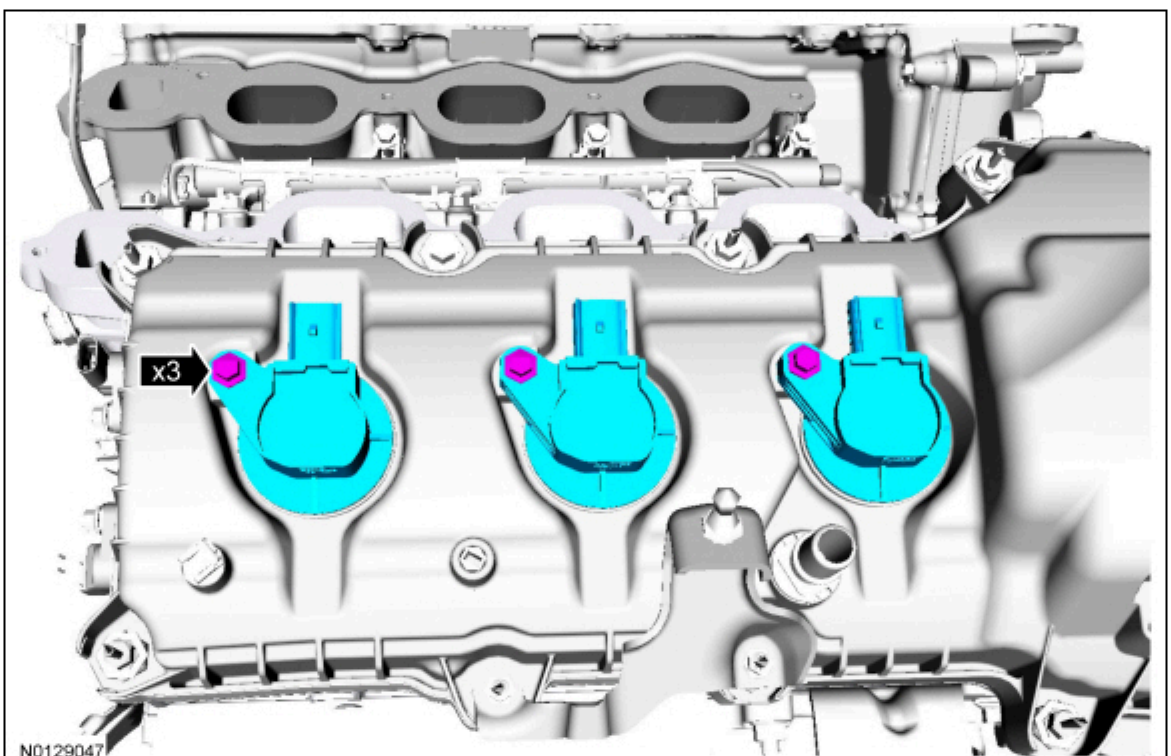
128. **NOTICE: Only use hand tools when installing the spark plugs, or damage may occur to the cylinder head or spark plug.**

    Install the 3 RH spark plugs.
    - Tighten to 15 Nm (133 lb-in).

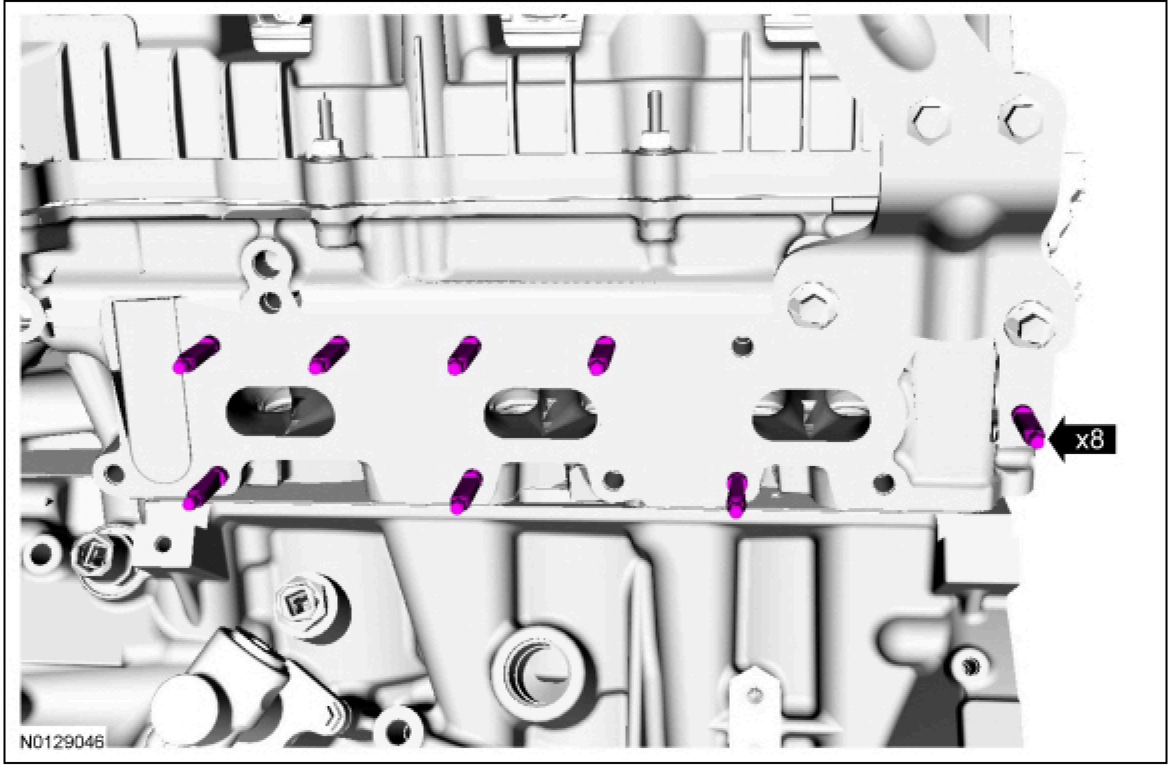
129. Inspect the coil seals for rips, nicks or tears. Remove and discard any damaged coil seals.
    - To install, slide the new coil seal onto the coil until it is fully seated at the top of the coil.



130. Install the 3 RH ignition coil-on-plug assemblies and the 3 bolts.
    - Tighten to 7 Nm (62 lb-in) then an additional 45 degrees.



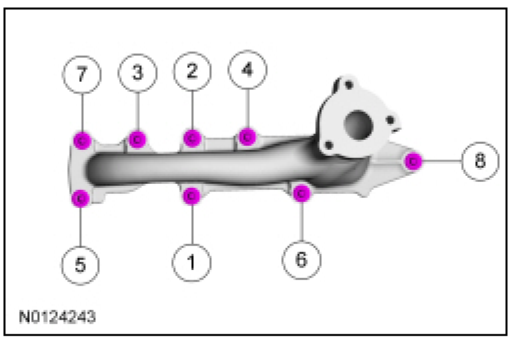
131. Install the 8 new LH exhaust manifold studs.
    - Tighten to 12 Nm (106 lb-in).



132. **NOTICE: Failure to tighten the exhaust manifold nuts to specification a second time will cause the exhaust manifold to develop an exhaust leak.**

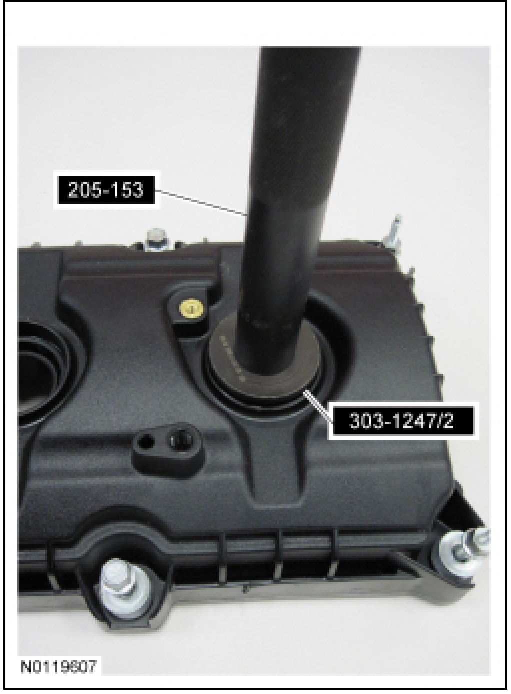
Using a new gasket, install the LH exhaust manifold and 8 new nuts. Tighten in 2 stages in the sequence shown.

- Stage 1: Tighten to 19 Nm (168 lb-in).
- Stage 2: Tighten to 25 Nm (18 lb-ft).



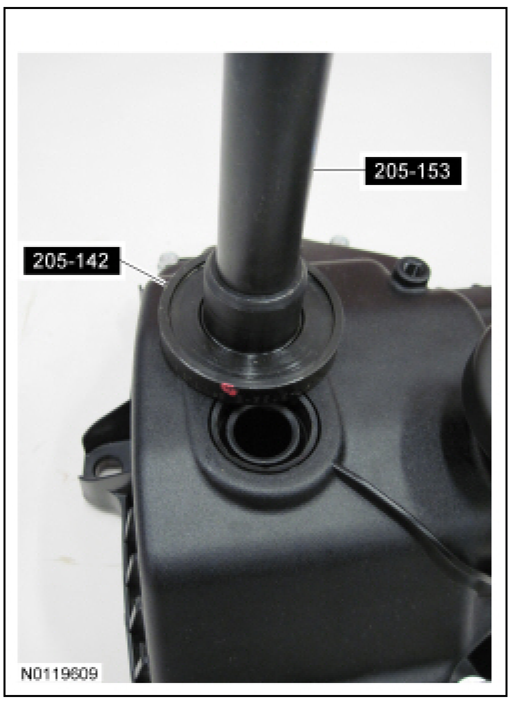
133. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

Using the VCT Spark Plug Tube Seal Installer and Handle, install new spark plug tube seals.



134. **NOTE:** *Installation of new seals is only required if damaged seals were removed.*

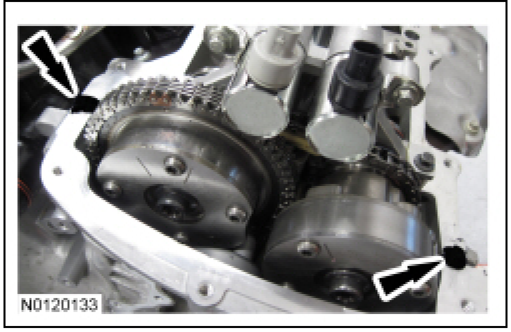
Using the Differential Bearing Cone Installer and Handle, install new VCT solenoid seal(s).



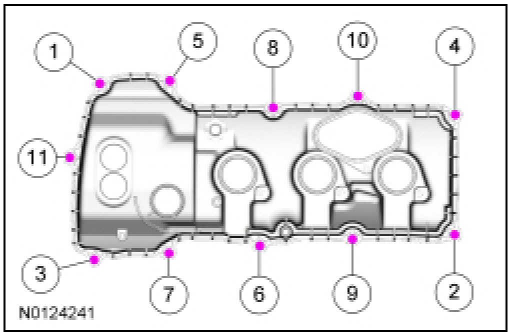
135. **NOTICE: Failure to use the correct Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.**

NOTE: *If the valve cover is not installed and the fasteners tightened within 4 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.*

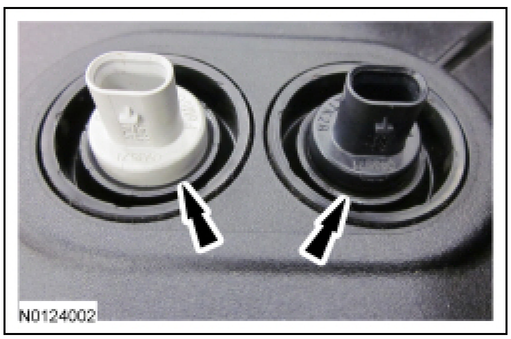
Apply an 8 mm (0.31 in) bead of Motorcraft® High Performance Engine RTV Silicone to the engine front cover-to-LH cylinder head joints.



136. Using a new gasket, install the LH valve cover and tighten the 11 stud bolts in the sequence shown.
    - Tighten to 10 Nm (89 lb-in).



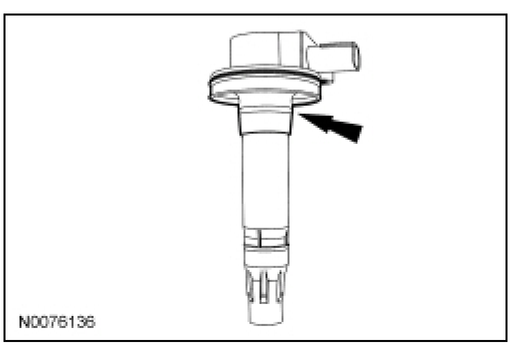
137. Ensure the VCT seals in the valve cover are below the top of the VCT oil control solenoid electrical connector or the VCT seal may leak oil.



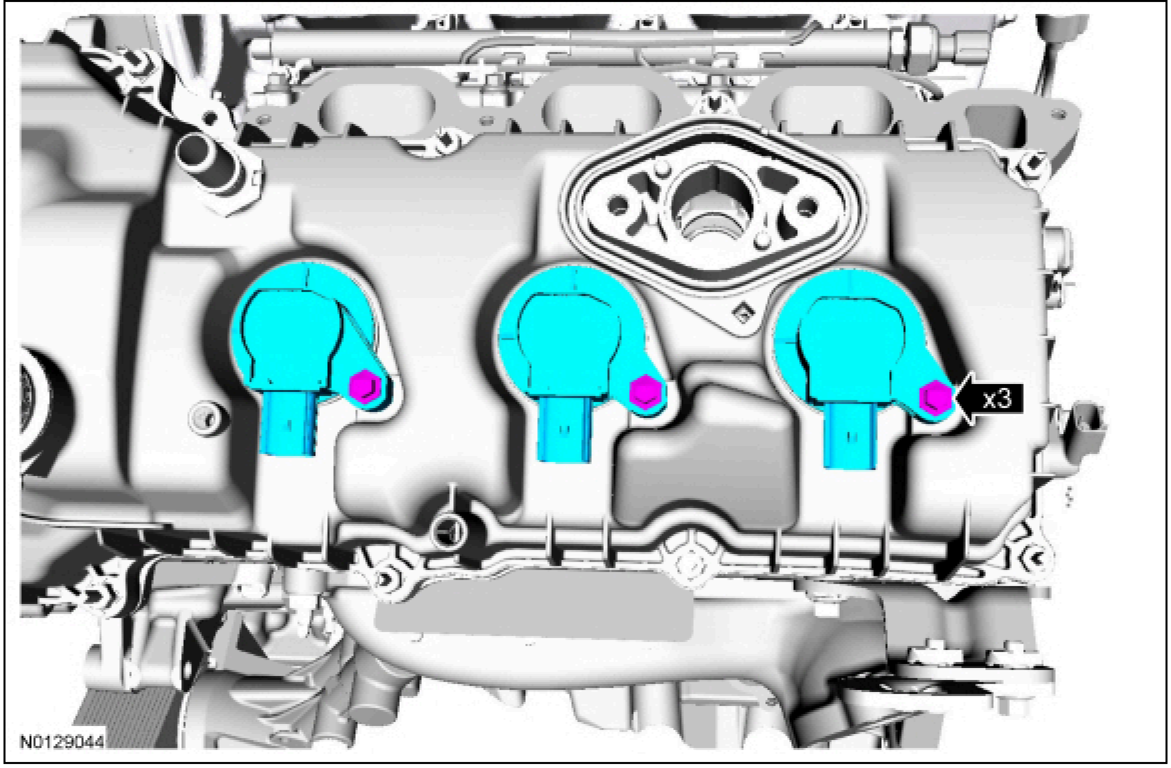
138. **NOTICE: Only use hand tools when installing the spark plugs, or damage may occur to the cylinder head or spark plug.**

    Install the 3 LH spark plugs.
    - Tighten to 15 Nm (133 lb-in).

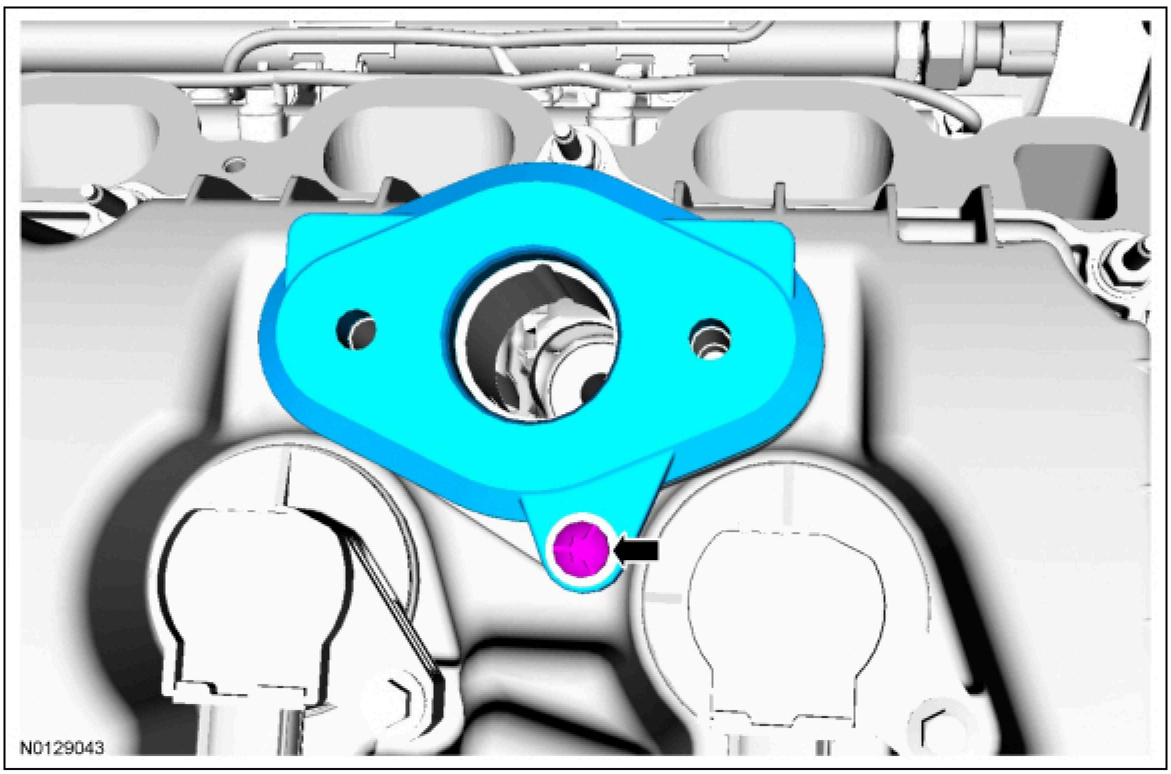
139. Inspect the coil seals for rips, nicks or tears. Remove and discard any damaged coil seals.
    - To install, slide the new coil seal onto the coil until it is fully seated at the top of the coil.



140. Install the 3 LH ignition coil-on-plug assemblies and the 3 bolts.
    - Tighten to 7 Nm (62 lb-in) then an additional 45 degrees.



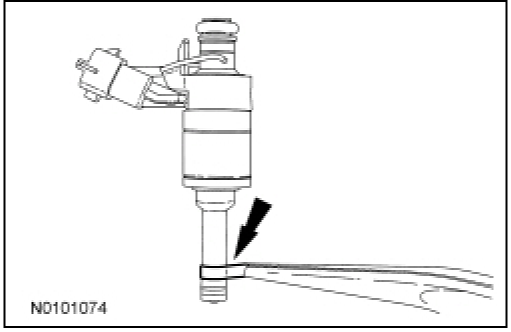
141. Using new gaskets, install the fuel injection pump mounting plate and bolt.
    - Tighten finger-tight.



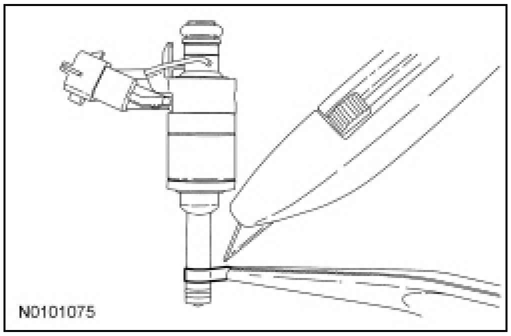
142. **NOTICE: Do not attempt to cut the lower Teflon® seal without pulling it away from the fuel injector or damage to the fuel injector may occur.**

    NOTE: *Be very careful when removing the lower Teflon® seals, not to scratch, nick or gouge the fuel injectors.*

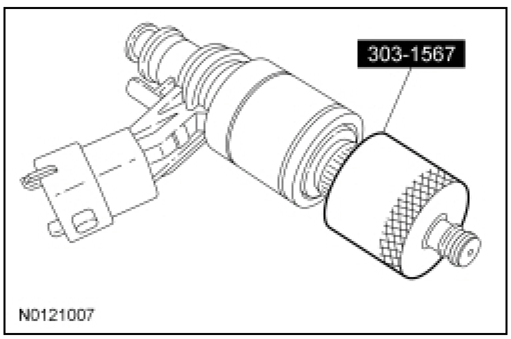
    Pull the Teflon® seal away from the fuel injector with narrow tip pliers.



143. Carefully cut and discard the lower fuel injector Teflon® seals.

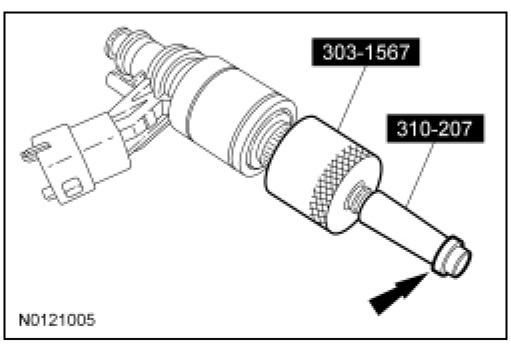


144. Place the Teflon® Seal Sizer, knurled side out, over the fuel injector tip until the fuel injector seal recess is exposed.



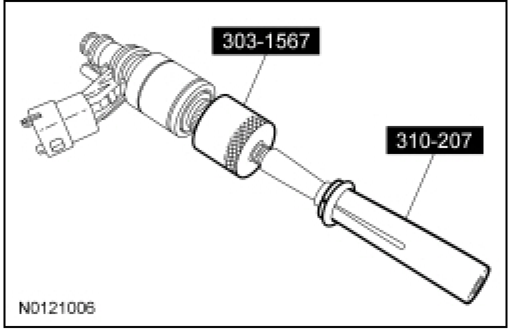
145. **NOTE:** *Do not lubricate the new lower Teflon® fuel injector seals.*

Install the new lower Teflon® seals on the narrow end of the Arbor (part of the Fuel Injector Seal Installer), then install the Arbor on the fuel injector tip.



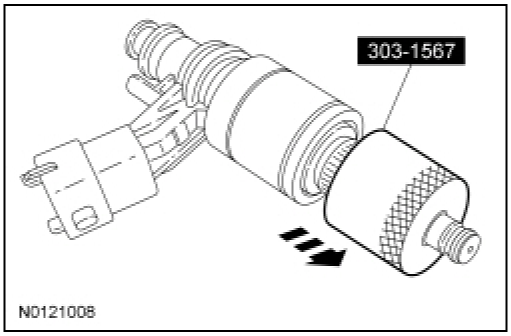
146. **NOTICE: Once the Teflon® seal is installed on the Arbor, it should immediately be installed onto the fuel injector to avoid excessive expansion of the seal.**

Using the Pusher Tool (part of the Fuel Injector Seal Installer), slide the Teflon® seals off of the Arbor and into the groove on the fuel injector.



147. **NOTE:** *Make sure the Teflon® seal is fully seated in the groove on the fuel injector before sizing the seal.*

Slide the Teflon® Seal Sizer off the end of the fuel injector to size the seal.

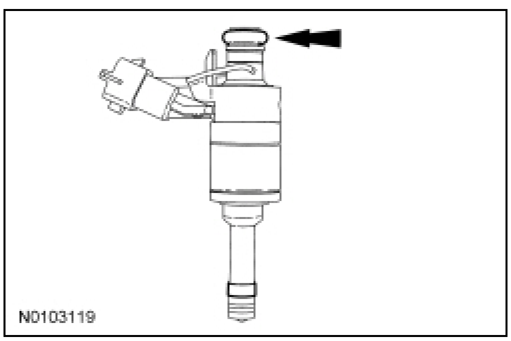


148. **NOTICE: Use fuel injector O-ring seals that are made of special fuel-resistant material. The use of ordinary O-ring seals may cause the fuel system to leak. Do not reuse the O-rings.**

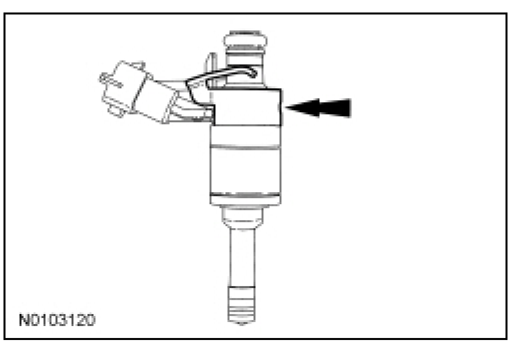
NOTE: *Apply clean engine oil to the 6 new upper fuel injector O-ring seals.*

NOTE: *Inspect the fuel injector support disk and replace if necessary.*

Install the 6 new upper fuel injector O-ring seals.



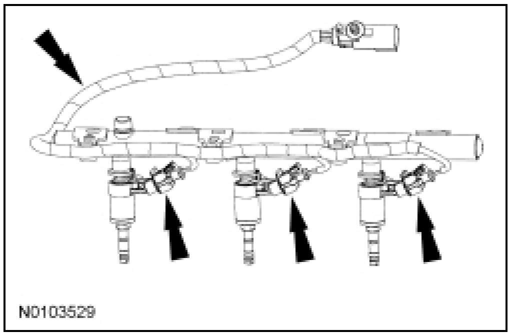
149. Install the 6 new fuel injector clips.



150. **NOTE:** *RH shown, LH similar.*

NOTE: *The anti rotation device on the fuel injector has to slip into the groove of the fuel rail cup.*

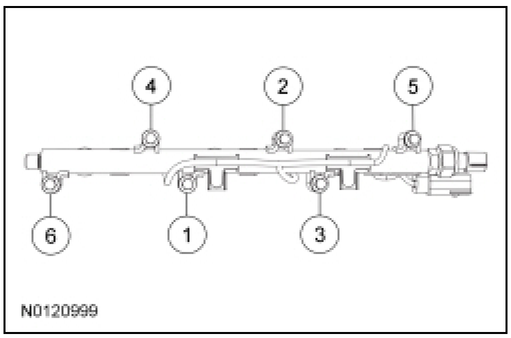
Install the 6 fuel injectors into the fuel rails and connect the fuel injector electrical connector.



151. **NOTICE: It is very important to visually inspect the routing of the fuel charge wire harness to make sure that they will not be pinched or damaged between the fuel rail and the cylinder head during installation.**

NOTE: *Tighten the bolts using a method that draws the fuel rail evenly to the cylinder head, preventing a rocking motion.*

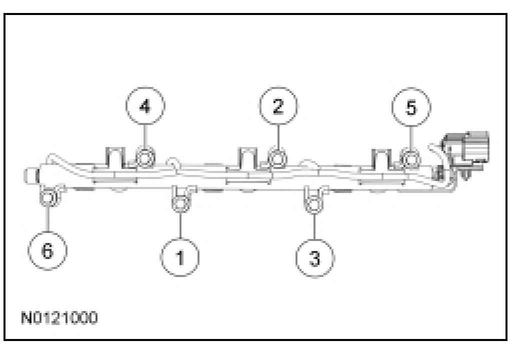
Install the RH fuel rail assembly and the 6 new bolts. Tighten in 3 stages in the sequence shown.
- Stage 1: Push down on the fuel rail face above the fuel injectors.
- Stage 2: Tighten to 10 Nm (89 lb-in).
- Stage 3: Tighten an additional 45 degrees.



152. **NOTICE: It is very important to visually inspect the routing of the fuel charge wire harness to make sure that they will not be pinched or damaged between the fuel rail and the cylinder head during installation.**

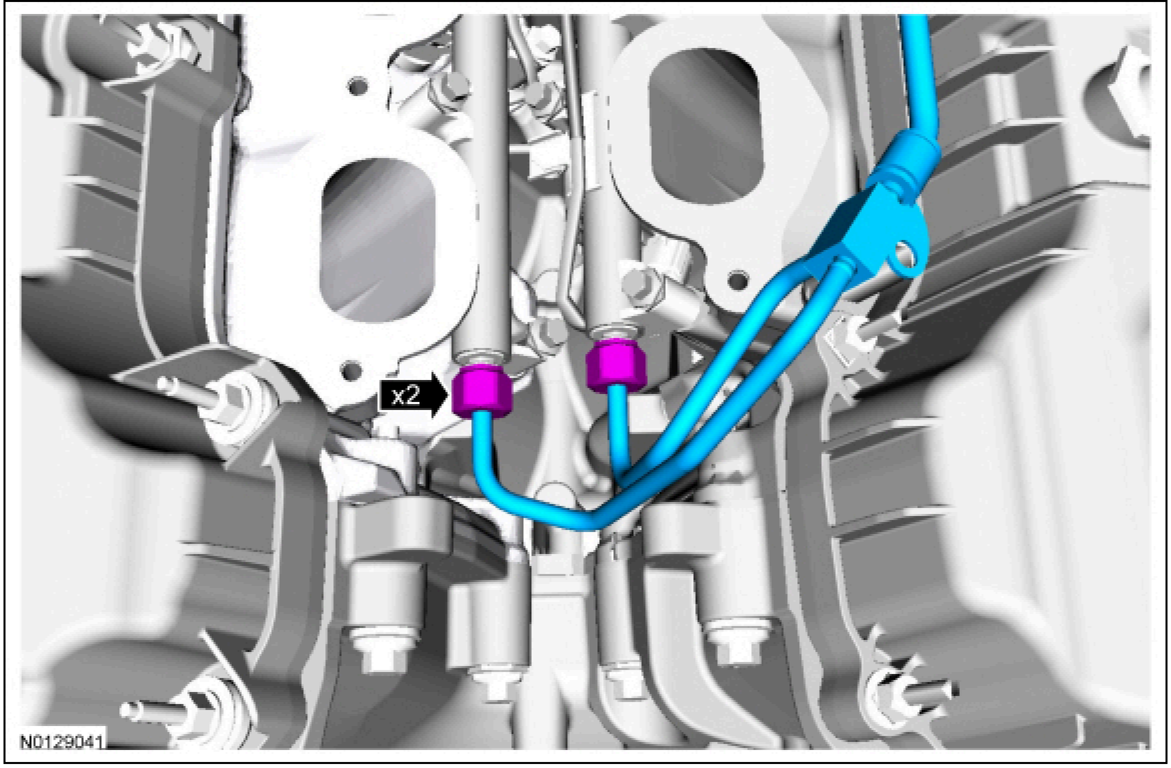
NOTE: *Tighten the bolts using a method that draws the fuel rail evenly to the cylinder head, preventing a rocking motion.*

Install the LH fuel rail assembly and the 6 new bolts. Tighten in 3 stages in the sequence shown.
- Stage 1: Push down on the fuel rail face above the fuel injectors.
- Stage 2: Tighten to 10 Nm (89 lb-in).
- Stage 3: Tighten an additional 45 degrees.

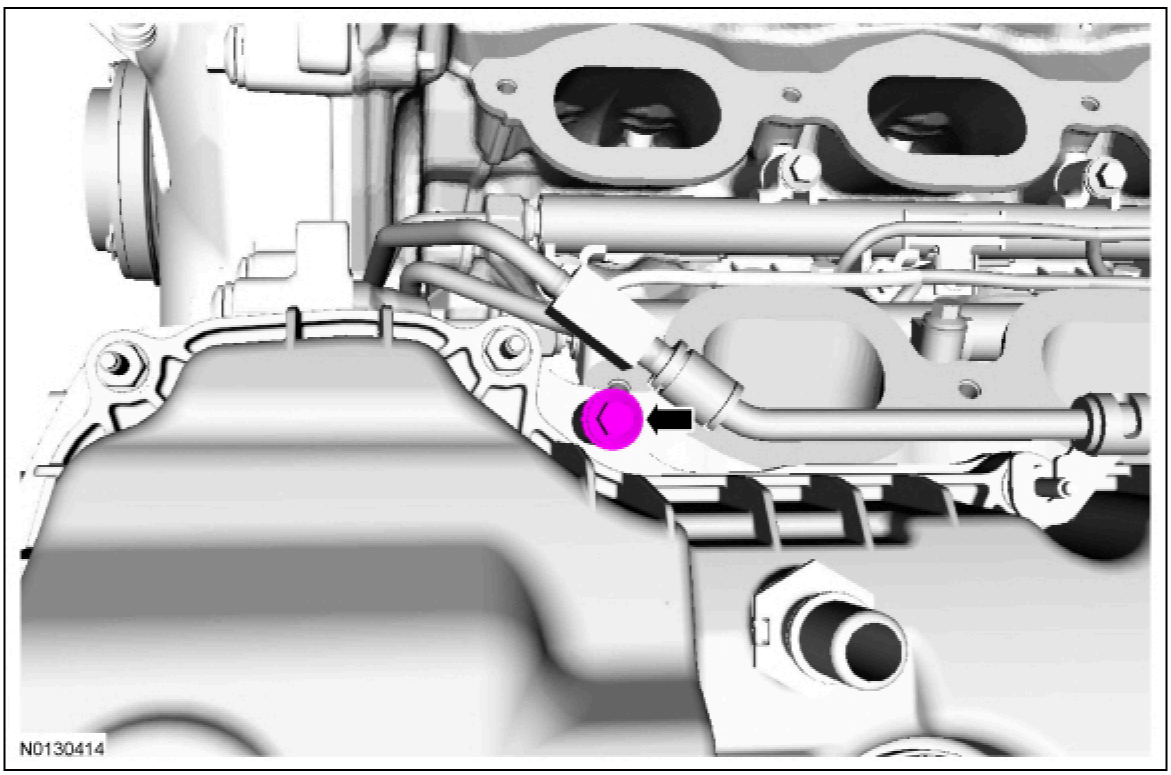


153. **NOTE:** *Apply clean engine oil to the threads of the 2 high-pressure fuel tube flare nuts.*

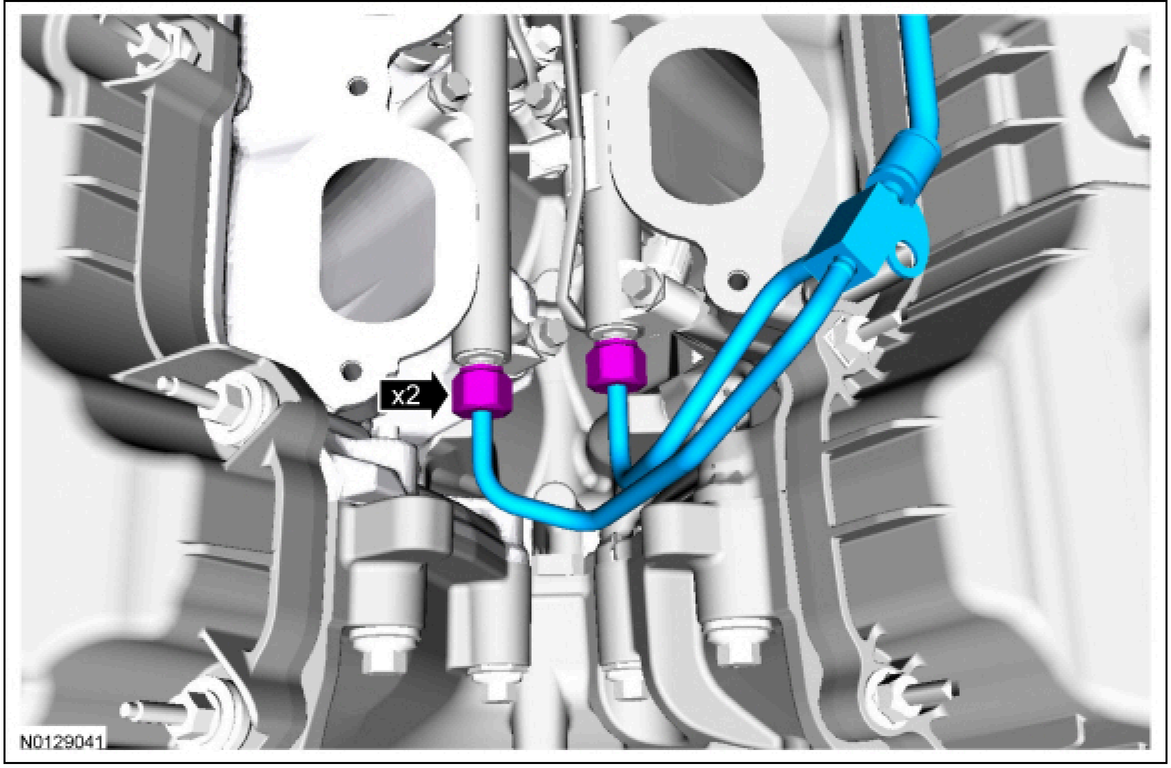
Position the new high-pressure fuel tube and hand start the flare nuts in 2 stages.
- Stage 1: Hand tighten the high-pressure fuel tube-to-RH fuel rail flare nut.
- Stage 2: Hand tighten the high-pressure fuel tube-to-LH fuel rail flare nut.



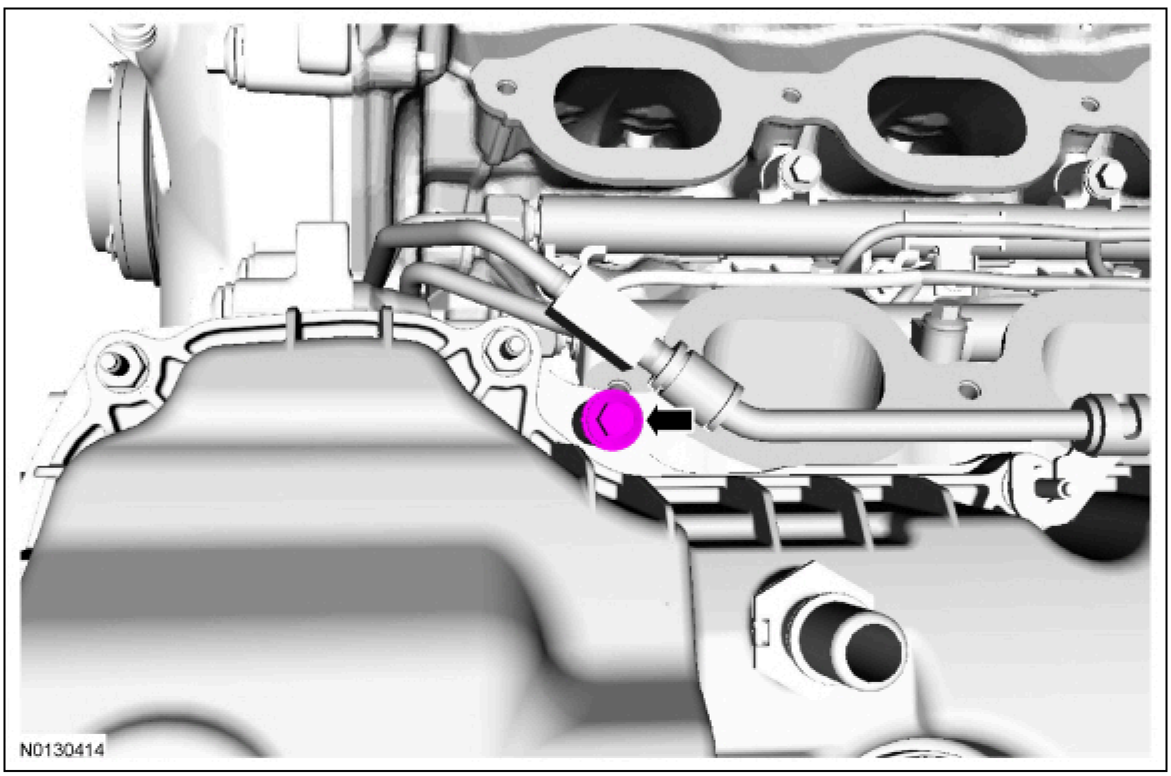
154. Loosely install the bolt for the high-pressure fuel tube.



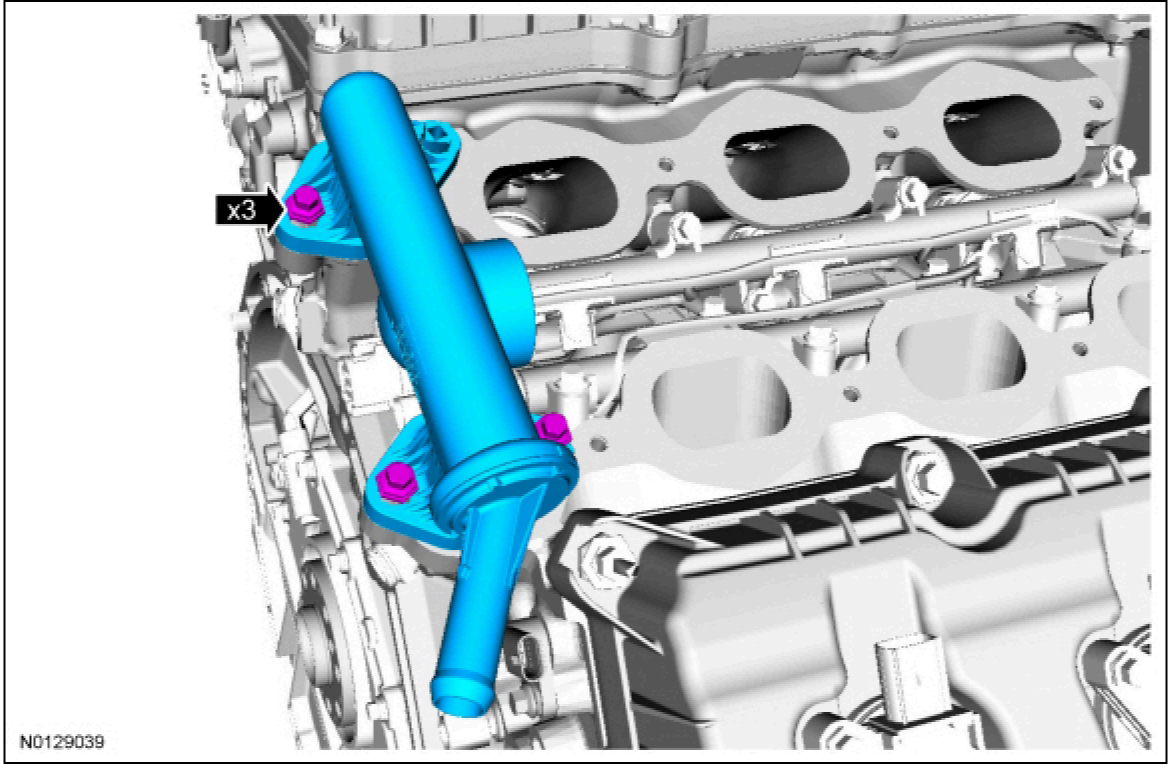
155. Connect the high-pressure fuel tube to the fuel rails. Tighten in 4 stages.
  - Stage 1: Tighten the high-pressure fuel tube-to-RH fuel rail flare nut to 15 Nm (133 lb-in).
  - Stage 2: Tighten an additional 30 degrees or 50 Nm (37 lb-ft), whichever comes first.
  - Stage 3: Tighten the high-pressure fuel tube-to-LH fuel rail flare nut to 15 Nm (133 lb-in).
  - Stage 4: Tighten an additional 30 degrees or 50 Nm (37 lb-ft), whichever comes first.



156. Tighten the bolt for the high-pressure fuel tube.
    - Tighten the bolt to 10 Nm (89 lb-in) then an additional 45 degrees.

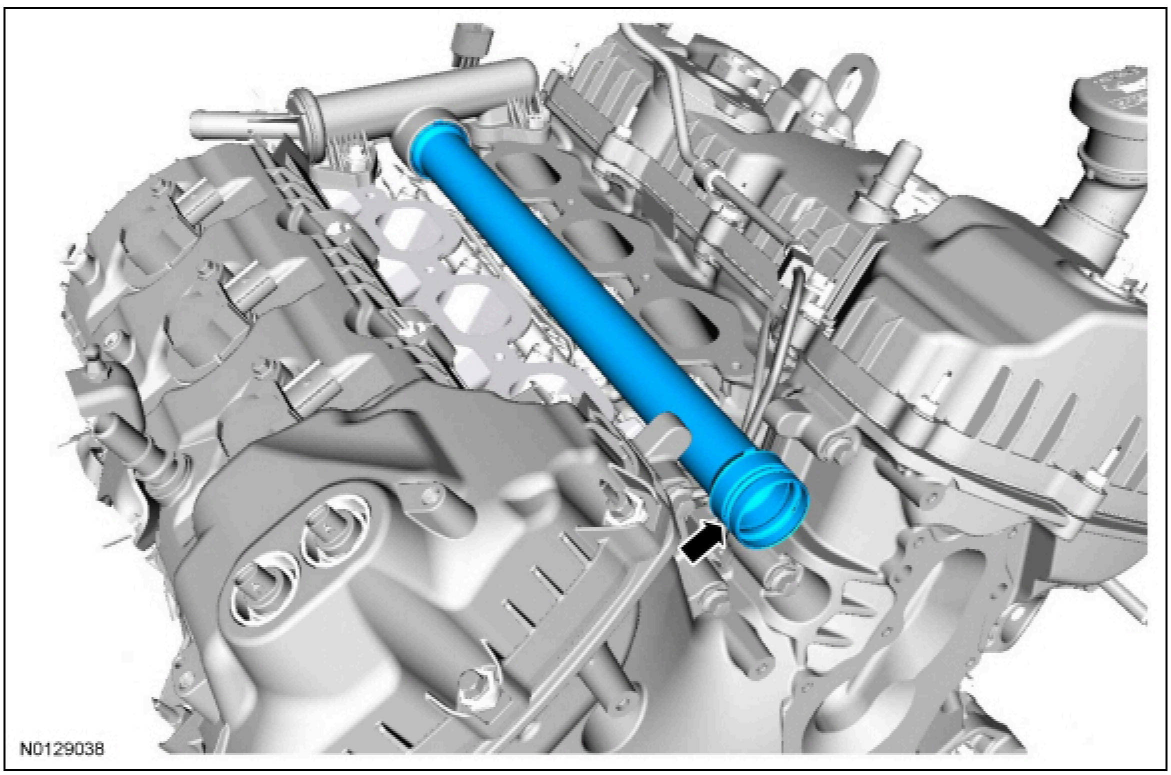


157. Using new gaskets, install the coolant crossover manifold assembly and the 3 bolts.
    - Tighten to 10 Nm (89 lb-in) then an additional 45 degrees.



158. **NOTE:** *Lubricate the O-ring seal with clean engine coolant.*

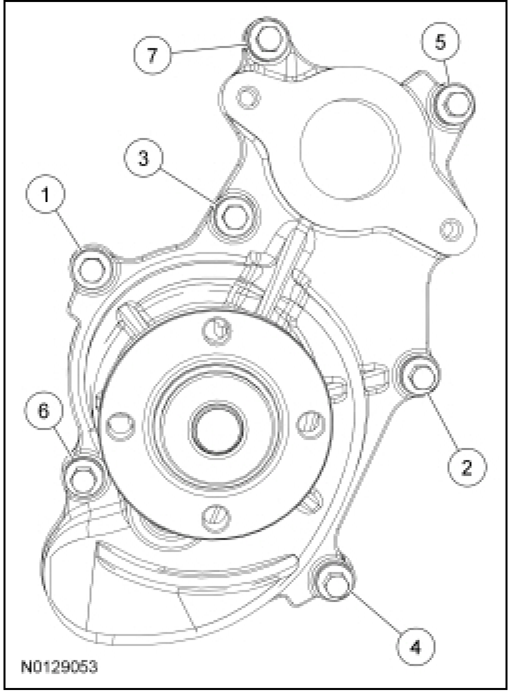
Using new O-ring seals, install the coolant inlet pipe.



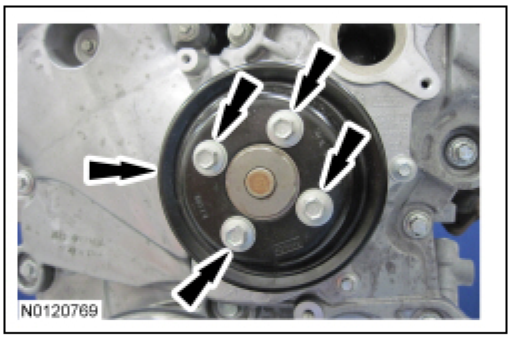
159. **NOTE:** *Lubricate the O-ring seal with clean engine coolant.*

Using new coolant pump gasket and coolant pump O-ring seal, install the coolant pump and the 7 bolts and tighten in 2 stages in the sequence shown.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten an additional 45 degrees.

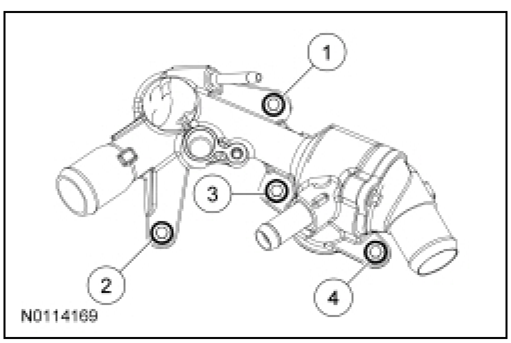


160. Install the coolant pump pulley and the 4 bolts.
    - Tighten to 24 Nm (18 lb-ft).



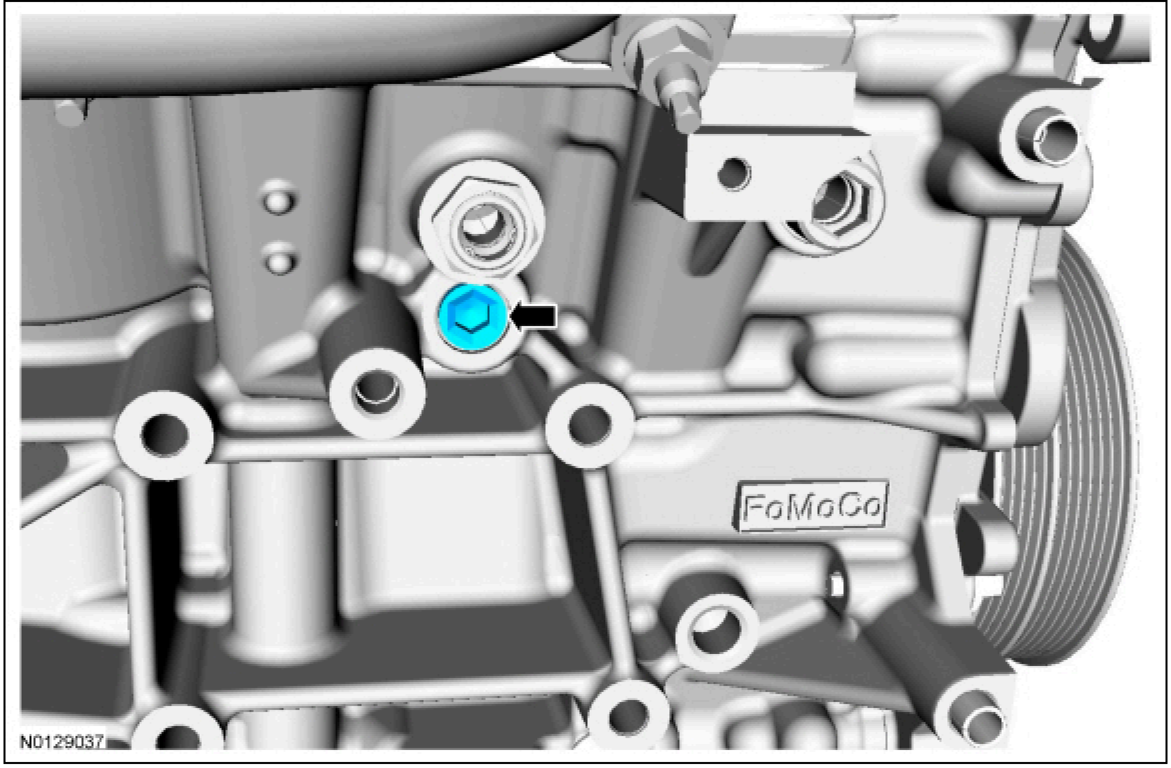
161. **NOTE:** *Lubricate the coolant inlet tube O-ring seal with clean engine coolant*.

    Install the thermostat housing and the 4 bolts and tighten in 2 stages in the sequence shown.
    - Stage 1: Tighten to 8 Nm (71 lb-in).
    - Stage 2: Tighten an additional 45 degrees.



162. **NOTE:** *Apply thread sealant with PTFE to the threads prior to installing the cylinder block drain plug*.

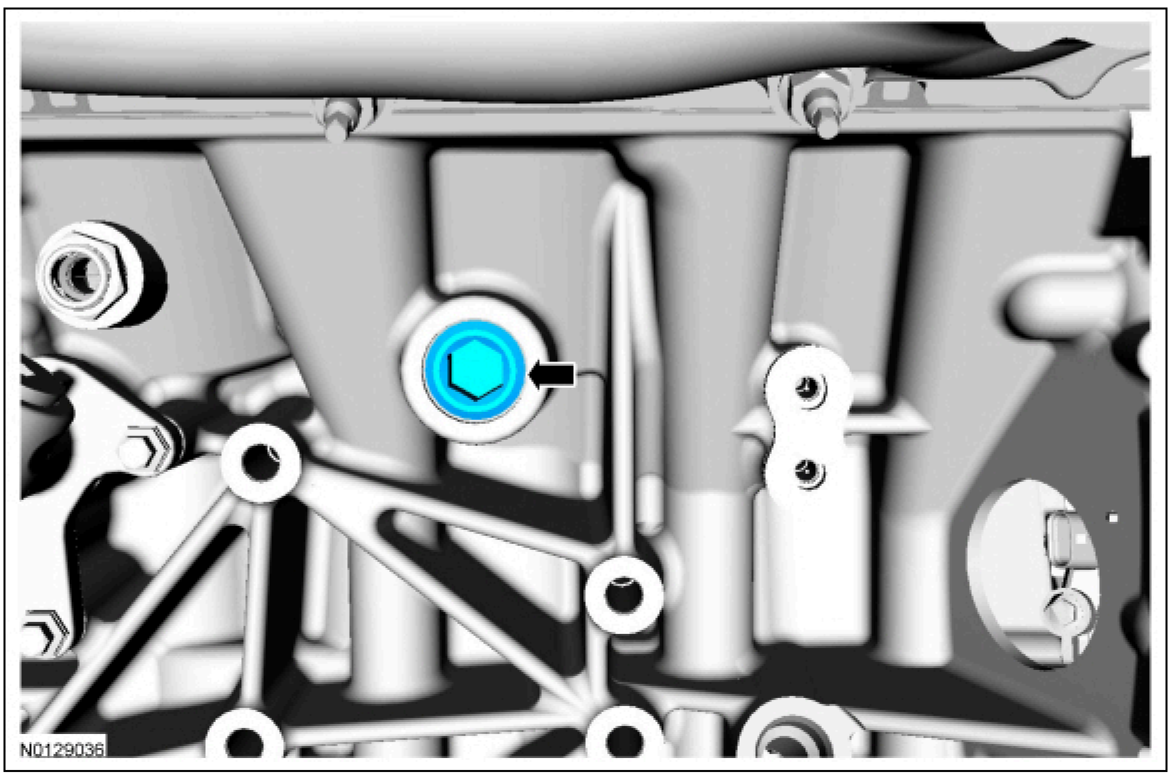
    Install the RH cylinder block drain plug.
    - Tighten to 20 Nm (177 lb-in) then an additional 180 degrees.



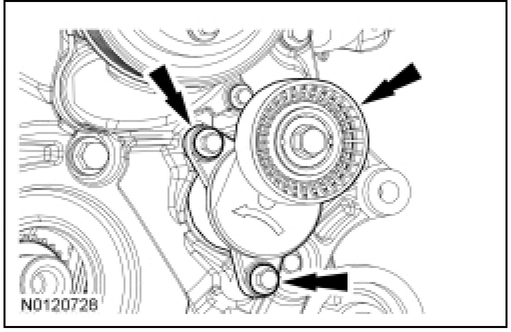
163. **NOTE:** *Apply thread sealant with PTFE to the threads prior to installing the cylinder block drain plug.*

Install the LH cylinder block drain plug.

- Tighten the cylinder block drain plug to 10 Nm (89 lb-in) then an additional 720 degrees.

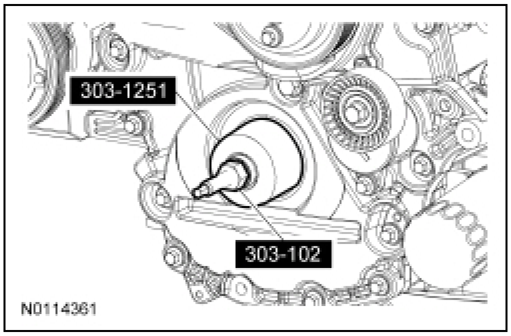


164. Install the accessory drive belt tensioner and the 2 bolts.
    - Tighten to 25 Nm (18 lb-ft).



165. **NOTE:** *Apply clean engine oil to the crankshaft front seal bore in the engine front cover.*

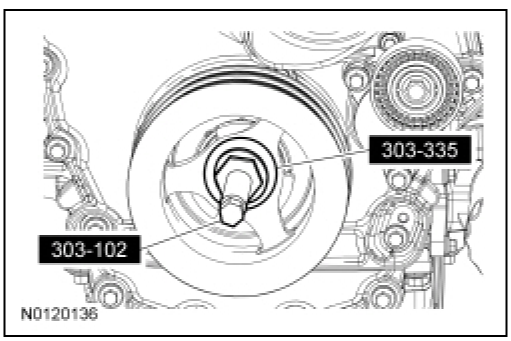
Using the Front Crankshaft Seal Installer and Crankshaft Vibration Damper Installer, install a new crankshaft front seal.



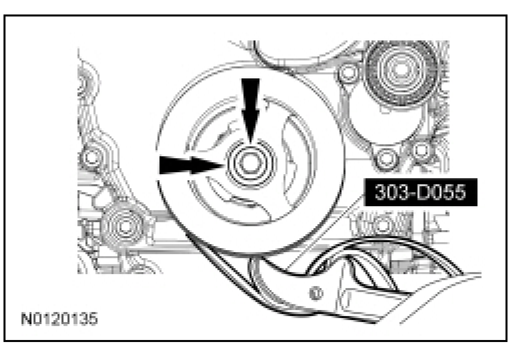
166. Lubricate the crankshaft front seal inner lip with clean engine oil.

167. **NOTE:** *Lubricate the outside diameter sealing surfaces with clean engine oil.*

Using the Front Cover Oil Seal Installer and Crankshaft Vibration Damper Replacer, install the crankshaft pulley.

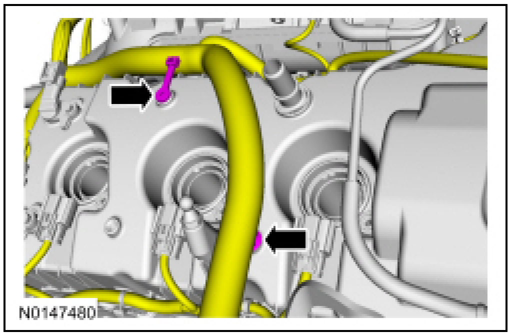


168. Using the Strap Wrench, install the crankshaft pulley washer and new bolt and tighten in 4 stages.
   - Stage 1: Tighten to 120 Nm (89 lb-ft).
   - Stage 2: Loosen one full turn.
   - Stage 3: Tighten to 50 Nm (37 lb-ft).
   - Stage 4: Tighten an additional 90 degrees.

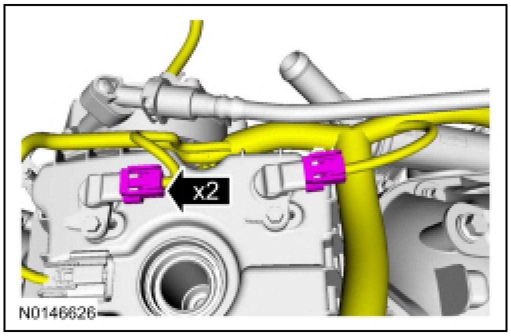


169. Position the engine wire harness on the engine. Attach all of the wiring harness retainers to the RH valve cover and stud bolts.

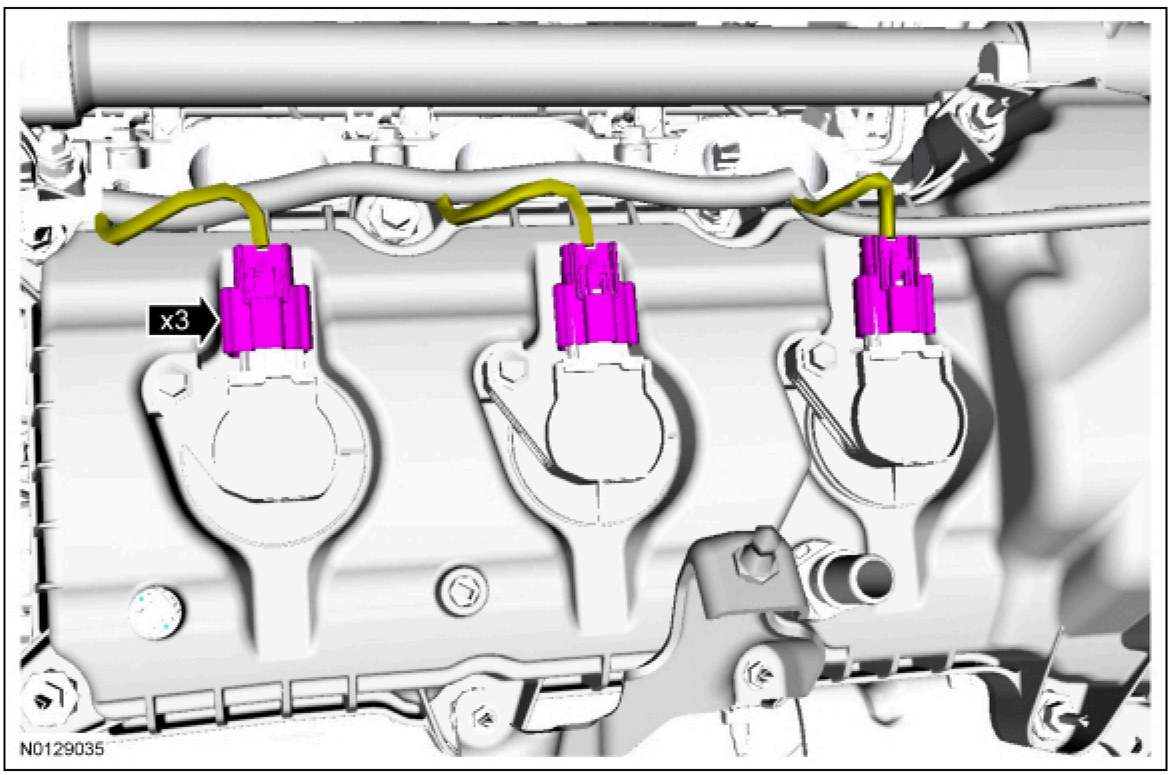
170. Connect the wire harness retainer and install the tie strap.



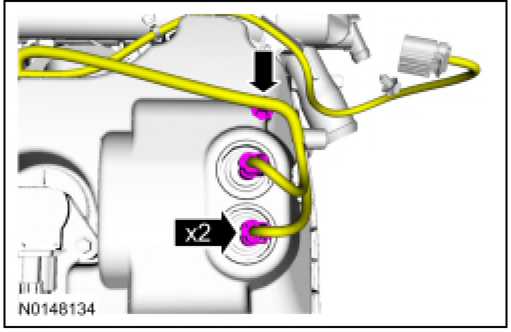
171. Connect the 2 RH CMP electrical connectors.



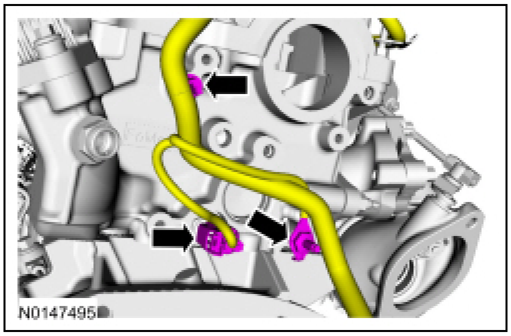
172. Connect the 3 RH ignition coil-on-plug electrical connectors.



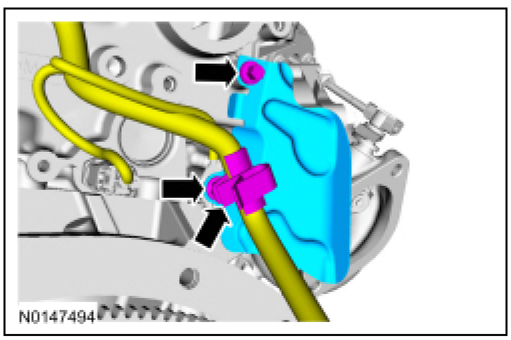
173. Connect the 2 RH VCT solenoid electrical connectors. Install the tie strap.



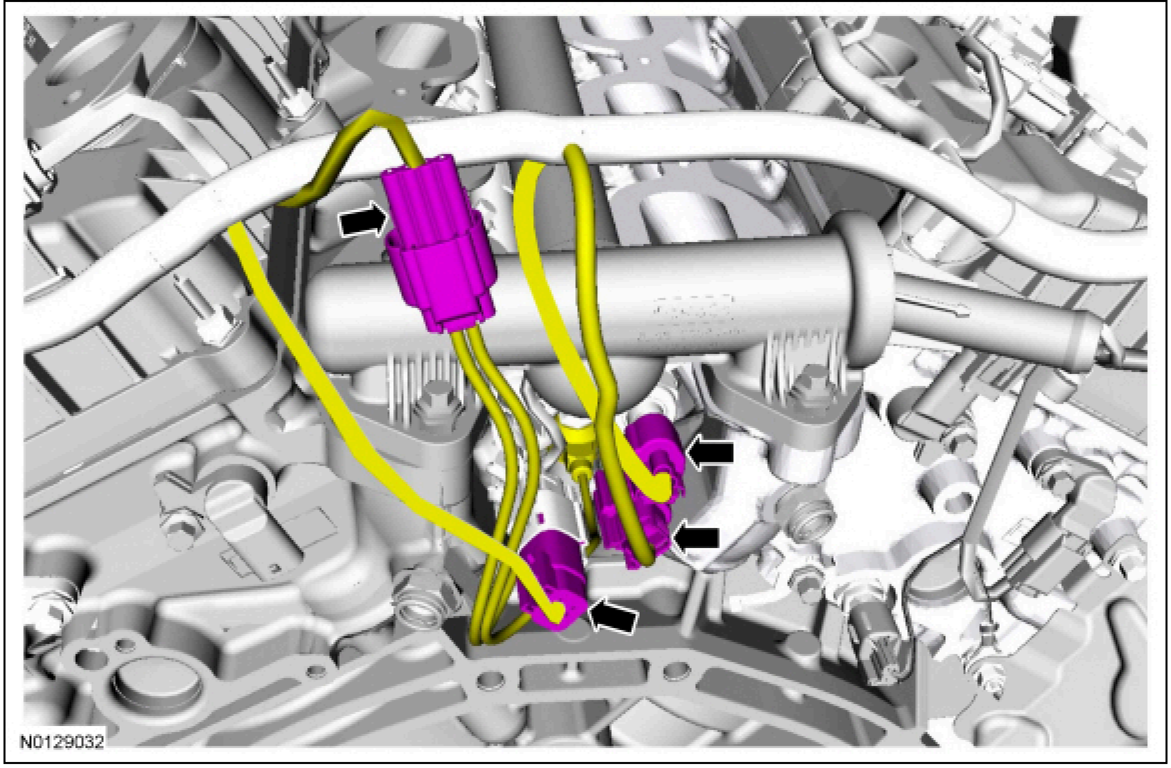
174. Connect the CHT sensor electrical connector. Install the ground stud bolt and the tie strap.
    - Tighten to 10 Nm (89 lb-in) then an additional 45 degrees.



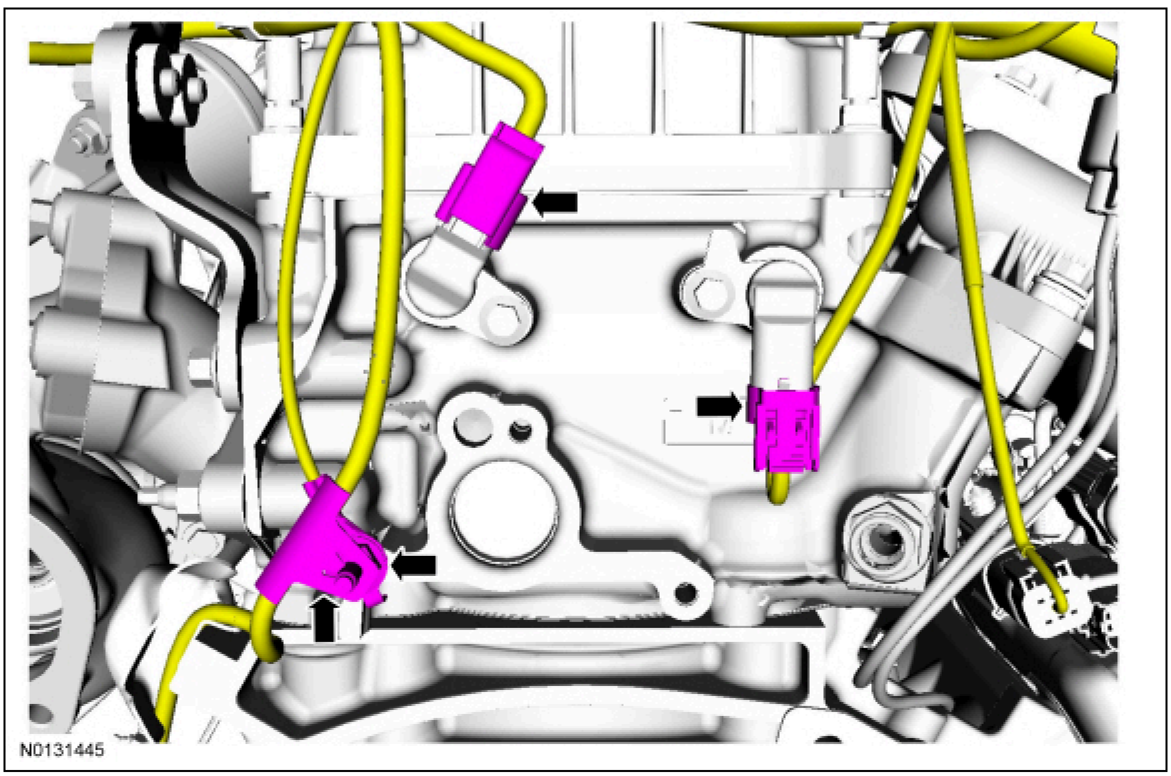
175. Install the CMP heat shield, bolt and the nut. Connect the wire retainer.
    - Tighten the bolt to 10 Nm (89 lb-in).
    - Tighten the nut to 10 Nm (89 lb-in).



176. Connect the 2 fuel charge harness and the Fuel Rail Pressure (FRP) electrical connectors. Connect the KS sensor electrical connector.

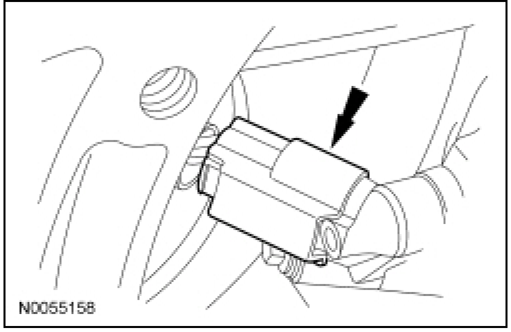


177. Connect the 2 LH CMP sensor electrical connectors. Install the ground stud bolt. Install the CKP wire retainer.
    - Tighten to 10 Nm (89 lb-in) then an additional 45 degrees.

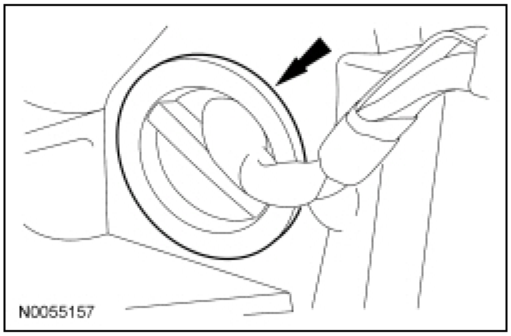


178. Attach all of the wiring harness retainers to the LH valve cover and stud bolts.

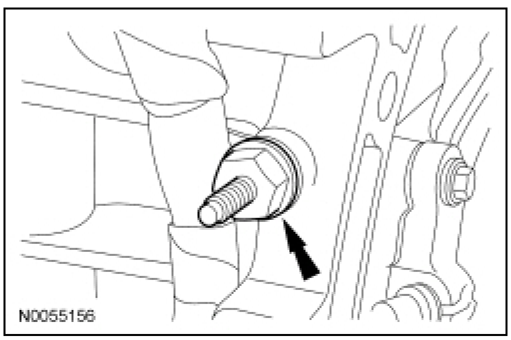
179. Connect the CKP sensor electrical connector.



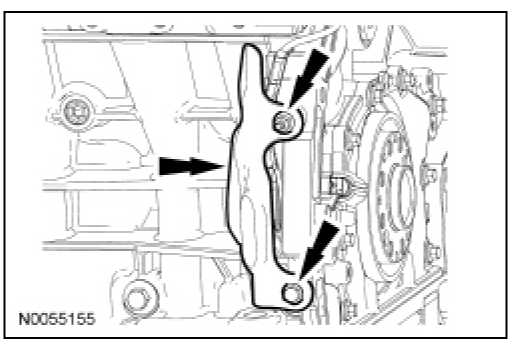
180. Install the wiring harness grommet.



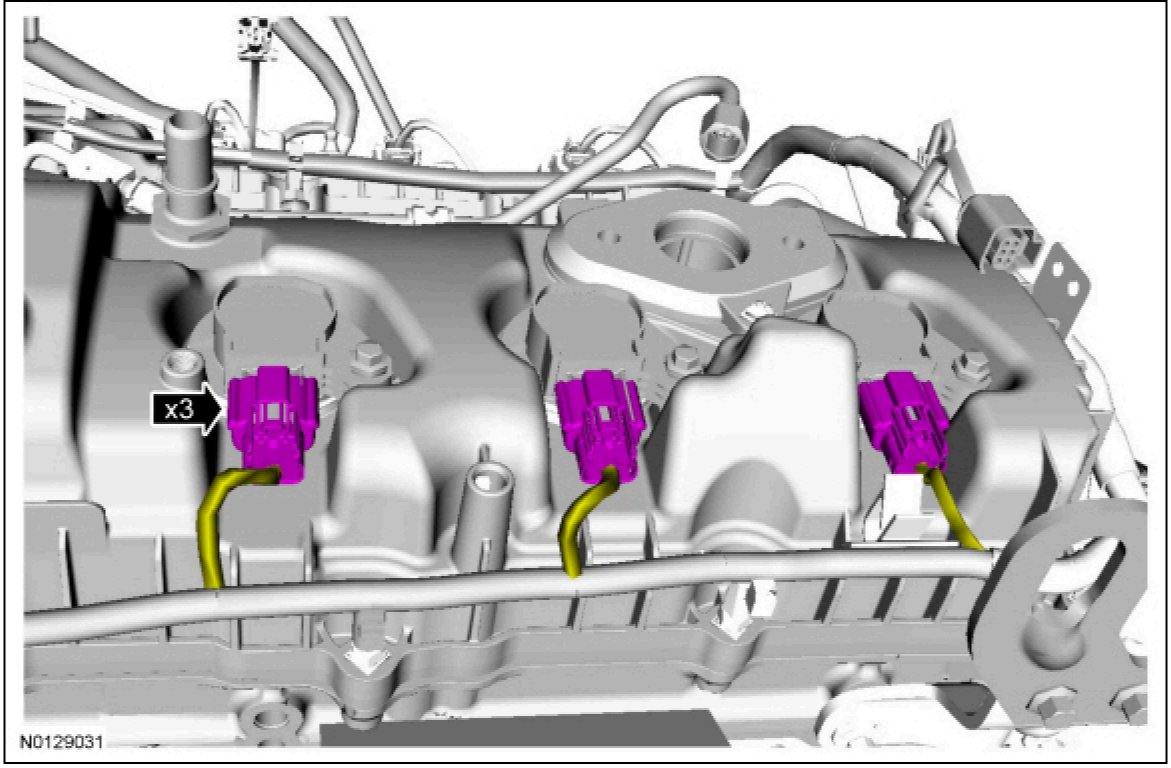
181. Install the wiring harness retainer stud bolt.
    - Tighten to 10 Nm (89 lb-in).



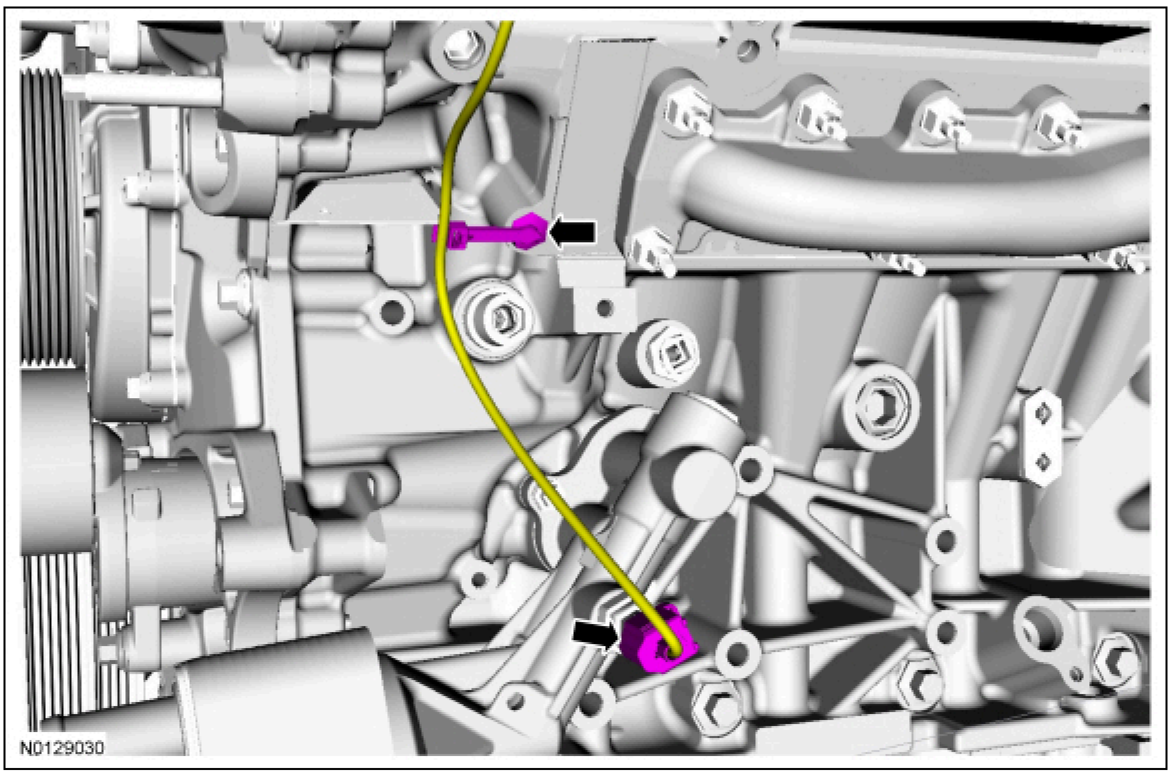
182. Install the CKP heat shield, the nut and the bolt.
    - Tighten to 10 Nm (89 lb-in).



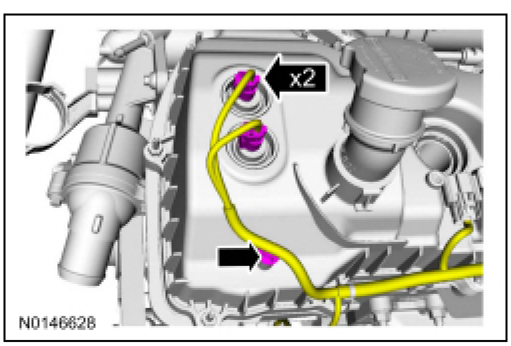
183. Connect the 3 LH ignition coil-on-plug electrical connectors.



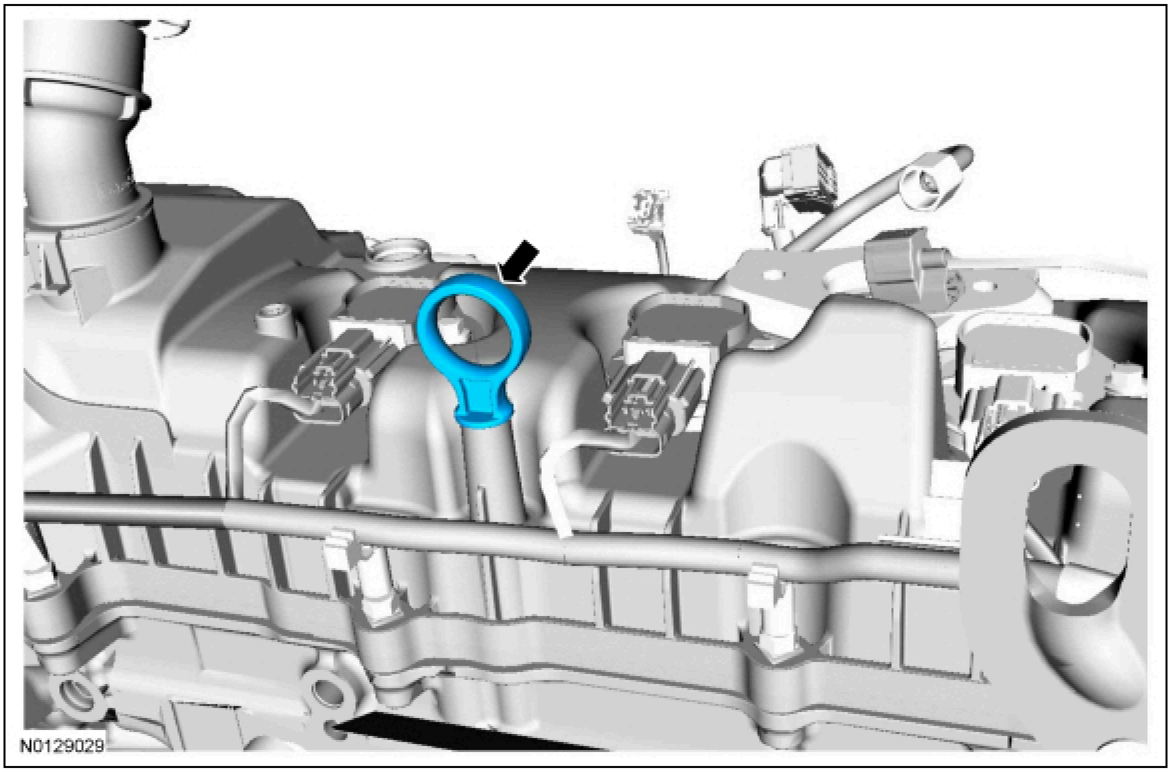
184. Connect the Engine Oil Pressure (EOP) switch electrical connector.
    - Attach the EOP switch wiring harness retainer to the cylinder head.



185. Connect the 2 LH VCT solenoid electrical connectors. Install the tie strap.

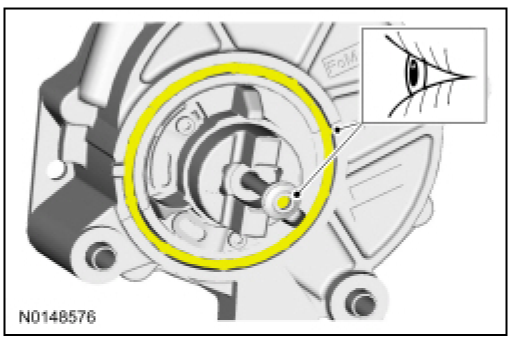


186. Install the engine oil level indicator.



187. **NOTICE: Make sure the oil tube filter screen is free of contaminants prior to the installation of the vacuum pump or damage to components may occur.**

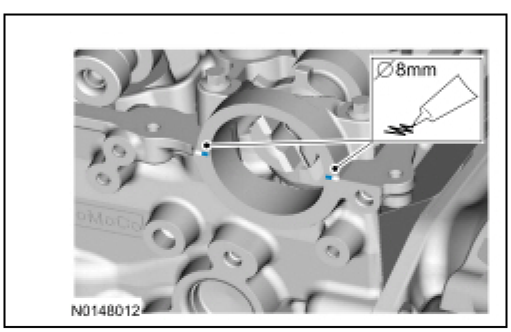
Visual check. Install a new gasket if necessary.



188. **NOTICE: If the brake vacuum pump is not secured within 10 minutes of sealant application, the sealant must be removed and the sealing area cleaned with metal surface prep. Failure to follow this procedure can cause future oil leakage.**

NOTE: *Make sure the mating surfaces are clean and free of foreign material.*

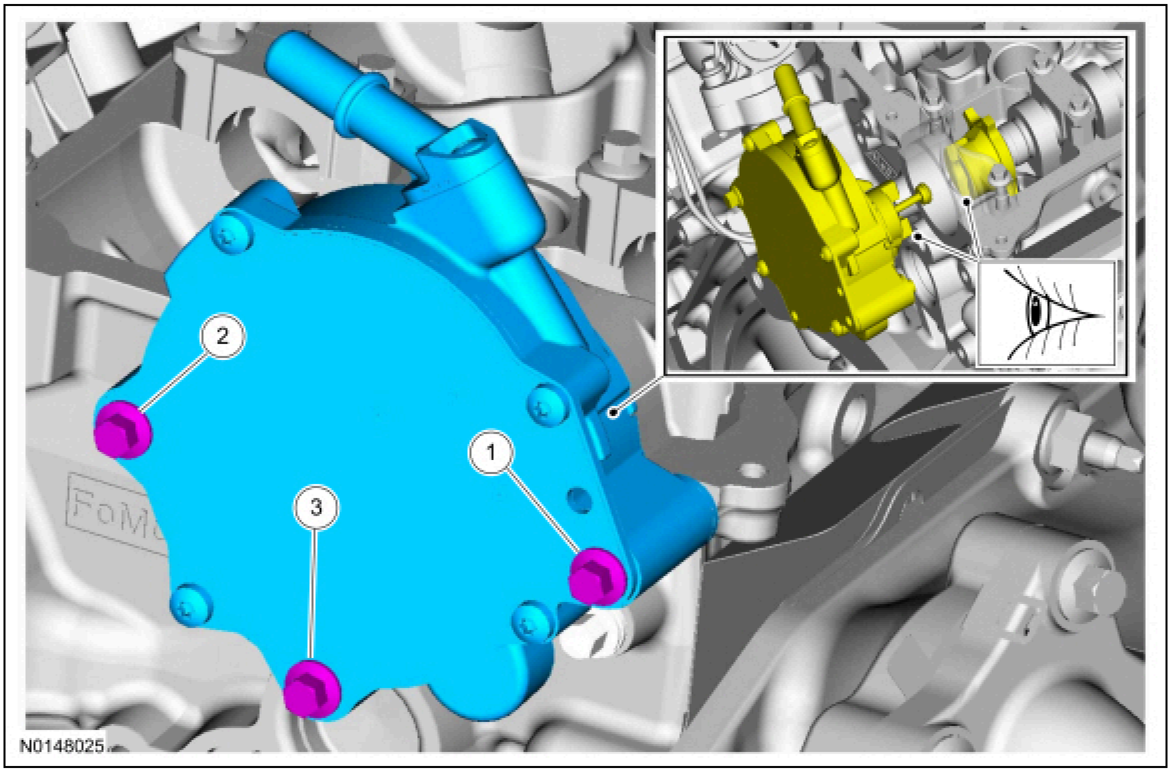
Apply a bead of the specified diameter from the specified tube.



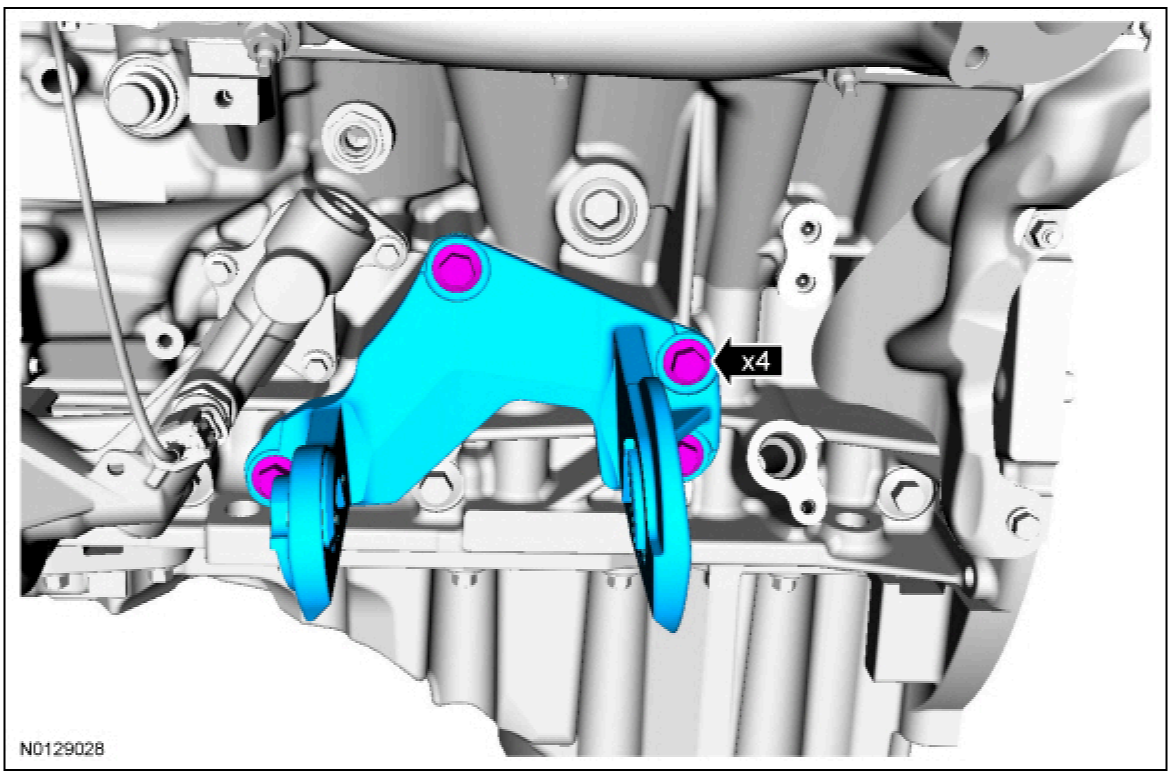
189. **NOTE:** *Manually align the brake vacuum pump drive key with the camshaft slot before installation.*

Visual check.

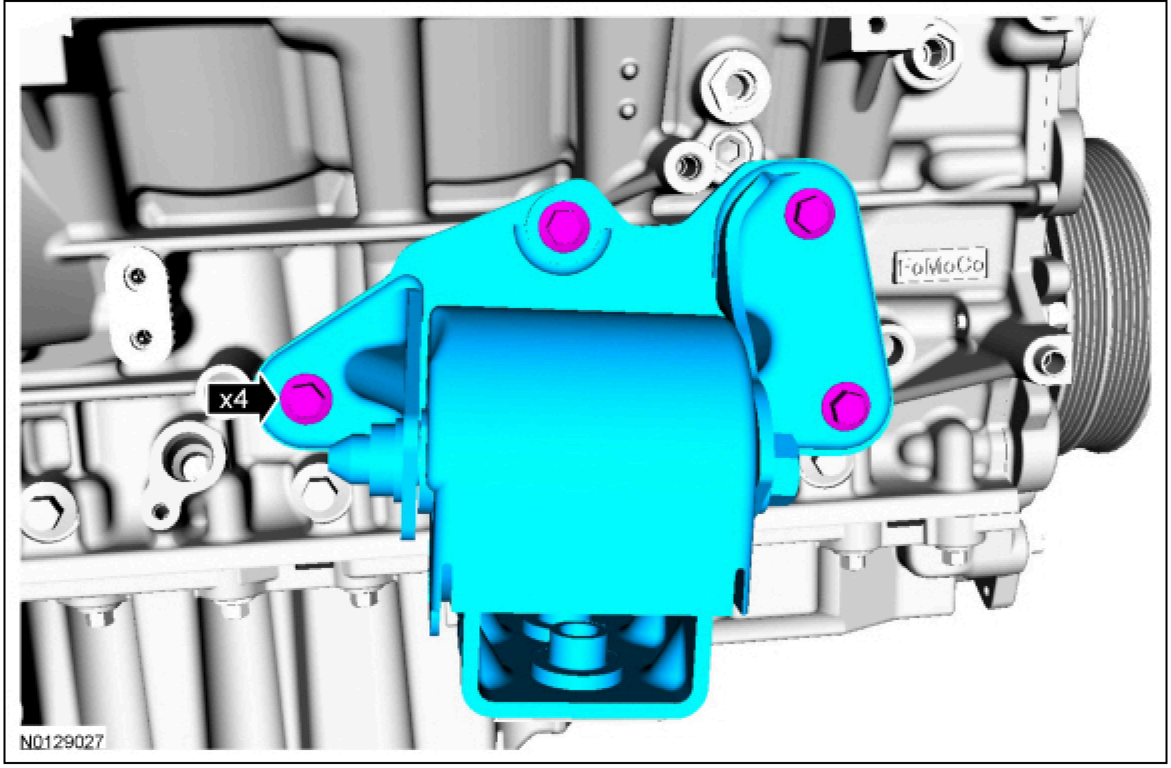
- Tighten in the sequence shown to 10 Nm (89 lb-in) then an additional 45 degrees.



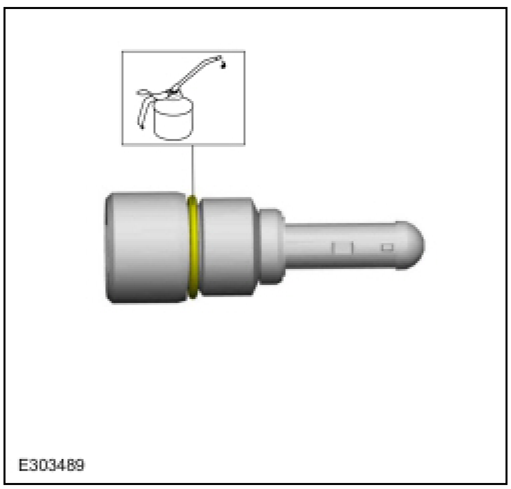
190. Install the LH engine support insulator-to-cylinder block bracket and the 4 new bolts.
    - Tighten to 63 Nm (46 lb-ft).



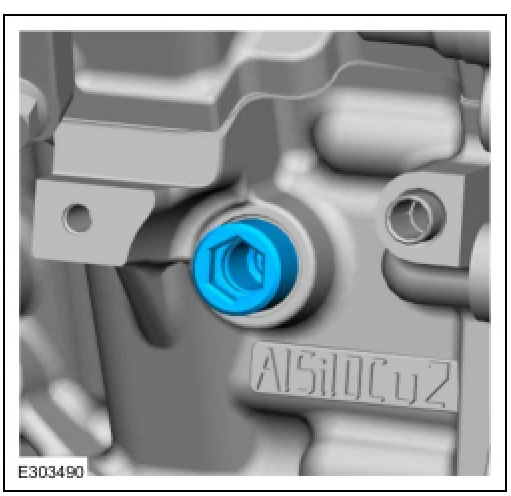
191. Install the RH engine support insulator and the 4 new bolts.
    - For standard payload package, tighten to 63 Nm (46 lb-ft).
    - For heavy duty payload package, tighten to 77 Nm (57 lb-ft).



192. Lubricate the turbocharger fitting and oil filter assembly O-ring with clean engine oil.



193. Install the turbocharger fitting and oil filter assembly.
    - Tighten to 32 Nm (24 lb-ft).

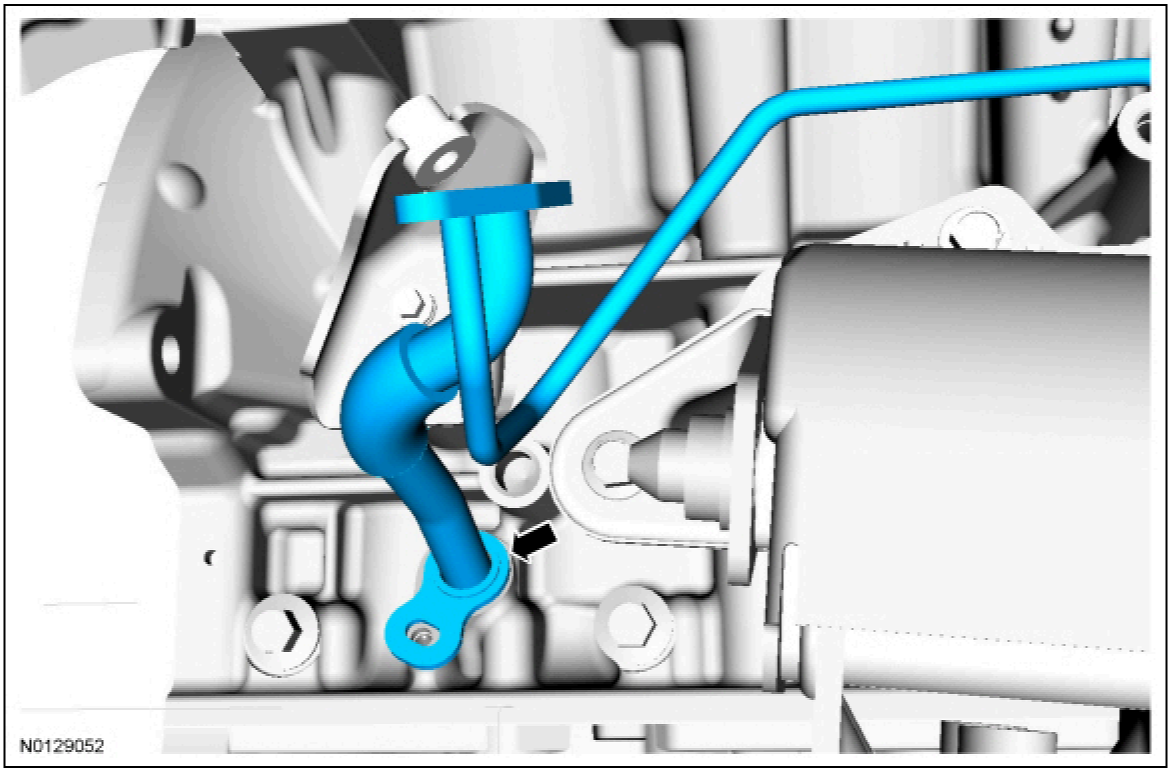


194. **NOTE:** *Lubricate the cylinder block bore with clean engine oil.*

    NOTE: *Apply clean engine oil to the O-ring seal.*

NOTE: *Make sure the turbocharger oil supply tube in positioned in the cylinder block during the installation of the turbocharger oil return tube.*

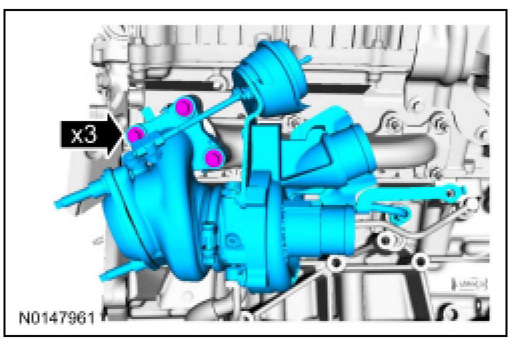
Using a new O-ring seal, install the RH turbocharger oil return tube.



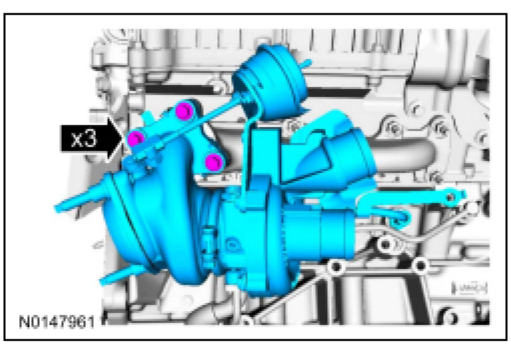
195. **NOTE:** *Install the RH turbocharger coolant supply tube in the turbocharger prior to installation.*

Using a new turbocharger exhaust manifold gasket, install the RH turbocharger and the 3 bolts.

- Tighten finger-tight.

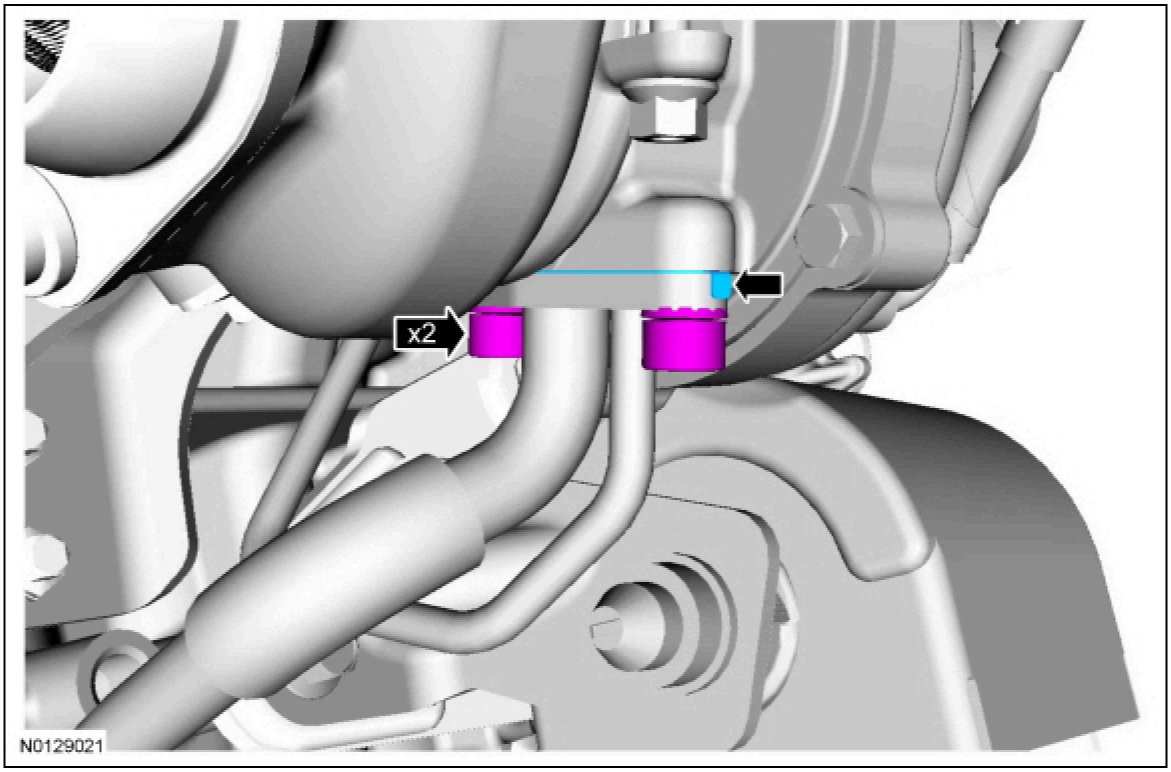


196. Tighten the RH turbocharger mounting bolts.
   - Tighten to 32 Nm (24 lb-ft).

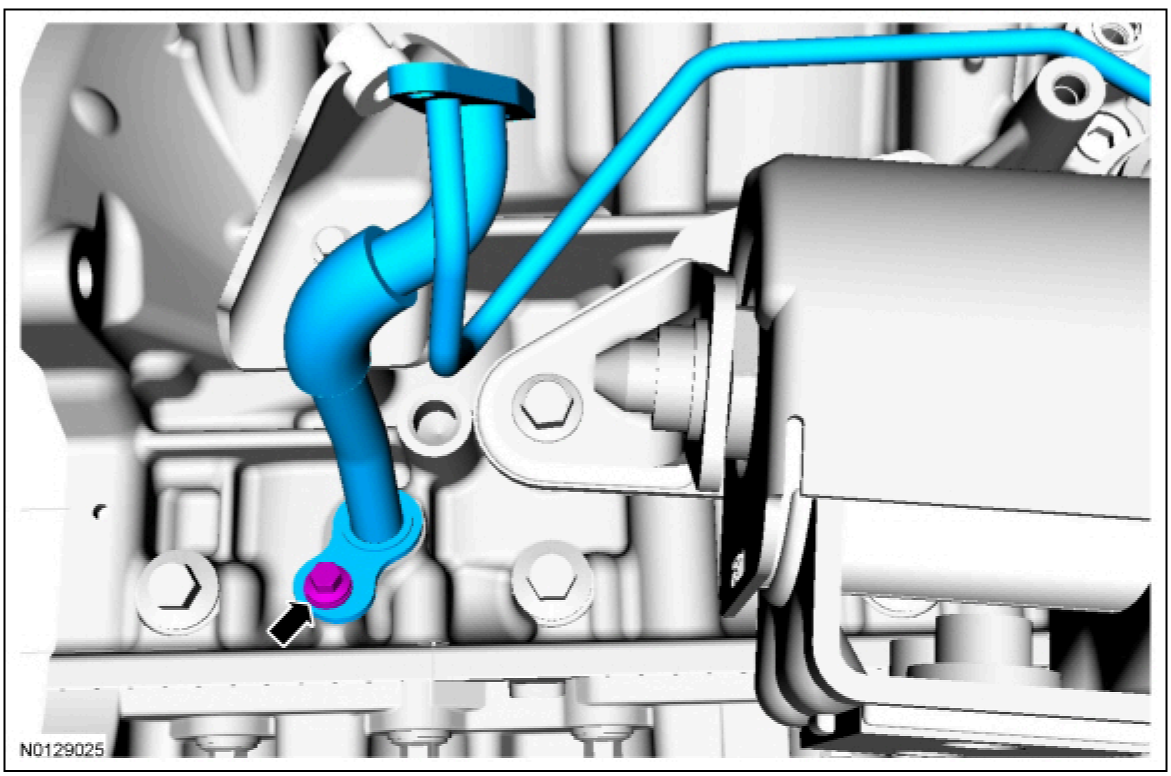


197. Using a new turbocharger oil return tube gasket, install the 2 bolts at the turbocharger. Tighten in the following stages.
   - Stage 1: Install the bolt for the oil pressure tube side halfway.
   - Stage 2: Install the bolt for the oil drain tube side and tighten to 10 Nm (89 lb-in).
   - Stage 3: Tighten the bolt for the oil drain tube side an additional 30 degrees.

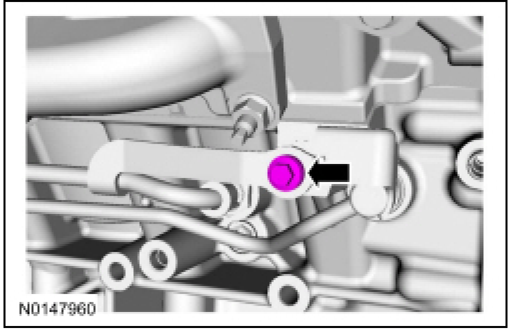
- Stage 4: Tighten the bolt for the oil pressure tube side to 10 Nm (89 lb-in).
- Stage 5: Tighten the bolt for the oil pressure tube side an additional 30 degrees.



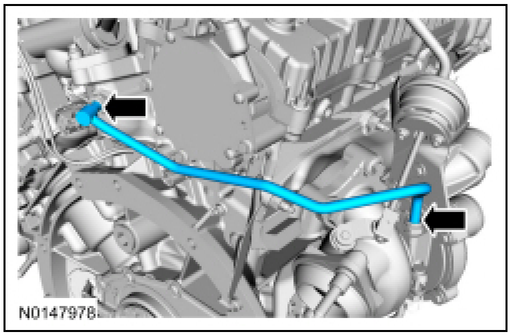
198. Install the RH turbocharger oil return tube bolt at the cylinder block. Tighten in the following stages.
    - Stage 1: Tighten to 8 Nm (71 lb-in).
    - Stage 2: Tighten an additional 30 degrees.



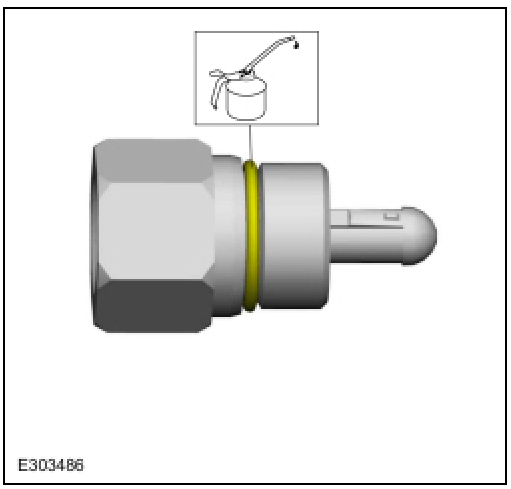
199. Install the bolt for the RH turbocharger coolant and oil tubes. Tighten in the following stages.
    - Stage 1: Tighten to 8 Nm (71 lb-in).
    - Stage 2: Tighten an additional 30 degrees.



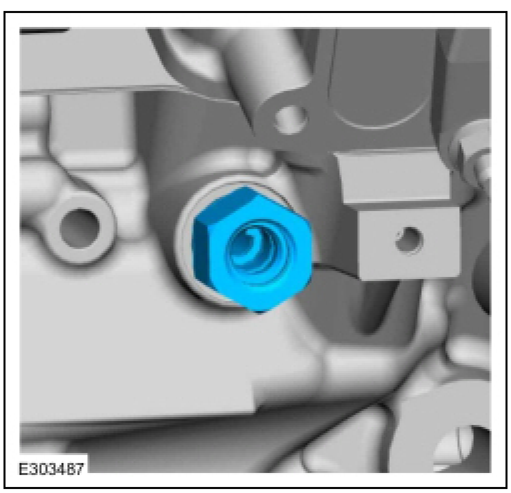
200. Install the RH turbocharger coolant return tube.



201. Lubricate the turbocharger fitting and oil filter assembly O-ring with clean engine oil.



202. Install the turbocharger fitting and oil filter assembly.
- Tighten to 48 Nm (35 lb-ft).

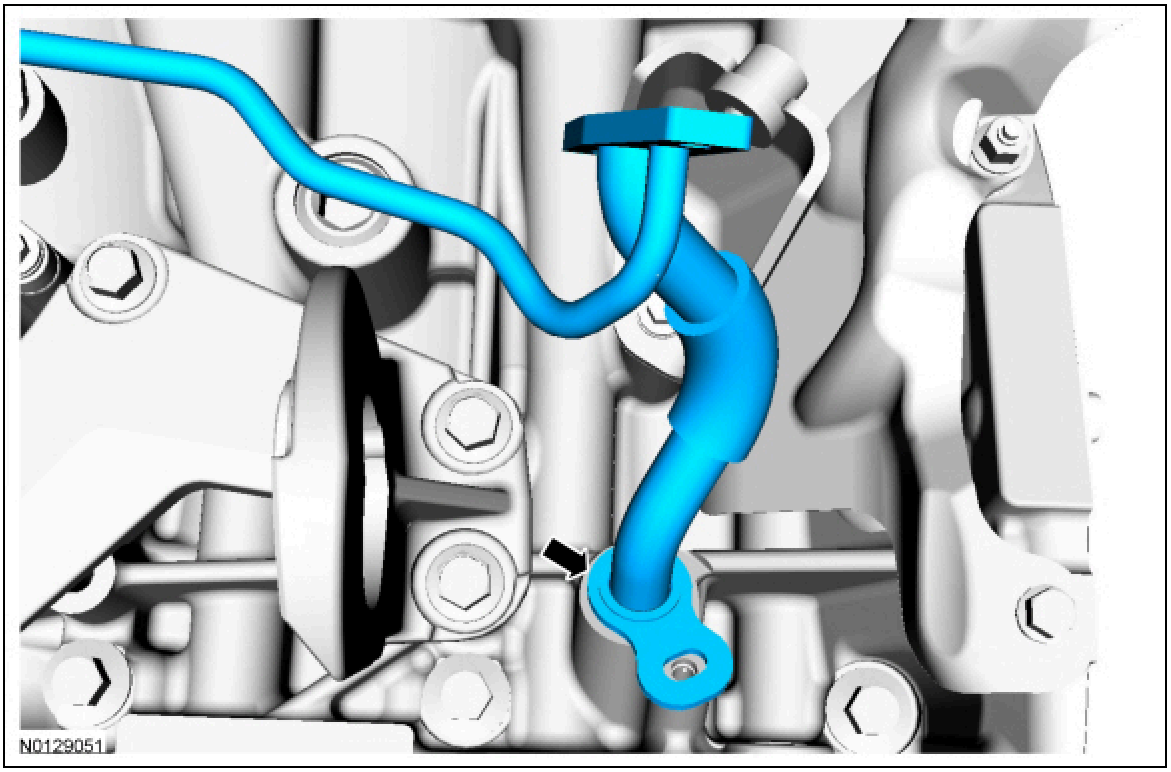


203. **NOTE:** *Lubricate the cylinder block bore with clean engine oil.*

NOTE: *Apply clean engine oil to the O-ring seal.*

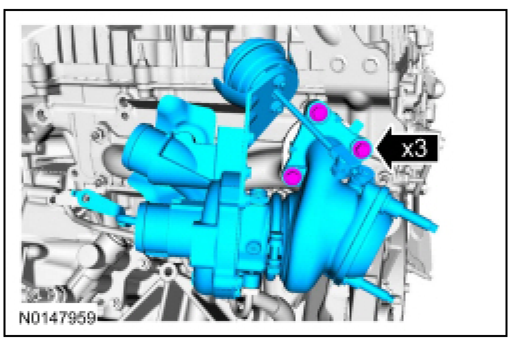
NOTE: *Make sure the turbocharger oil supply tube in positioned in the cylinder block during the installation of the turbocharger oil return tube*.

Using a new O-ring seal, install the LH turbocharger oil return tube.

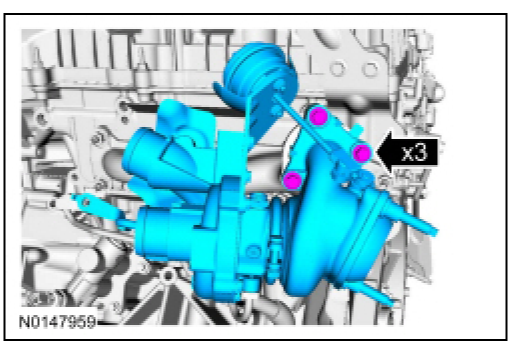


204. **NOTE:** *Install the LH turbocharger coolant supply tube in the turbocharger prior to installation*.

Using a new turbocharger exhaust manifold gasket, install the LH turbocharger and the 3 bolts.
- Tighten finger-tight.

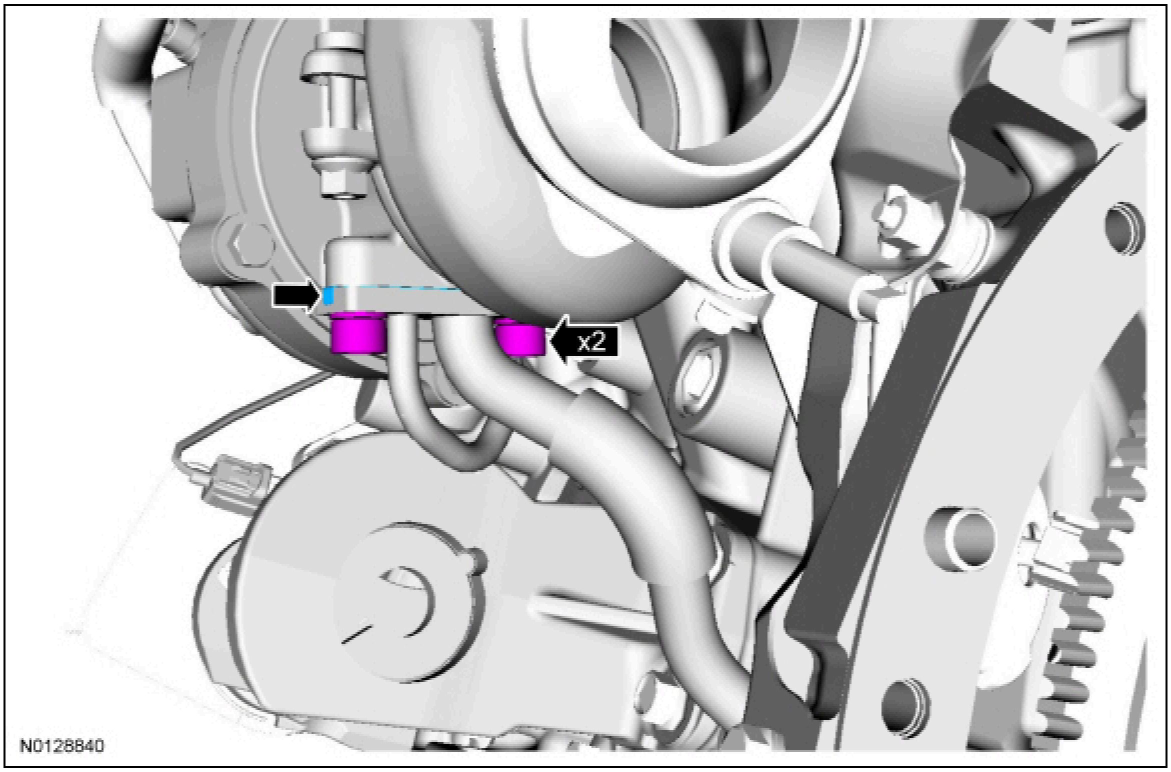


205. Tighten the LH turbocharger mounting bolts.
- Tighten to 32 Nm (24 lb-ft).

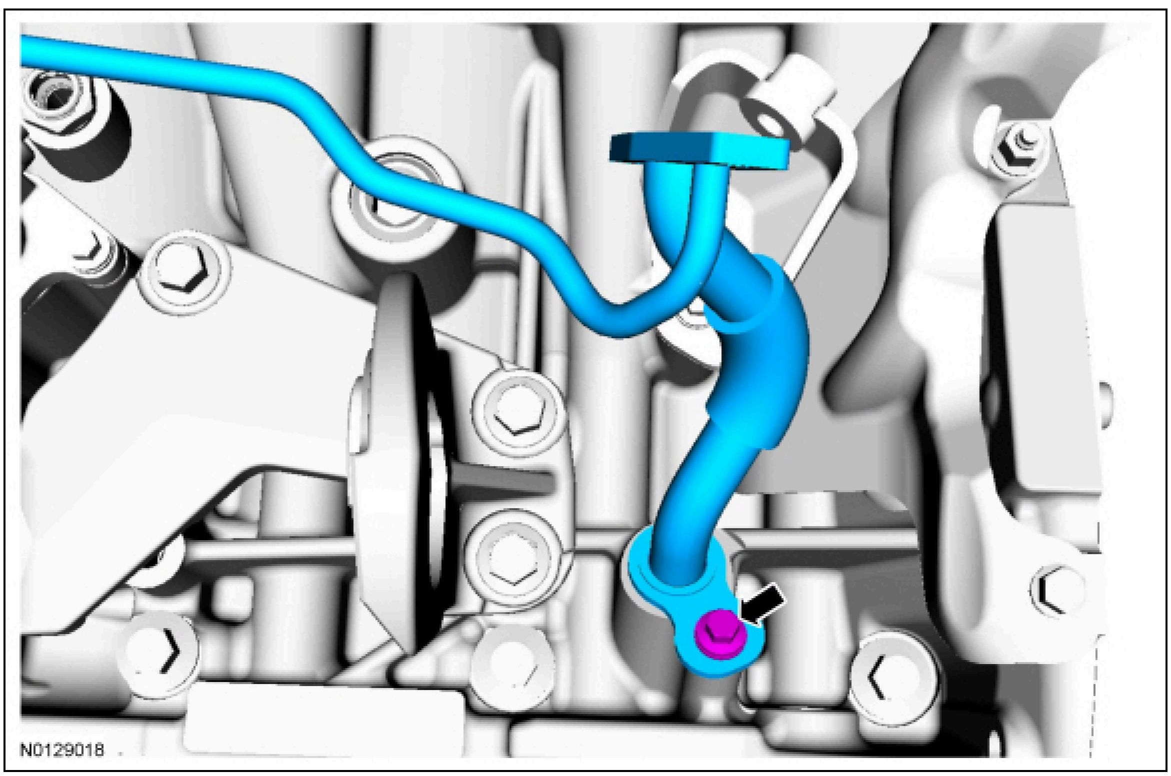


206. Using a new turbocharger oil return tube gasket, install the 2 bolts at the turbocharger. Tighten in the following stages.
- Stage 1: Install the bolt for the oil pressure tube side halfway.
- Stage 2: Install the bolt for the oil drain tube side and tighten to 10 Nm (89 lb-in).
- Stage 3: Tighten the bolt for the oil drain tube side an additional 30 degrees.

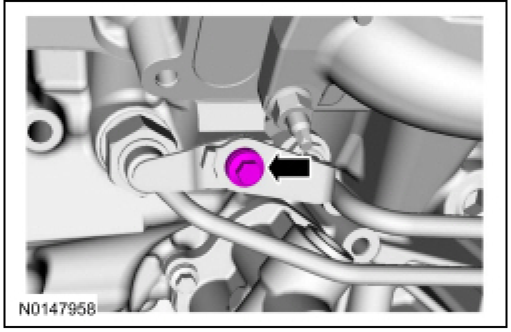
- Stage 4: Tighten the bolt for the oil pressure tube side to 10 Nm (89 lb-in).
- Stage 5: Tighten the bolt for the oil pressure tube side an additional 30 degrees.



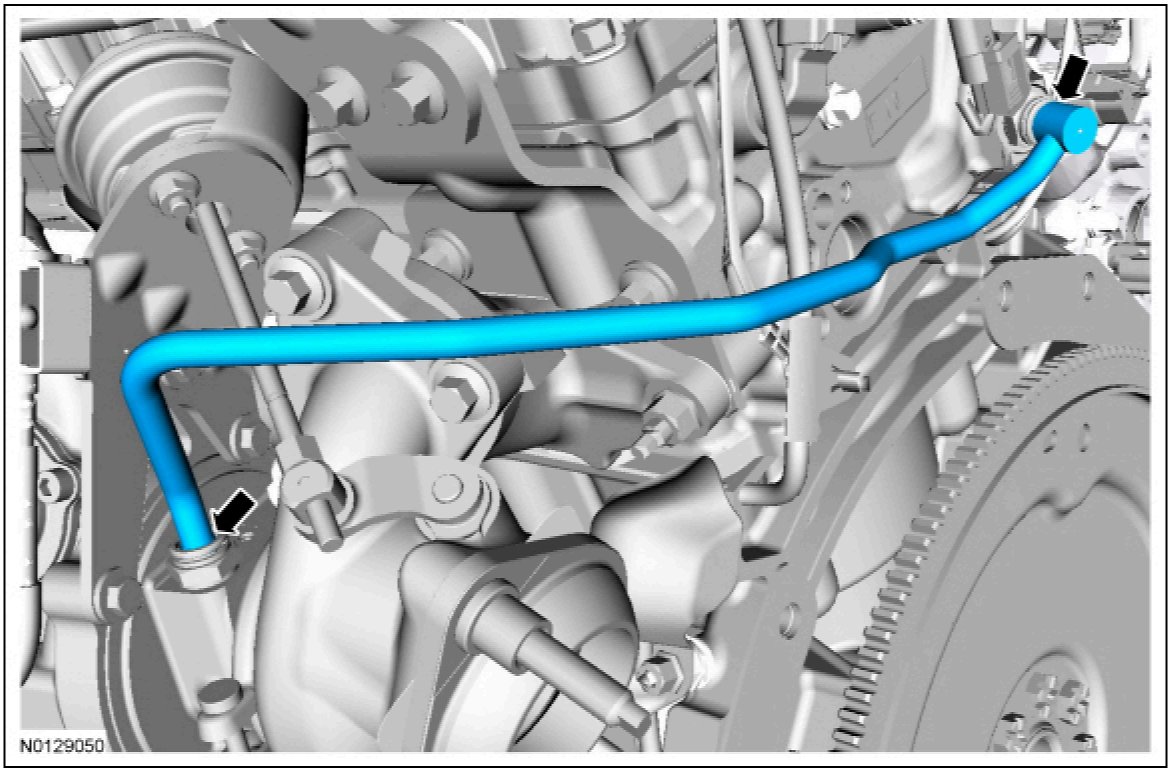
207. Install the LH turbocharger oil return tube bolt at the cylinder block. Tighten in the following stages.
    - Stage 1: Tighten to 8 Nm (71 lb-in).
    - Stage 2: Tighten an additional 30 degrees.



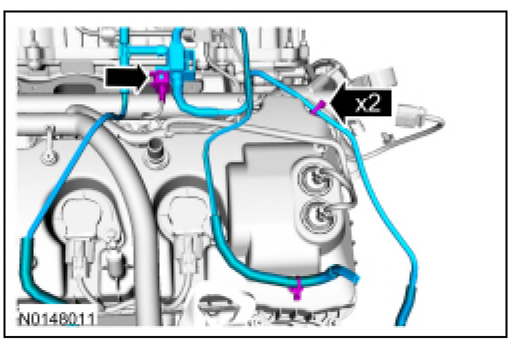
208. Remove the bolt for the LH turbocharger coolant and oil tubes. Tighten in the following stages.
    - Stage 1: Tighten to 8 Nm (71 lb-in).
    - Stage 2: Tighten an additional 30 degrees.



209. Install the LH turbocharger coolant return tube.

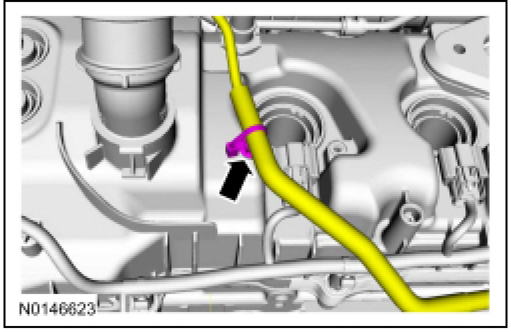


210. Install the wastegate hose assembly and 2 tie straps. Connect the turbocharger pressure regulator electrical connector.



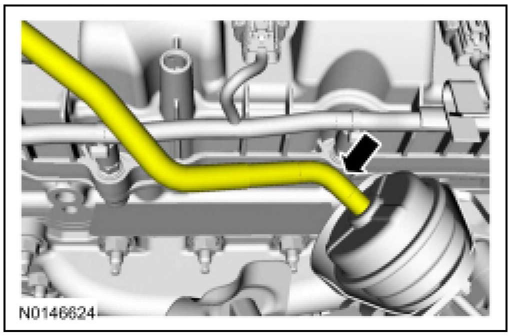
211. **NOTE:** *LH shown, RH similar.*

Position the wastegate hoses and install the 2 tie straps.

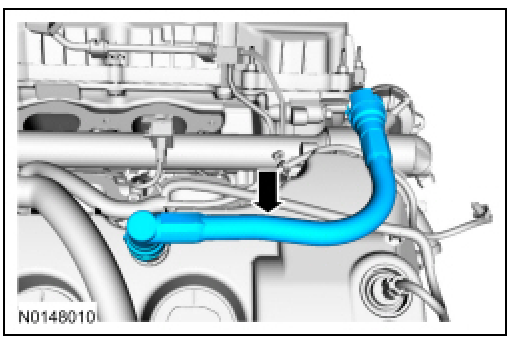


212. **NOTE:** *LH shown, RH similar.*

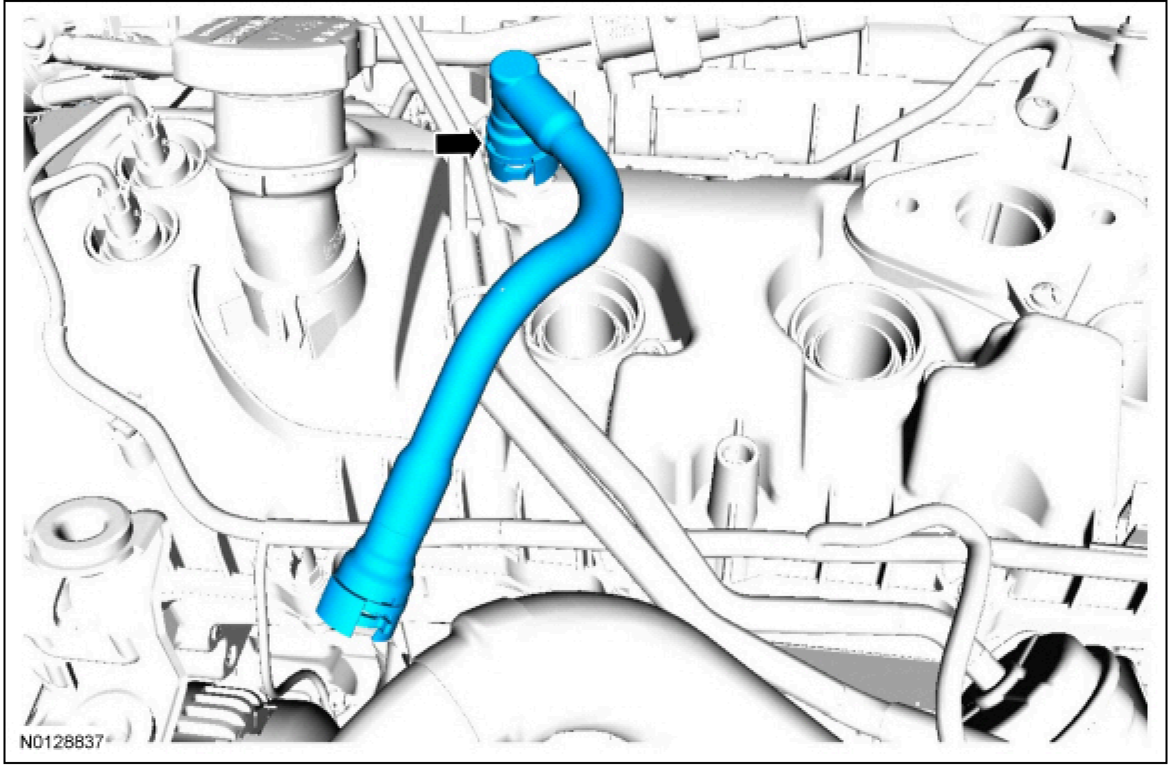
Connect the 2 wastegate hoses at the turbochargers.



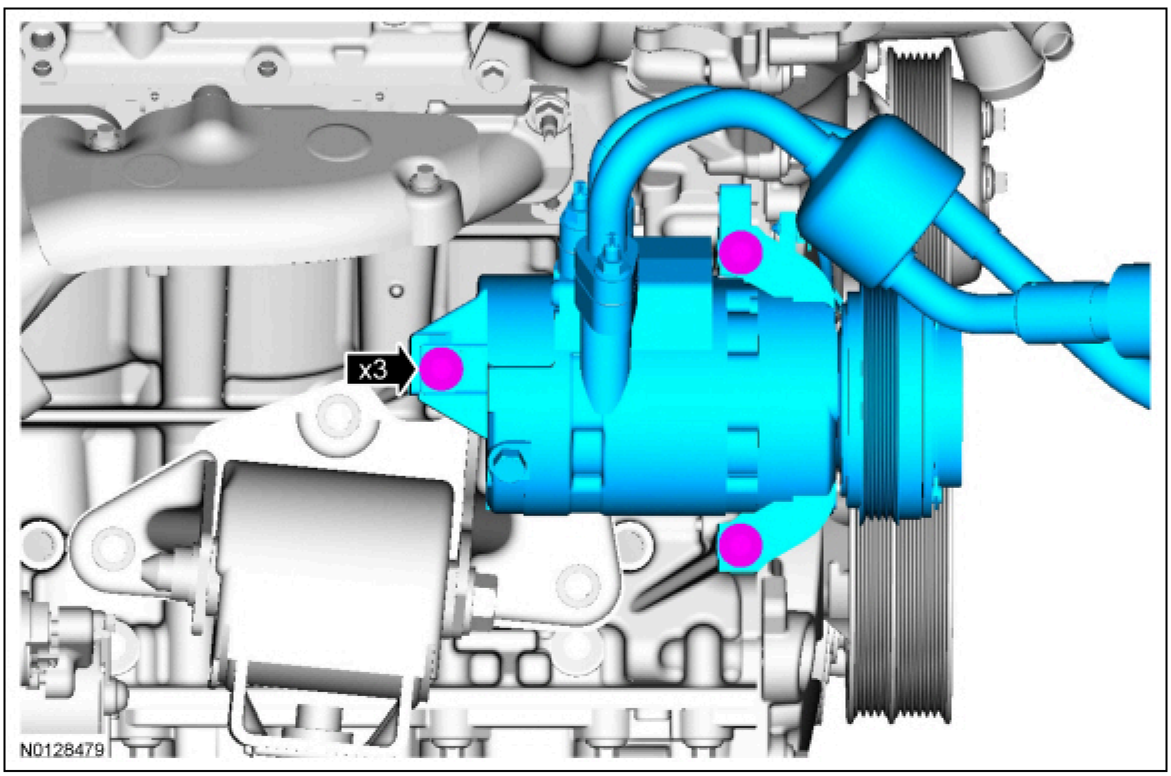
213. Install the RH PCV tube. For additional information, refer to Section 310-00.



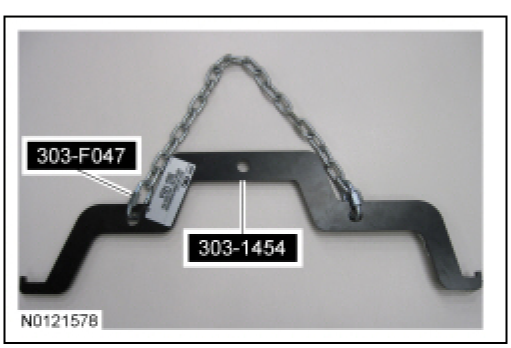
214. Install the LH crankcase vent tube. For additional information, refer to Section 310-00.



215. Install the A/C compressor and the 3 bolts.
    - Tighten to 25 Nm (18 lb-ft).

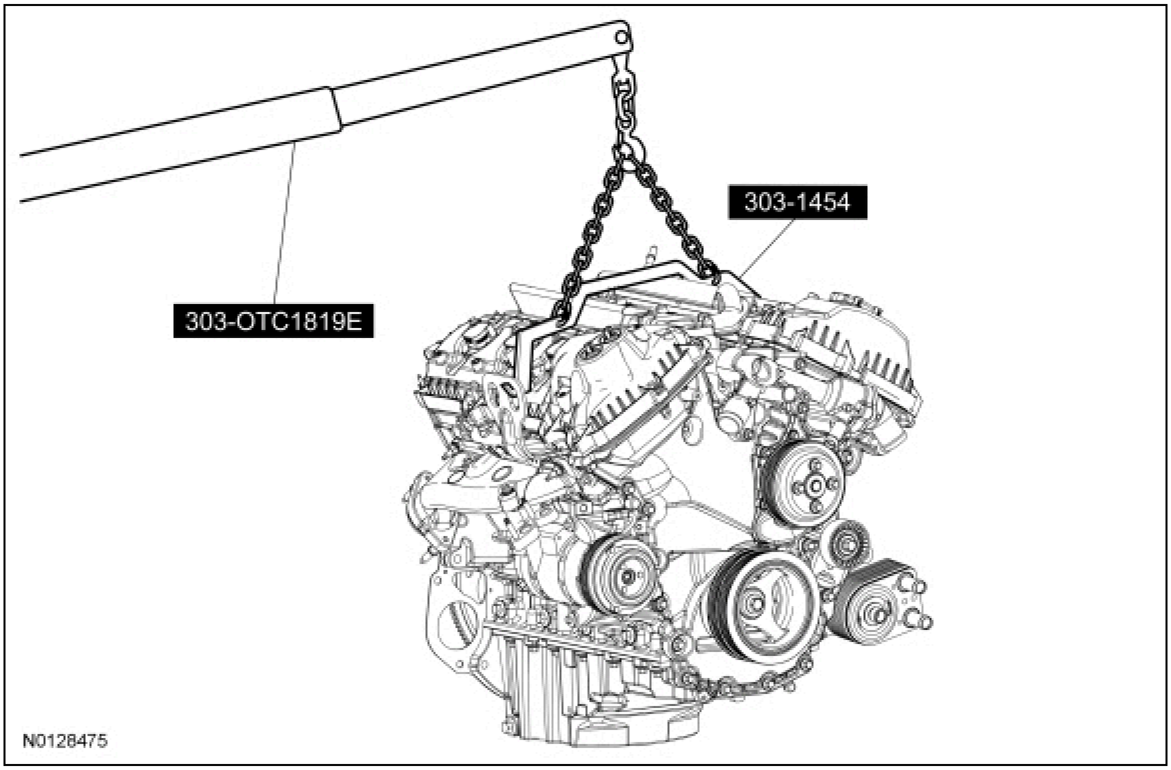


216. Using part of the Engine Lifting Bracket and chain, assemble the Engine Spreader Bar as shown.

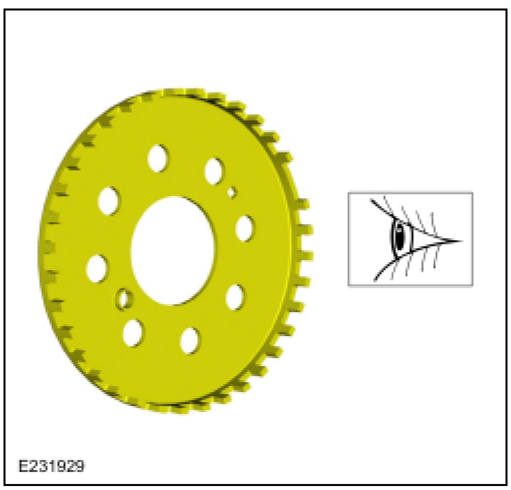


217. **NOTE:** *Use a commercially available quick link on the right side between the lifting eye and the engine spreader bar.*

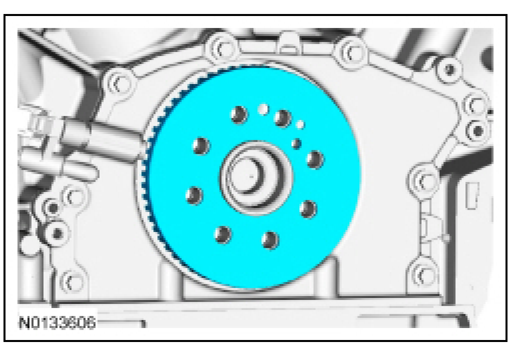
Using the Floor Crane and Engine Spreader Bar, remove the engine from the stand.



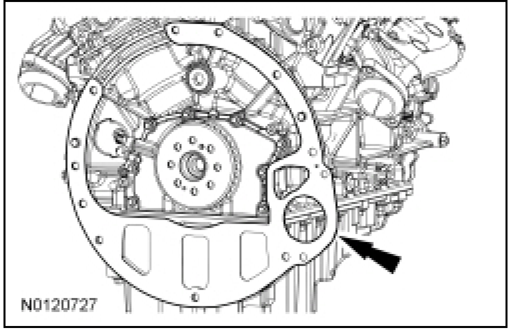
218. Inspect the ignition pulse ring for damage. If the ignition pulse ring has been dropped or has any visual damage, it must be replaced.



219. Install the ignition pulse ring.



220. Install the spacer plate.



221. Install the flexplate and the 8 bolts and tighten is sequence shown.
    - Tighten to 80 Nm (59 lb-ft).